Theta correlations, Laguerre signs, and hypercomplex jets of the theta heat flow

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Abstract

The physical first jet of the theta heat flow has energy $E = H^2 + H_x^2$. Its evolution at two stationary sets is governed by $\mathcal{L}_0 = H_x^2 - HH_{xx}$ and $\mathcal{L}_1 = H_{xx}^2 - H_xH_{xxx}$. We express both forms and E through one complete-source correlation family, retaining the full arithmetic interactions and the original cutoff errors. The octonionic observations reduce to planar dot products and determinants. A reflected monogenic extension preserves the physical four-jet without the auxiliary correction required by a stronger off-slice prescription. Exact calibration and logarithmic-norm identities clarify which information these geometric descriptions retain. For the theta source, both full Laguerre forms are strictly positive at sufficiently large height when $0 < t \le 1/2$ and $t\log(|x|/(4\pi)) \ge 61/24$, including the closed boundary; the starting height is unevaluated. Complete-source certificates give both stationary signs and $\sqrt{E} > 10^{-17}$ on $0 \le t \le 1/2$, $|x| \le 112$. Collision expansions and analytic rigidity turn prescribed all-cutoff lower bounds into a sufficient logarithmic-loss criterion. The reviewed octonionic, Fourier-kernel, and exceptional-theta theories identify possible tools, but do not supply the missing theta-specific signed estimate. The inner cusp and the finite transition remain open.

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1 The theta heat flow and the main result

The four real derivatives

$$h_j(t,x) = \partial_x^j H(t,x), \qquad j = 0,1,2,3, \tag{1.1}$$

form the smallest consecutive jet containing the collision detector (H, H_x) and both quadratic expressions

$$\mathcal{L}_0 = h_1^2 - h_0 h_2, \qquad \mathcal{L}_1 = h_2^2 - h_1 h_3. \tag{1.2}$$

Here "four jets" means these four consecutive derivative components. Their common scalar origin is essential: they are not four independent fields. The first two give the quaternionic observation $F = h_0 + \mathbf{j}h_1$ and its energy $E = h_0^2 + h_1^2$. Since $H_t = -H_{xx}$,

$$\frac{1}{2}E_t = -h_0h_2 - h_1h_3 = \begin{cases} \mathcal{L}_0, & h_1 = 0, \\ \mathcal{L}_1, & h_2 = 0. \end{cases} \tag{1.3}$$

The stationary sets in this identity are those of the original scalar function and its first derivative. Their signs will remain the same objects throughout the analytic and hypercomplex constructions.

1.1 The fixed source and normalization

The family under consideration is

$$\Phi(u) = \sum_{n\geq 1} \phi_n(u), \qquad \phi_n(u) = (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) e^{-\pi n^2 e^{4u}}, \tag{1.4}$$

$$H(t,z) = \int_0^\infty e^{tu^2} \Phi(u) \cos(zu)\, du, \qquad H(0,z) = \frac{1}{8}\xi\left(\frac{1}{2} + \frac{iz}{2}\right). \tag{1.5}$$

We use the convention $\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)/2$, agreeing with [3]. In particular the critical line corresponds to real z, and the heat equation has the backward sign

$$\partial_t H = -\partial_z^2 H. \tag{1.6}$$

For completeness, the normalization follows directly from the theta Mellin identity. Put $A(u) = e^u \sum_{n\geq 1} e^{-\pi n^2 e^{4u}}$. Poisson summation makes $P(u) = e^u + 2A(u)$ even, and differentiation

gives $\Phi = (P'' - P)/16 = (A'' - A)/8$. Splitting the theta Mellin integral at one and substituting $y = e^{4u}$ gives

$$\xi(1/2 + iz/2) = \frac{1}{2} - (1 + z^2) \int_0^\infty A(u) \cos(zu)\, du.$$

Evenness of P gives $A'(0) = -1/2$. Two integrations by parts now prove (1.5); all positive-infinity boundary terms vanish. This also proves that the complete source Φ is even. For $u \geq 0$ its terms are positive, and their successive majorant ratios yield

$$0 < \Phi(u) < 4\pi^2 e^{9u - \pi e^{4u}}. \tag{1.7}$$

Indeed the ratio of successive terms of $n^4 e^{-\pi n^2 e^{4u}}$ is at most $16e^{-3\pi} < 1/2$. The bound justifies differentiation of every order on compact sets of complex (t, z). Thus H is jointly entire, even in z, and real on real (t, z); (1.6) follows under the integral. For each fixed real time it is a real entire function of order at most one, as is also immediate from the same source bound by splitting the integral at $u = O(\log(2 + |z|))$.

1.2 Uniform positivity outside the inner region

The leading result concerns the dimensionless parameter $t \log |x|$, which remains nontrivial as time tends to zero while height increases.

Theorem 1.1 (Uniform four-jet asymptotic)**.** *There is a finite, presently unevaluated $X_* > 4\pi$ with the following property. Uniformly for*

$$x \geq X_*, \qquad 0 < t \leq \tfrac{1}{2}, \qquad \theta := tL \geq 61/24, \qquad L := \log \frac{x}{4\pi}, \tag{1.8}$$

there is an explicitly defined nonzero complex amplitude $Q = Q_t(x)$ such that, with $\omega = L/4$ and $\phi = \arg Q$,

$$\frac{1}{2|Q|} \left(h_0, \frac{h_1}{\omega}, \frac{h_2}{\omega^2}, \frac{h_3}{\omega^3} \right) = (\cos\phi, -\sin\phi, -\cos\phi, \sin\phi) + O(L^{-1}). \tag{1.9}$$

Consequently

$$h_0^2 + (h_1/\omega)^2 = 4|Q|^2(1 + O(L^{-1})), \tag{1.10}$$

$$\mathcal{L}_0 = 4|Q|^2\omega^2(1 + O(L^{-1})), \qquad \mathcal{L}_1 = 4|Q|^2\omega^4(1 + O(L^{-1})). \tag{1.11}$$

After increasing X_*, both Laguerre forms are strictly positive everywhere in (1.8), and $F(t, x) \neq 0$ there. Evenness gives the reflected statements at negative heights.

The theorem combines the effective approximation in [3, Theorem 1.3] with five exponent-pair estimates, uniformly in all four derivatives. Strict rational margins absorb the positive arithmetic losses and the complex-disc crossing, so the lower boundary 61/24 is included exactly. The starting height remains unevaluated; no optimality claim is made. Lemma 2.4 gives a conditional continuation to 5/2, while Proposition 10.1 records a weaker sufficient weighted-remainder hypothesis. Neither hypothesis is proved here. A different, more conservative region has a fully explicit height.

Proposition 1.2 (An explicit outer region)**.** *Set $X_{\exp} = 4\pi \exp(10^9)$. For $x \geq X_{\exp}$, $0 < t \leq 1/2$, and $tL \geq 5$, every coordinate error in (1.9) is at most $108000000/L \leq 0.108 < 1/8$. Consequently*

$$\frac{h_0^2 + (h_1/\omega)^2}{4|Q|^2} \geq \frac{1}{2}, \qquad \frac{\mathcal{L}_j}{4|Q|^2\omega^{2j+2}} \geq \frac{31}{64}, \quad j = 0, 1. \tag{1.12}$$

Reflection gives the negative-height statements.

The complete explicit estimates are in Appendix B. This enormous cutoff does not provide a practical bridge from height 112, and it is not an evaluated cutoff for the sharper 61/24 theorem.

The geometric interpretation is that a common nonzero amplitude controls all four jets, including at phases where individual leading components vanish. In quaternionic and octonionic coordinates, the same quadratic forms become signed geometric observations. Those constructions do not supply positivity by themselves; they identify precisely which part of the theta analysis is needed. A second, complementary result is the validated bounded cover

$$|F(t,x)| > 10^{-17} \quad (0 \le t \le 1/2,\ |x| \le 112), \tag{1.13}$$

together with both strict stationary signs there. Its complete-source proof and certificate structure are given in Section 7, including Theorem 7.3.

Real multiple zeros at positive times are the relevant obstruction: the classical heat-flow inputs imply

$$\mathrm{RH} \iff F(t,x) \neq 0 \quad \text{for every } t > 0 \text{ and } x \in \mathbb{R}. \tag{1.14}$$

Section 3 gives the reduction and the sufficient stationary bounded-loss criterion. Theorem 1.1 and (1.13) leave an inner small-time region and a finite transition strip. The remaining problem has two exact descriptions. Section 4 reduces the oriented octonionic observations to two-dimensional determinants and dot products, and gives a simpler reflected lift of the physical trace. Section 6 expresses both signs and the energy in one complete theta correlation family. Its hybrid version retains the full-amplitude cutoff used in Proposition 10.2. These reductions simplify the objects to be estimated; they add no unconditional sign region. The applicability of further octonionic analysis is assessed in Section 6.5.

2 Proof of the uniform outer-region theorem

2.1 A holomorphic comparison amplitude

Use the principal logarithm and set

$$M_0(s) = \frac{s(s-1)}{16}\pi^{-s/2}\sqrt{2\pi}\exp\left\{\left(\frac{s}{2}-\frac{1}{2}\right)\operatorname{Log}\frac{s}{2}-\frac{s}{2}\right\}, \tag{2.1}$$

$$\alpha(s) = \frac{M_0'(s)}{M_0(s)} = \frac{1}{2s}+\frac{1}{s-1}+\frac{1}{2}\operatorname{Log}\frac{s}{2\pi}, \qquad M_t(s) = e^{t\alpha(s)^2/4}M_0(s), \tag{2.2}$$

$$s_\pm(z) = \frac{1\pm iz}{2}, \qquad w_\pm(z) = s_\pm(z) + \frac{t}{2}\alpha(s_\pm(z)), \tag{2.3}$$

$$Q_\pm(z) = M_t(s_\pm(z))\zeta(w_\pm(z)), \qquad \mathcal{P}_t(z) = Q_+(z)+Q_-(z). \tag{2.4}$$

For real x we take $Q = Q_+(x)$; the other branch is its conjugate. The functions in these definitions are analytic on the small discs used below. Their arguments stay at imaginary parts approximately $\pm x/2$, away from all logarithmic cuts, while $\Re w_\pm > 1$ keeps the zeta arguments away from the pole.

Write $z = u+iv$, $L_u = \log(u/(4\pi))$, and

$$N = \left\lfloor \sqrt{u/(4\pi)+t/16} \right\rfloor, \qquad S_{t,N}(w) = \sum_{n\le N} e^{t\log^2 n/4} n^{-w}.$$

Polymath's estimate applies when $u \ge 200$, $0 \le v \le 1$, and $0 < t \le 1/2$. In the present notation it reads

$$H(t,z) = M_t(s_-(z))S_{t,N}(w_-(z)) + M_t(s_+(z))S_{t,N}(w_+(z)) + \mathcal{E}(t,z), \tag{2.5}$$

where $|\mathcal{E}| \le |M_t(s_-)|(e_A + e_B + e_{C,0})$. This is the exact two-branch expression of the cited theorem: its first exponent s_* equals w_- and its second exponent, after incorporating the factor n^v, is $\overline{s_*} + \kappa - v = w_+$. To retain the full imported error information, define

$$\gamma = \frac{M_t(s_+)}{M_t(s_-)}, \qquad \kappa = \frac{t}{2}\left(\alpha\left(\frac{1-v+iu}{2}\right) - \alpha\left(\frac{1+v+iu}{2}\right)\right).$$

The bounds needed from [3, eqs. (20)–(24)] are

$$|\gamma| \le e^{0.02v}(u/(4\pi))^{-v/2}, \qquad |\kappa| \le \frac{tv}{2(u-6)}, \tag{2.6}$$

$$e_A + e_B \le \sum_{n\le N}(1+|\gamma|N^{|\kappa|}n^v)e^{t\log^2 n/4}n^{-\Re w_-}\left[\exp\left\{\frac{(t^2/16)\log^2(u/(4\pi n^2)) + 0.626}{u-6.66}\right\} - 1\right], \tag{2.7}$$

$$e_{C,0} \le (u/(4\pi))^{-(1+v)/4}\exp\left\{-\frac{tL_u^2}{16} + \frac{1.24(3^v+3^{-v})}{N-0.125} + \frac{3|L_u + i\pi/2| + 10.44}{u-12}\right\}. \tag{2.8}$$

No derivatives of this finite approximation will be taken. Its cutoff can jump with u; the analytic model (2.4) will remove that difficulty before differentiation.

2.2 A maximal estimate for logarithmic-phase blocks

The gain over absolute summation comes from retaining the oscillation in n^{-iT} before estimating each arithmetic block.

Lemma 2.1 (Logarithmic-phase blocks)**.** *For $T \ge 1$, $M \ge 1$, and every interval I of integers contained in $(M, 2M]$,*

$$\left|\sum_{n\in I} n^{-iT}\right| \ll T^{1/6}M^{1/2} + T^{-1/6}M. \tag{2.9}$$

The implied constant is absolute. The same estimate holds with T replaced by $-T$, and is uniform over all shortened terminal blocks.

Proof. Apply the third derivative test, in the form of [4, Theorem 2 and its preceding proof], to $f(y) = -(T/(2\pi))\log y$. On $(M, 2M]$ its third derivative has magnitude comparable to T/M^3, with an absolute comparison ratio. For an interval of length $J \le M$, the test gives

$$J(T/M^3)^{1/6} + J^{1/2}(T/M^3)^{-1/6} \le T^{1/6}M^{1/2} + T^{-1/6}M.$$

Empty intervals contribute zero; one-point intervals are absorbed by increasing the absolute constant. Conjugation handles the opposite phase. □

For a continuously differentiable, possibly complex weight W, Abel summation on a shortened block is the exact identity

$$\sum_{M<n\le R} W(n)n^{-iT} = W(R)S_M(R) - \int_M^R W'(y)S_M(y)\,dy, \quad S_M(y) = \sum_{M<n\le y} n^{-iT}, \quad M \le R \le 2M. \tag{2.10}$$

Thus (2.9) is multiplied by $|W(R)| + \int_M^R |W'(y)|\,dy$. For a nonnegative decreasing weight, this total-variation factor is exactly $W(M)$. The maximal form of the lemma is needed because the last block ends at the actual cutoff N.

2.3 Five exponent pairs and a closed outer boundary

We strengthen the elementary third-derivative estimate above using five limiting exponent pairs. Write

$$\begin{aligned} P &= (13/84, 55/84), & AP &= (13/194, 76/97), \\ A^2P &= (13/414, 359/414), & D &= (18/199, 593/796), \\ R &= (4742/38463, 35731/51284). \end{aligned} \tag{2.11}$$

Here D is the name of the displayed pair, not an assertion that an arbitrary transformation called D preserves exponent pairs. Bourgain's theorem supplies P with arbitrarily small positive losses [8, Theorem 6]; Trudgian and Yang supply D, R and the A-process [9, Section 1.1, Lemma 1.1, and eq. (1.10)],

$$A(k,l) = \left(\frac{k}{2k+2}, \frac{k+l+1}{2k+2}\right).$$

Their exponent-pair definition is uniform over arbitrary subintervals. For $f(y) = (T/(2\pi))\log y$, its first derivative is exactly the model monomial with exponent parameter one, so all derivative-approximation errors are zero. Conjugation handles the opposite phase. Consequently, for each fixed $\eta > 0$, every pair (k,l) in (2.11) gives the complete maximal estimate

$$\sup_{I \subset (M,2M]} \left| \sum_{n \in I} n^{-iT} \right| \ll_\eta T^{k+\eta} M^{l-k}, \qquad M \geq 1, \quad T \geq 2\pi M. \tag{2.12}$$

The supremum is over integer intervals, including shortened terminal blocks. Bourgain's proper-subinterval completion may cost a logarithm; choose a smaller initial positive loss to absorb it into the displayed η. No zero-loss endpoint theorem is used.

For clarity, the descendants of P also have admissibly small losses. Under equal positive increments ϵ in both coordinates, the increments in the coordinates of $A(k,l)$ are

$$\frac{\epsilon}{2(1+k)(1+k+\epsilon)}, \qquad \frac{\epsilon(1+k-l)}{2(1+k)(1+k+\epsilon)}.$$

They are nonnegative and at most $\epsilon/2$ for the pairs used here; a second application remains continuous. If the final coordinate increments are at most η, their contribution to the logarithm of the bound is at most $\eta(\log T - \log M) + \eta \log M = \eta \log T$. Thus (2.12) includes all losses and is a full bound in the indicated range.

Set

$$\theta_* := \frac{61}{24}, \qquad \eta = \rho := \frac{1}{100000}.$$

For (t,x) in (1.8), put

$$D_x = \{z : |z - x| \leq L^{-1}\}, \qquad L_u = \log(u/(4\pi)), \qquad \theta_u = tL_u, \quad z = u + iv. \tag{2.13}$$

In this proof write N_{RS} for the integer N in (2.5); it will not be confused with the source cutoff introduced in Section 5. Uniformly on D_x, the explicit formula (2.2) gives, for either branch $w = \sigma + i\tau = w_\pm(z)$,

$$\begin{aligned} &L_u = L + O((xL)^{-1}), \quad |\tau| \asymp u, \quad \log N_{\mathrm{RS}} \leq L_u/2 + O(u^{-1}), \\ &\theta_u \geq \theta_* - \rho, \qquad \sigma \geq \tfrac{1}{2} + \tfrac{\theta_u}{4} - \delta_x, \qquad 0 \leq \delta_x = O(L^{-1}) \leq \rho. \end{aligned} \tag{2.14}$$

These estimates hold after one fixed increase of X_* and are uniform for $0 < t \leq 1/2$, including growing $\theta_u \leq L_u/2$. In particular $\sigma \geq 9/8$ there. The real disc is allowed to cross slightly below θ_*; the following strict margins pay that displacement.

For $\beta = (\log M)/L_u$ define the unperturbed weighted exponent

$$F_{k,l}(\beta;\theta) = k + (l-k-\tfrac{1}{2})\beta - \frac{\theta}{4}\beta(1-\beta). \tag{2.15}$$

At $\theta = \theta_*$ it is convex, with second derivative $61/48$. The following endpoint values are exact.

Pair	Interval for β	Left value	Right value
A^2P	$[1/5, 3/10]$	$-43/13800$	$-869/662400$
AP	$[3/10, 3/8]$	$-459/310400$	$-145/198656$
D	$[3/8, 81/200]$	$-215/407552$	$-21807/254720000$
R	$[81/200, 3/7]$	$-4368359/49232640000$	$-1/1176$
P	$[3/7, 1/2]$	$-1/1176$	$-11/2688$

The intervals cover $[1/5, 1/2]$ without gaps. Convexity therefore gives, with the corresponding pair in each interval,

$$F_{k,l}(\beta;\theta_*) \le -\mu_*, \qquad \mu_* := \frac{21807}{254720000}. \tag{2.16}$$

The positive exponent-pair loss costs at most η, the real-part error at most $\delta_x/2$, and the decrease of θ_u below θ_* at most $\rho/16$. For the tiny possible extension $\beta \le 1/2 + O((uL_u)^{-1})$, continue to use P. Its principal change beyond $1/2$ is $\theta_u(\beta - 1/2)^2/4$, which tends to zero uniformly even when θ_u grows; the other perturbations do also. Increase X_* until this extension costs at most ρ. The surviving margin is

$$\mu_* - \rho(1 + \tfrac{1}{2} + \tfrac{1}{16} + 1) = \frac{76399}{1273600000} > \frac{1}{20000}. \tag{2.17}$$

Replacing $T = |\tau|$ by a constant multiple of u changes bounded factors only. All these losses are paid at the actual lower real boundary $\theta = 61/24$.

2.4 Uniform replacement of the weighted sums at the closed boundary

Lemma 2.2 (Weighted logarithmic moments)**.** *Under* (2.14)*, for every fixed nonnegative integer* r*,*

$$\sum_{n \le N_{\mathrm{RS}}} (\log n)^r e^{t\log^2 n/4} n^{-w} - (-1)^r \zeta^{(r)}(w) = O_r(L^{-1}), \qquad z \in D_x. \tag{2.18}$$

For $r = 0$ *the factor at* $n = 1$ *is interpreted as one. The implied constant is uniform throughout* (1.8)*.*

Proof. Write $\ell = L_u$ and split at $Y = e^{\ell/5}$. For $n \le Y$, use $e^q - 1 \le qe^q$ and $e^{t\log^2 n/4} \le n^{\theta_u/20}$. The small-index discrepancy is at most

$$\frac{t}{4} \sum_{n \ge 2} (\log n)^{r+2} n^{-(1/2 + \theta_u/5 - \delta_x)}. \tag{2.19}$$

Put $\theta_u = \theta_* - \rho + s$ with $s \ge 0$. Since

$$\frac{1}{2} + \frac{\theta_*}{5} = \frac{121}{120}, \qquad \frac{1}{120} - \frac{\rho}{5} - \rho > \frac{1}{240},$$

the exponent in (2.19) is at least $1 + 1/240 + s/5$. Hence, with the fixed finite constant $D_r = \sum_{n\ge 2} (\log n)^{r+2} n^{-1-1/240}$, that discrepancy is bounded by

$$\frac{D_r}{4\ell}(\theta_* - \rho + s) 2^{-s/5} = O_r(\ell^{-1}). \tag{2.20}$$

The displayed numerator is bounded on the whole half-line $s \geq 0$; this proves uniformity for growing θ_u, rather than merely for time parameters in a fixed compact substrip. Since $\sigma \geq 9/8$, the omitted ordinary zeta tail satisfies

$$\sum_{n>Y} (\log n)^r n^{-\sigma} = O_r(\ell^r e^{-\ell/40}). \tag{2.21}$$

Partition $Y < n \leq N_{\mathrm{RS}}$ into the exact dyadic intervals $(M, \min(2M, N_{\mathrm{RS}})]$, starting at $M = Y$. The weight

$$W_r(y) = (\log y)^r e^{t \log^2 y/4} y^{-\sigma}$$

is nonnegative and, for sufficiently large ℓ, decreasing, because

$$y(\log W_r)' = \frac{r}{\log y} + \frac{t}{2} \log y - \sigma \leq \frac{5r}{\ell} - \frac{1}{2} + \delta_x + O(u^{-1}) < 0. \tag{2.22}$$

Apply (2.10) and (2.12), choosing the pair by the left endpoint $\beta = (\log M)/\ell$. The weighted block is at most a constant times $W_r(M) T^{k+\eta} M^{l-k}$. A block may cross a boundary in the table: the estimate is still valid, because (2.12) already controls every shortened interval at its given scale M. Equations (2.16)–(2.17) therefore yield

$$\sum_{Y<n\leq N_{\mathrm{RS}}} (\log n)^r e^{t \log^2 n/4} n^{-w} = O_r(\ell^{r+1} e^{-\ell/20000}). \tag{2.23}$$

There are $O(\ell)$ blocks, including the shortened final block. Combining (2.19), (2.21), and (2.23) proves the lemma. □

2.5 Complete value errors and derivative control

The positive sums in the imported errors require no further arithmetic cancellation. For $n \leq N_{\mathrm{RS}}$,

$$e^{t \log^2 n/4} n^{-\theta_u/4} = \exp\{-\tfrac{t}{4} \log n(\ell - \log n)\} \ll 1, \qquad n^{\delta_x} \ll 1.$$

On the upper half-disc (2.6) bounds $|\gamma|$ and $N_{\mathrm{RS}}^{|\kappa|}$ uniformly. The factor n^v replaces the first branch's real part $(1+v)/2$ by $(1-v)/2$, already covered in (2.14). Thus the complete positive weight sum in (2.7) is $O(\sum_{n\leq N_{\mathrm{RS}}} n^{-1/2}) = O(u^{1/4})$. Its per-index error multiplier is $O(L^2/u)$, uniformly for $t \leq 1/2$. The correction exponent in (2.8) tends uniformly to zero. Consequently all three imported errors obey

$$e_A + e_B = O(L^2 u^{-3/4}), \qquad e_{C,0} \ll e^{-\ell/4 - \theta_u \ell/16} = O(u^{-2/5}), \tag{2.24}$$

where $1/4 + (\theta_* - \rho)/16 > 2/5$.

Set $\Gamma = |M_t(s_+(x))|$. The logarithmic z-derivative of either $M_t(s_\pm(z))$ is $O(L)$, so integration over a segment of length at most $1/L$ gives

$$|M_t(s_\pm(z))| \asymp \Gamma \qquad (z \in D_x). \tag{2.25}$$

Combining (2.5), Lemma 2.2 at $r = 0$, and (2.24)–(2.25) gives

$$H(t,z) - \mathcal{P}_t(z) = O(\Gamma/L), \qquad z \in D_x. \tag{2.26}$$

Initially the imported approximation applies to the upper half-disc; Schwarz symmetry supplies the lower half.

Absolute convergence of the Dirichlet series for ζ and $1/\zeta$ in $\Re w \geq 9/8$ gives

$$\zeta(9/8)^{-1} \leq |\zeta(w)| \leq \zeta(9/8). \tag{2.27}$$

Thus $Q \neq 0$ and $|Q| \asymp \Gamma$. The complete difference in (2.26) is holomorphic on a neighborhood of the closed disc. Cauchy's formula, with time fixed, yields

$$\partial_x^k (H(t,\cdot) - \mathcal{P}_t)(x) = O_k(|Q| L^{k-1}), \qquad k = 0,1,2,3. \tag{2.28}$$

At $k = 0$ this means $O(|Q|/L)$. The moving Riemann–Siegel floor has been eliminated before differentiation.

2.6 The common phase and the two signs

At the real center the explicit carrier satisfies

$$\alpha(s_+(x)) = L/2 + i\pi/4 + O(x^{-1}), \qquad \alpha^{(m)}(s_+(x)) = O_m(x^{-m}), \quad m \geq 1. \tag{2.29}$$

Fixed-order logarithmic derivatives of ζ are uniformly bounded in $\Re w \geq 9/8$. Therefore

$$(\log Q_+)' = \frac{i}{2}\left[\alpha + \frac{t}{2}\alpha\alpha' + \left(1 + \frac{t}{2}\alpha'\right)\frac{\zeta'(w_+)}{\zeta(w_+)}\right] = i\omega + O(1), \tag{2.30}$$

$$(\log Q_+)'' = O(1), \qquad (\log Q_+)''' = O(1). \tag{2.31}$$

Here all α terms are evaluated at $s_+(x)$ and time is fixed. The estimates remain uniform as $\theta = tL$ grows because $t \leq 1/2$. The exponential derivative formulas give

$$Q_+^{(k)}(x) = Q\{(i\omega)^k + O(L^{k-1})\}, \qquad k = 1,2,3;$$

the real-axis derivatives of Q_- are their conjugates. Equation (2.28) proves (1.9) with $O(L^{-1})$, throughout the closed domain (1.8).

Write $P_k = h_k/(2|Q|\omega^k)$ and $P = (c,-s,-c,s) + e$, where $c^2 + s^2 = 1$. If $|e_k| \leq d$, direct expansion gives

$$P_0^2 + P_1^2 \geq 1 - 4d, \qquad P_1^2 - P_0P_2,\ P_2^2 - P_1P_3 \geq 1 - 4d - d^2. \tag{2.32}$$

For example the first signed form equals $1 - 2se_1 + e_1^2 - ce_2 + ce_0 - e_0e_2$; retain its nonnegative square and bound the other terms. Increasing X_* until $\omega \geq 1$ and $d \leq 1/8$ proves Theorem 1.1 and the source-defined envelopes

$$\begin{aligned} E \geq 2|Q|^2 &\geq \frac{2\Gamma^2}{\zeta(9/8)^2}, \\ \mathcal{L}_0 \geq \frac{31}{16}|Q|^2\omega^2, &\qquad \mathcal{L}_1 \geq \frac{31}{16}|Q|^2\omega^4. \end{aligned} \tag{2.33}$$

The endpoint $\theta = 61/24$ is included because the arithmetic margins were strictly positive before the disc and exponent-pair losses were paid. This strengthens the earlier $8/3+\varepsilon$ result, and in particular covers its formerly excluded boundary $8/3$. The height X_* remains unevaluated. Appendix B retains the independent explicit height under the stronger condition $tL \geq 5$.

Corollary 2.3 (Asymptotic collision localization)**.** *If $H(T_m, a_m) = H_x(T_m, a_m) = 0$, $0 < T_m \leq 1/2$, and $|a_m| \to \infty$, then every sufficiently large m satisfies*

$$T_m \log\frac{|a_m|}{4\pi} < \frac{61}{24}.$$

In particular the limsup of these products is at most $61/24$.

Proof. For $|a_m| \geq X_*$, a product at least $61/24$ contradicts the energy lower envelope (2.33), using reflection when $a_m < 0$. □

This localizes any hypothetical collision sequence; it does not assert that collisions exist or impose this threshold below the unevaluated height cutoff.

2.7 A conditional continuation to five halves

The largest blocks alone do not control the remaining interval:

$$F_P(1/2;5/2) = -\frac{1}{672}, \qquad F_D(2/5;5/2) = \frac{9}{3980} > 0.$$

At $\beta = 2/5$, D is the best of the five displayed pairs. The positive upper exponent prevents these particular estimates from proving a small remainder there. It does not prove growth of the actual sum or failure of a theta sign. Absolute treatment of low indices at $\theta = 5/2$ requires a split $Y = e^{\alpha\ell}$ with $\alpha < 1/5$, so merely moving that split beyond the failing middle block is not a repair.

Lemma 2.4 (Conditional middle-block transfer)**.** *Suppose there are fixed constants $C, \kappa > 0$ and T_0 such that for $T \ge T_0$, $\beta = \log M/\log T \in [7/25, 47/100]$, and every integer interval $I \subset (M, 2M]$,*

$$\left|\sum_{n\in I} n^{-iT}\right| \le CT^{\beta/2+(5/8)\beta(1-\beta)-\kappa}. \tag{MB}$$

Then, for sufficiently large x, uniformly for $0 < t \le 1/2$ and the entire closed strip $5/2 \le tL \le 8/3$, the four-jet asymptotic (1.9) holds with $O(L^{-1})$, and $E, \mathcal{L}_0, \mathcal{L}_1$ are strictly positive. The two cutoff conclusions of Corollary 5.4 also hold there, with $\zeta(11/10)$ replacing $\zeta(9/8)$ in the relative coefficient. The hypothesis (MB) is not established here.

Proof. Use the same shrinking disc and put $Y = e^{19\ell/100}$. At the real lower boundary $\theta = 5/2$, the exponent in the absolute small-index discrepancy is

$$\frac{1}{2} + \frac{(5/2)(1-19/100)}{4} = \frac{161}{160} > 1.$$

For sufficiently large height the disc perturbations leave it at least $1 + 1/320$. Since θ is bounded on this strip, the discrepancy is $O_r(L^{-1})$. Also $\Re w \ge 11/10$, so the ordinary zeta tail is $O_r(\ell^r e^{-19\ell/1000})$.

For the high blocks outside the assumed middle range, use A^2P on $[19/100, 2/7]$ and P on $[23/50, 1/2]$. Their exact endpoint values at $\theta = 5/2$ are

$$\begin{array}{c|cc} A^2P & -3293/3312000 & -1/4508 \\ P & -41/84000 & -1/672. \end{array}$$

Convexity gives strict savings on both full intervals, preserved by sufficiently small positive exponent-pair losses. The middle range in (MB) overlaps these intervals by $1/175$ and $1/100$, respectively. Those fixed overlaps absorb the difference between $\log T = \ell + O(1)$ and the scale ℓ used on the complex disc.

For a remaining block, write $b = \log M/\ell$, while the hypothesis uses $\beta = \log M/\log T$. Multiplication by the actual decreasing weight and Abel summation give, apart from fixed powers of ℓ, a logarithm bounded by

$$-\kappa\ell - \frac{\theta_u - 5/2}{4} b(1-b)\ell + O(1).$$

The $O(1)$ includes the logarithmic-coordinate change and the real-part error, since $\delta_x \ell = O(1)$. On this fixed strip the disc's possible displacement below $\theta = 5/2$ tends to zero. Thus the bound is at most $-\kappa\ell/2$ after increasing the starting height. Summing the $O(\ell)$ blocks proves Lemma 2.2 on the entire closed strip, conditionally on (MB).

All imported errors remain negligible; in particular $e_{C,0} = O(u^{-2/5})$ since $1/4 + (5/2)/16 = 13/32 > 2/5$, with room for the vanishing disc perturbation. The analytic comparison, Cauchy

estimates, and phase-uniform algebra above apply with the zero-free half-plane $\Re w \geq 11/10$. They give $E \geq 2\Gamma^2/\zeta(11/10)^2$ and the same two positive Laguerre envelopes. The proof of Corollary 5.4 then gives both cutoff conclusions. □

Together with Theorem 1.1, the hypothesis would cover all $tL \geq 5/2$ in the allowed time range at sufficiently large height. It is a sufficient arithmetic condition, not a necessary condition or an RH-equivalent reformulation. A direct estimate for the complete weighted or signed physical observation could bypass it. Neither this conditional transfer nor the unconditional theorem settles the inner cusp or evaluates the finite transition to the asymptotic region.

3 Why positive-time collisions are the obstruction

The preceding asymptotic result is useful because a failure of RH forces a finite collision of real zeros at a positive time. We record the precise reduction, including the compactness input, before formulating the signed estimate still needed outside the proved regions.

We use the normalization

$$H(0,z) = \frac{1}{8}\xi\left(\frac{1}{2} + \frac{iz}{2}\right), \qquad H_t = -H_{zz}.$$

The positive, rapidly decreasing theta source makes H jointly entire in (t,z), even in z, and real on the real (t,z) plane. Moreover, $H(t,iy) > 0$ for real t, y, so no time slice is identically zero and every zero has finite multiplicity.

The classical de Bruijn–Newman theory gives a finite real threshold Λ such that all zeros of $H(t,\cdot)$ are real exactly when $t \geq \Lambda$; the endpoint is included [1, 2]. In this normalization $\Lambda \leq 1/2$ and RH is equivalent to $\Lambda \leq 0$. De Bruijn's strip contraction keeps all zeros in $|\operatorname{Im} z| \leq 1$ for $t \geq 0$. We also use the uniform confinement theorem of [3, Theorem 1.5(i)]: there is an absolute constant $C > 0$ such that, for $0 < t \leq 1/2$, a nonreal zero satisfies

$$|\operatorname{Re} z| < e^{C/t}. \tag{3.1}$$

Uniformity in time, rather than a separate uncontrolled radius for each slice, is what prevents a first collision from escaping to infinity.

Lemma 3.1 (Local splitting at a multiple zero)**.** *Let W be real analytic near (T,a), solve $W_t = -W_{xx}$, and have an exact spatial zero of multiplicity $m \geq 2$ there. Its nearby zeros have leading locations*

$$a + \sqrt{2(t-T)}\,\lambda_k + O(|t-T|), \qquad k = 1, \dots, m,$$

where the λ_k are the distinct real roots of the probabilists' Hermite polynomial He_m*. In particular, W has a nonreal zero at times immediately before T.*

Proof. Put $W_q = \partial_x^q W(T,a)$. The convergent heat Taylor series is

$$W(T+\delta, a+w) = \sum_{p,q\geq 0} \frac{(-\delta)^p w^q}{p!q!} W_{2p+q}.$$

With $\delta = \eta^2$ and $w = \sqrt{2}\eta\lambda$, division by η^m extends analytically to $\eta = 0$, where its value is $2^{m/2} W_m \,\mathrm{He}_m(\lambda)/m!$. The roots of He_m are real and simple: its Rodrigues formula gives orthogonality to all lower-degree polynomials for the positive Gaussian weight, and the polynomial formed from

its real sign-changing factors would contradict that orthogonality unless there were m such roots. The simple roots therefore continue by the implicit function theorem. For $t < T$, take $\eta = i\sqrt{T-t}$; at least one Hermite root is nonzero, so its leading imaginary displacement cannot be cancelled by the $O(T-t)$ remainder. □

Proposition 3.2 (Finite collision criterion for RH)**.** *RH holds if and only if*

$$H(t,x) = H_x(t,x) = 0 \quad \text{has no solution with } 0 < t \le 1/2,\ x \in \mathbb{R}. \tag{3.2}$$

Equivalently, $E(t,x) = h_0(t,x)^2 + h_1(t,x)^2$ is positive throughout this region. No uniform positive lower bound for E is required. Equivalently, there is no such joint zero at any positive time.

Proof. If RH holds, all zeros are real at all nonnegative times. A multiple zero at a positive time would contradict Lemma 3.1.

Conversely, if RH fails then $0 < \Lambda \le 1/2$. Choose $t_k \uparrow \Lambda$ with $t_k \ge \Lambda/2$, and a nonreal zero z_k at each t_k. The strip bound and (3.1) put these zeros in the fixed compact set $|\operatorname{Im} z| \le 1$, $|\operatorname{Re} z| \le e^{2C/\Lambda}$. A subsequence converges to a zero a of $H(\Lambda,\cdot)$, which is real by the threshold theorem. If that zero were simple, the local implicit zero branch would be unique; conjugation symmetry would make it real for real times on both sides of Λ. This contradicts $z_k \notin \mathbb{R}$. Thus $H(\Lambda,a) = H_x(\Lambda,a) = 0$. The source positivity gives $H(\Lambda,0) > 0$, so $a \neq 0$. □

3.1 A sufficient estimate only on two stationary sets

Recall

$$\mathcal{L}_j = h_{j+1}^2 - h_j h_{j+2}, \qquad j = 0, 1.$$

On a smooth stationary curve $h_{j+1}(t,\chi(t)) = 0$, the heat equation gives

$$\frac{1}{2}\frac{d}{dt} h_j(t,\chi(t))^2 = \mathcal{L}_j(t,\chi(t)).$$

The following result includes singular stationary sets and all finite multiplicities. Its losses may grow with height or as $t \downarrow 0$; only local boundedness through each possible positive-time collision is required.

Lemma 3.3 (A stationary predecessor of an even zero)**.** *Suppose $W_t = -W_{xx}$ and W has exact spatial multiplicity $2r$ at (T,a), where $r \ge 1$. With $W_q = \partial_x^q W(T,a)$, there is a real analytic function $\chi(h)$ for small $h \ge 0$ such that*

$$W_x(T-h,\chi(h)) = 0, \qquad \chi(h) = a - \frac{W_{2r+1}}{rW_{2r}}h + O(h^2),$$

$$W(T-h,\chi(h)) = \frac{W_{2r}}{r!}h^r + O(h^{r+1}), \qquad W_{xx}(T-h,\chi(h)) = \frac{W_{2r}}{(r-1)!}h^{r-1} + O(h^r).$$

Proof. Use the heat Taylor series above with $\delta = -h$ and $w = hy$. The function $h^{-r}W_x(T-h,a+hy)$ extends analytically to $h = 0$, where it equals $W_{2r}y/(r-1)! + W_{2r+1}/r!$. Its y derivative is nonzero. The implicit function theorem gives $y(h)$, and substitution in the same series gives the two displayed asymptotics. □

Theorem 3.4 (Stationary bounded-loss criterion)**.** *Suppose there are nonnegative functions K_0, K_1, locally bounded on $(0,1/2] \times \mathbb{R}$, for which*

$$\begin{aligned} h_1 = 0 \quad &\Longrightarrow \quad \mathcal{L}_0 \ge -K_0 E, \\ h_2 = 0 \quad &\Longrightarrow \quad \mathcal{L}_1 \ge -K_1 E. \end{aligned} \tag{3.3}$$

Then RH holds. Local boundedness here includes a neighborhood of every candidate joint zero, using a left neighborhood in time at $t = 1/2$.

Proof. Suppose H has a real multiple zero at (T, a) in this time range, of exact multiplicity $m \geq 2$. If $m = 2r$, apply Lemma 3.3 to $W = H$. On its predecessor branch $h_1 = 0$, $E = H^2 > 0$, and

$$-\frac{\mathcal{L}_0}{E} = \frac{H_{xx}}{H} = \frac{r}{h} + O(1), \qquad h = T - t \downarrow 0.$$

This contradicts the first inequality with locally bounded K_0. If $m = 2r + 1$, apply the lemma to $W = H_x$. Then $h_2 = 0$, $H_x = ch^r + O(h^{r+1})$, and $H_{xxx} = rch^{r-1} + O(h^r)$, with $c \neq 0$. The heat Taylor series also gives $H = O(h^{r+1})$ on this branch, because its displacement from a is $O(h)$. Consequently

$$E = H_x^2(1 + O(h^2)), \qquad -\frac{\mathcal{L}_1}{E} = \frac{r}{h} + O(1),$$

contradicting the second inequality. All divisions occur on proved nonzero predecessor branches, not at the supposed collision. Proposition 3.2 now applies. □

Both levels matter: the heat polynomial $H(t, x) = x^3 - 6(t - T)x$ has $\mathcal{L}_0 = 3x^4 + 36(t - T)^2 \geq 0$ everywhere, but has a triple zero at $(T, 0)$. Its second stationary loss diverges on approaching that collision from earlier times. Thus positivity of the first observation alone is insufficient for arbitrary multiplicity.

3.2 The finite part of the collision pole and delayed cancellation

The preceding collision argument uses only the leading pole. Its next coefficient also identifies cancellations that a delayed source estimate must preserve. All statements in this subsection are conditional on a finite collision; they do not assert that the theta family has one.

Proposition 3.5 (Finite part on the stationary predecessor)**.** *Suppose $H(T, \cdot)$ has exact multiplicity $m = 2r + j$ at x_*, where $T > 0$, $r \geq 1$, and $j \in \{0, 1\}$. Put*

$$A = h_m(T, x_*) \neq 0, \qquad B = h_{m+1}(T, x_*), \qquad D = h_{m+2}(T, x_*),$$
$$k = A/r!, \qquad \alpha_* = \frac{D}{(r+1)A} - \frac{B^2}{2rA^2}.$$

On the predecessor $h_{j+1}(T - s, \chi(s)) = 0$ of Lemma 3.3, as $s \downarrow 0$,

$$\begin{aligned} \chi(s) &= x_* - \frac{B}{rA}s + O(s^2), \\ h_j(T - s, \chi(s)) &= ks^r(1 + \alpha_* s + O(s^2)), \\ \frac{\mathcal{L}_j}{E} &= -\frac{r}{s} - \alpha_* + O(s), \qquad E \sim k^2 s^{2r}, \qquad \mathcal{L}_j \sim -rk^2 s^{2r-1}. \end{aligned} \tag{3.4}$$

For odd multiplicity, the more precise comparison is

$$\frac{H}{H_x} = -\frac{B}{r(r+1)A}s + O(s^2), \qquad E = h_1^2\left(1 + \frac{B^2 s^2}{r^2(r+1)^2A^2} + O(s^3)\right). \tag{3.5}$$

Proof. For $W = h_j$, the heat Taylor expansion at the collision is

$$W(T - s, x_* + y) = \sum_{p,q \geq 0} \frac{s^p y^q}{p!q!} h_{j+2p+q}(T, x_*).$$

The implicit-function construction in Lemma 3.3 gives $y = -Bs/(rA) + O(s^2)$. The terms of order s^{r+1} in W are $Ds^{r+1}/(r+1)!$, $Bs^r y/r!$, and $As^{r-1}y^2/[2(r-1)!]$. Their sum, divided by ks^r, is $\alpha_* s$, proving the second expansion. Along the branch, $dW(T-s,\chi(s))/ds = W_{xx}$ exactly, so $W_{xx}/W = r/s + \alpha_* + O(s)$. For $j=0$, $E = W^2$ and the result follows. For $j=1$, the same series for H gives

$$H(T-s,\chi(s)) = -\frac{B}{r(r+1)r!}s^{r+1} + O(s^{r+2}),$$

which proves (3.5). Since $\mathcal{L}_1 = -WW_{xx}$ on this branch, the relative $O(s^2)$ energy correction changes the displayed quotient only by $O(s)$. □

The moving energy derivative must be distinguished from the fixed-position one. With $\xi(t) = \chi(T-t)$,

$$\frac{d}{dt}E(t,\xi(t)) = \begin{cases} 2\mathcal{L}_0, & h_1 = 0, \\ 2\mathcal{L}_1 + 2\xi' h_0 h_1, & h_2 = 0. \end{cases}$$

The extra term divided by E is $O(s)$ on the odd predecessor. The exact branch identity without that term is $\frac{1}{2}\frac{d}{dt}h_1^2 = \mathcal{L}_1$. For odd multiplicity, the surviving real zero has displacement $-Bs/((r+1)A) + O(s^2)$ from x_*: substituting $y = sz$ in the heat series for H and applying the implicit function theorem gives this coefficient. Its displacement from the stationary predecessor is $Bs/[r(r+1)A] + O(s^2)$, consistently with (3.5).

Fix $a = \pi/16$, $b = 1/4$, and $\mu = a^2 + b^2$, and retain $C_j = -h_{j+2} - 2ah_{j+1} - \mu h_j$. For any declared delay put

$$G_j = C_j(t-\delta, x), \qquad P_j = h_j G_j, \qquad Q_j = h_j(C_j - G_j/2).$$

At current stationarity the exact combination is

$$\frac{P_j/2 + Q_j}{E} = \frac{\mathcal{L}_j}{E} - \mu\frac{h_j^2}{E} = -\frac{r}{s} - (\alpha_* + \mu) + O(s). \tag{3.6}$$

Thus changing the delay changes the two summands but not the observation whose adverse part must be controlled.

Proposition 3.6 (Stronger poles introduced by a positive delay)**.** *Suppose the prescribed delay is analytic near the candidate event, $\delta_* := \delta(T, x_*) > 0$, and put $g = C_j(T - \delta_*, x_*)$. If $g \neq 0$, then on the predecessor*

$$\frac{P_j}{E} = \frac{g}{k}s^{-r}(1 + O(s)), \qquad \frac{Q_j}{E} = -\frac{P_j}{2E} - \frac{r}{s} - (\alpha_* + \mu) + O(s).$$

For $r > 1$, the two products therefore have opposite leading poles of order s^{-r}. Their intervening negative powers also cancel in (3.6). If $A_{\min} = (-P_j/E)_+$ and $B_{\min} = (-Q_j/E)_+$ are used only as diagnostic quantities on this nonzero branch, then for $r > 1$ and $g \neq 0$

$$A_{\min}/2 + B_{\min} \sim \frac{|g/k|}{2}s^{-r}.$$

Both split ratios are $O(s^{-1})$ if and only if $G_j = O(s^{r-1})$. For $r = 1$, writing $\gamma = g/k$ (including $g = 0$),

$$P_j/E = \gamma/s + O(1), \qquad Q_j/E = -(1 + \gamma/2)/s + O(1).$$

The necessary leading conditions for $P_j \geq 0$ and $Q_j \geq -KE$ with locally bounded K are respectively $\gamma \geq 0$ and $\gamma \leq -2$, and are incompatible.

Proof. Analyticity gives $G_j = g + O(s)$, while $h_j = ks^r(1 + O(s))$ and $E = h_j^2(1 + O(s^2))$; the even case has the exact equality $E = h_j^2$. These facts prove the first expansion, and the second follows from (3.6). Taking the negative part according to the sign of g/k proves the diagnostic formula when $r > 1$. Finally $P_j/E = G_j/[ks^r](1 + O(s))$ proves the equivalence; the combined identity gives the same bound for Q_j/E. The case $r = 1$ follows without assuming $g \neq 0$. □

For comparison, a diagnostic delay $\delta = \rho s + O(s^2)$ with fixed $\rho \geq 0$ gives, from the leading heat polynomial,

$$\frac{P_j}{E} = -\frac{r(1+\rho)^{r-1}}{s} + O(1), \qquad \frac{Q_j}{E} = \frac{r\{(1+\rho)^{r-1}/2 - 1\}}{s} + O(1).$$

If only $\delta/s \to \rho$ is known, replace $O(1)$ by $o(s^{-1})$. These rules refer to an unknown collision time and are stress tests, not proposed source protocols. A split can even create an adverse pole on a favorable branch: for $H = y^4 - 12uy^2 + 12u^2$, $u = t - T > 0$, at $y = 0$,

$$E = 144u^4, \quad \mathcal{L}_0 = 288u^3 > 0, \quad \frac{P_0}{E} = -\frac{2\delta + \mu\delta^2}{u^2} + \frac{2 + 2\mu\delta}{u} - \mu$$

for a fixed $0 < \delta < T$, whereas $P_0/(2E) + Q_0/E = 2/u - \mu$. This is a polynomial control outside the theta class.

3.3 Logarithmic losses and local analytic rigidity

For prescribed $M > 0$ and finite $q \geq 0$, define

$$\mathfrak{f}_q(e) = e[1 + \log_+(M/e)]^q \quad (e > 0), \qquad \mathfrak{f}_q(0) = 0, \qquad \log_+ v = \max\{0, \log v\}.$$

Proposition 3.7 (A logarithmic stationary-loss criterion)**.** *Suppose on the two stationary sets that $\mathcal{L}_j \geq -K_j\mathfrak{f}_q(E)$, with nonnegative K_j locally bounded through every candidate positive-time collision. Then no such collision occurs in a domain containing the required predecessor neighborhoods. If the domain covers $(0, 1/2] \times \mathbb{R}$, RH follows.*

Proof. On the predecessor of a zero of multiplicity $2r + j$, $E \sim k^2 s^{2r}$ and $-\mathcal{L}_j/E = r/s + O(1)$. Hence

$$\frac{-\mathcal{L}_j}{\mathfrak{f}_q(E)} = \frac{r(1 + o(1))}{s[2r\log(1/s)]^q} \longrightarrow \infty,$$

contradicting the local bound for K_j. Nontriviality of the theta family ensures finite spatial multiplicity. All divisions are on the nonzero predecessor, and Proposition 3.2 supplies the final RH implication. □

A loss by a fixed smaller energy power does not have the same all-multiplicity implication: $-\mathcal{L}_j/E^p$ has order $s^{2r(1-p)-1}$, which diverges only if $p > 1 - 1/(2r)$. At equality its limit is $r|k|^{1/r}$, and below equality it tends to zero. Also, a loss coefficient integrable in time along the predecessor would suffice to exclude the event: any coefficient allowing it must dominate $r/s + O(1)$, whose integral diverges. This is a branchwise requirement; two-dimensional local integrability alone gives no such control on the stationary curve. Analyticity is essential for the logarithmic statement when $q > 1$. With $\beta = 1/(q - 1)$, the smooth function $y(t) = \exp[-(T - t)^{-\beta}]$, continued by zero for $t \geq T$, satisfies $y' = -\beta y[\log(1/y)]^q$ and reaches zero. It is not analytic at T and is not a theta or backward-heat example.

The logarithmic allowance is useful quantitatively, but on analytic stationary germs it does not enlarge the qualitative class of locally bounded losses.

Lemma 3.8 (Analytic rigidity of logarithmic losses)**.** *Let S be semianalytic near a finite point p, and let $E \geq 0$ and L be real analytic in an ambient neighborhood. Suppose, on S near p,*

$$L \geq -KE[1 + \log_+(M/E)]^q,$$

where $K \geq 0$ is bounded, $M > 0$ and $q \geq 0$ are fixed and finite, and the right side is zero at $E = 0$. On a smaller neighborhood in S there is a finite $C \geq 0$ such that $L \geq -CE$. Conversely this linear bound implies the logarithmic bound with $K = C$.

Proof. If no such C exists, there are $z_n \in S$ tending to p with $E(z_n) > 0$, $-L(z_n) > 0$, and $E(z_n)/[-L(z_n)] \to 0$. The semianalytic set

$$Z = \{(z, u) : z \in S,\ E(z) > 0,\ -L(z) > 0,\ u > 0,\ E(z) = u[-L(z)]\}$$

therefore has $(p, 0)$ in its closure. Real analytic curve selection [10, (7.2), p. 40] supplies an analytic arc $(z(\rho), u(\rho))$ into Z for $\rho > 0$. Boundedness of L and $E = u(-L)$ imply $E(p) = 0$. Since $u(0) = 0$ and $E(z(0)) = 0$, but both functions are positive for $\rho > 0$, they have finite orders $u(\rho) = a\rho^k(1 + o(1))$ and $E(z(\rho)) = b\rho^m(1 + o(1))$, with $a, b > 0$ and integers $k, m \geq 1$. The assumed estimate would force

$$u(\rho)^{-1} \leq K[1 + \log_+(M/E(z(\rho)))]^q = O((1 + |\log \rho|)^q),$$

contradicting its order ρ^{-k}. At $E = 0$ the assumed estimate already gives $L \geq 0$. The converse is immediate. □

The theta stationary sets are analytic, including their singular points. Restrictions by $t \leq 1/2$ or by a closed $t \log(x/(4\pi))$ strip are locally semianalytic for $x > 4\pi$, so the lemma applies there as well. It yields an existence constant on a germ, not a source formula for that constant, nor uniform control at infinity or as $t \downarrow 0$. In particular it does not authorize defining a proof coefficient from an observed quotient $-\mathcal{L}_j/E$. The exact record `numerics/COLLISION_BEHAVIOR_CHECKS_20261014.json` contains 1072 named assertions over 60 rational jet cases of multiplicities 2 through 13, including 30 delay comparisons. Its driver `numerics/check_collision_behavior.py` independently forms the physical numerators and denominators and checks the finite-part coefficients. These finite polynomial checks support the algebra; they do not prove analytic branch existence, the logarithmic limit, or a theta-specific sign estimate. Review 20 records the internal audit.

4 The four jets as quaternionic and octonionic fields

The physical geometry is planar. The vectors $v_j = (h_j, h_{j+1})$ satisfy

$$E = |v_0|^2, \qquad \mathcal{L}_j = -\det(v_j, v_{j+1}), \qquad \tfrac{1}{2}E_t = -v_0 \cdot v_2.$$

Quaternionic fields extend these overlapping pairs away from the real slice; octonionic associators preserve their signed dot products and determinants. We first give this scalar dictionary, then construct two monogenic extensions with different off-slice traces. The constructions organize the observations but impose no additional theta inequality.

4.1 A quaternionic heat field with independent external time

Use Hamilton units $\mathbf{ij} = \mathbf{k}$, $\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = -1$, and identify $\mathbb{C}$ with $\mathbb{C}_\mathbf{i} = \mathbb{R} + \mathbf{i}\mathbb{R}$. Complex coefficients are placed on the right of $\mathbf{j}$. Thus

$$(a + \mathbf{j}b)(c + \mathbf{j}d) = ac - \overline{b}d + \mathbf{j}(bc + \overline{a}d), \qquad \overline{a + \mathbf{j}b} = \overline{a} - \mathbf{j}b, \tag{4.1}$$

for $a, b, c, d \in \mathbb{C}_\mathbf{i}$, and $|a + \mathbf{j}b|^2 = |a|^2 + |b|^2$.

Write the independent quaternion coordinates as

$$q = x_0 + \mathbf{i}x_1 + \mathbf{j}x_2 + \mathbf{k}x_3 = z_0 + \mathbf{j}z_1, \qquad z_0 = x_0 + \mathbf{i}x_1, \quad z_1 = x_2 - \mathbf{i}x_3.$$

The unnormalized left Fueter operator and its conjugate are

$$D_q = \partial_{x_0} + \mathbf{i}\partial_{x_1} + \mathbf{j}\partial_{x_2} + \mathbf{k}\partial_{x_3}, \qquad \overline{D}_q = \partial_{x_0} - \mathbf{i}\partial_{x_1} - \mathbf{j}\partial_{x_2} - \mathbf{k}\partial_{x_3}.$$

A differentiable field is *left Fueter regular* if $D_q F = 0$. The convention using the factor $1/2$ has the same kernel; all units here multiply on the left. For the classical theory see [5].

Let $H(\tau, z)$ be the jointly entire theta heat family, satisfying $H_\tau = -H_{zz}$. With $t \in \mathbb{R}$ an independent external parameter, set

$$\tau = t + \overline{z_1} = t + x_2 + \mathbf{i}x_3.$$

$$G_m(t, q) = \partial_z^m H(\tau, z_0) + \mathbf{j}\partial_z^{m+1} H(\tau, z_0), \qquad F = G_0. \tag{4.2}$$

The coordinate z_1 supplies a complex heat-time displacement; both entries of G_m have the same spectral argument.

Proposition 4.1 (Regular lift and jet identities)**.** *Each G_m is globally real analytic and left Fueter regular in q. For every $m \geq 0$,*

$$D_q G_m = 0, \qquad \overline{D}_q G_m = 2G_{m+1}, \qquad \partial_t G_m = -G_{m+2}. \tag{4.3}$$

$$\begin{aligned} \partial_{x_0} G_m &= G_{m+1}, & \partial_{x_1} G_m &= G_{m+1}\mathbf{i}, \\ \partial_{x_2} G_m &= -G_{m+2}, & \partial_{x_3} G_m &= -G_{m+2}\mathbf{i}. \end{aligned} \tag{4.4}$$

Proof. For complex-valued U, V, direct use of (4.1) gives

$$\tfrac{1}{2} D_q(U + \mathbf{j}V) = U_{\overline{z}_0} - V_{z_1} + \mathbf{j}(V_{z_0} + U_{\overline{z}_1}).$$

The derivatives are componentwise Wirtinger derivatives. Taking $U = \partial_z^m H(\tau, z_0)$ and $V = \partial_z^{m+1} H(\tau, z_0)$, the first two terms vanish by holomorphy and the final pair cancel by the heat equation. Joint entireness proves global real analyticity. Differentiating the declared arguments proves (4.4) and the external time identity. Finally $D_q + \overline{D}_q = 2\partial_{x_0}$ proves the remaining assertion. □

4.2 The planar observations and their exact identities

On the physical slice $q = x \in \mathbb{R}$, write $h_m = h_m(t, x) = \partial_x^m H(t, x)$. The jets are real and

$$F = h_0 + \mathbf{j}h_1, \qquad F_x = G_1 = h_1 + \mathbf{j}h_2, \qquad F_{xx} = G_2 = h_2 + \mathbf{j}h_3 = -F_t. \tag{4.5}$$

Thus the four physical jets occur in three consecutive overlapping quaternion pairs. The basic observations are

$$E = |F|^2 = h_0^2 + h_1^2, \qquad \mathcal{L}_j = h_{j+1}^2 - h_j h_{j+2} \quad (j = 0, 1). \tag{4.6}$$

In particular $F(t,x) = 0$ means exactly that $H(t,\cdot)$ has a multiple real zero at x. The scalar positive-time collision criterion therefore translates to

$$\mathrm{RH} \quad \Longleftrightarrow \quad F(t,x) \neq 0 \quad (t > 0,\ x \in \mathbb{R}).$$

This uses the scalar collision theorem for the theta family. It makes no assertion about zeros at nonreal quaternion arguments.

The signed products and the heat equation give, on this slice,

$$\overline{G_j} G_{j+1} = h_j h_{j+1} + h_{j+1} h_{j+2} - \mathbf{j}\mathcal{L}_j, \tag{4.7}$$

$$E_t = -2(h_0 h_2 + h_1 h_3), \qquad (\partial_t + \partial_x^2) E = 2(h_1^2 + h_2^2). \tag{4.8}$$

Consequently

$$h_1 = 0 \implies E_t = 2\mathcal{L}_0, \qquad h_2 = 0 \implies E_t = 2\mathcal{L}_1. \tag{4.9}$$

These formulas preserve the two different stationary conditions. The positive source term in the backward-heat identity (4.8) does not give a forward minimum principle.

For $a = A + \mathbf{j}B$ and $b = C + \mathbf{j}D$ in this physical plane, put

$$a \cdot b = AC + BD, \qquad a \wedge b = AD - BC.$$

Then $\bar{a}b = a \cdot b + \mathbf{j}(a \wedge b)$, and the planar Gram identity gives

$$|a|^2 |b|^2 = (a \cdot b)^2 + (a \wedge b)^2. \tag{4.10}$$

In particular, for any physical residual $R \in \mathbb{R} + \mathbf{j}\mathbb{R}$,

$$E|F_x|^2 = \tfrac{1}{4} E_x^2 + \mathcal{L}_0^2, \qquad E|R|^2 = (F \cdot R)^2 + (F \wedge R)^2. \tag{4.11}$$

A lower bound for $F \cdot R$ need not control the transverse component $F \wedge R$. This distinction will matter when a full residual-norm bound is replaced by a one-sided signed estimate.

There is also a quotient-free relation among the consecutive pairs. Writing $\mathfrak{d} = h_0 h_3 - h_1 h_2$, the two-by-three determinant identity gives

$$\mathcal{L}_0 F_t = \mathcal{L}_1 F + \mathfrak{d} F_x. \tag{4.12}$$

Indeed, its real and $\mathbf{j}$ components are respectively $-\mathcal{L}_0 h_2 = \mathcal{L}_1 h_0 + \mathfrak{d} h_1$ and $-\mathcal{L}_0 h_3 = \mathcal{L}_1 h_1 + \mathfrak{d} h_2$. These polynomial identities remain valid when a jet or a Laguerre form vanishes.

On any interval where $h_j + i h_{j+1} \neq 0$, the same orientation has the scalar phase description

$$\partial_x \arg(h_j + i h_{j+1}) = -\frac{\mathcal{L}_j}{h_j^2 + h_{j+1}^2}. \tag{4.13}$$

This formula describes phase away from a common zero; the polynomial observations above remain the appropriate quantities at a candidate collision.

The quadratic form itself has only three real coordinates, with

$$\mathcal{L}_j = h_{j+1}^2 + \tfrac{1}{4}(h_j - h_{j+2})^2 - \tfrac{1}{4}(h_j + h_{j+2})^2. \tag{4.14}$$

Its inertia is $(2,1)$. Each frequency kernel $(y+z)^2$ and $(y-z)^2$, considered separately, uses only the features $1, y, y^2$, since $(y \pm z)^2 = y^2 + z^2 \pm 2yz$. Each corresponding finite frequency matrix therefore has rank at most three; no such rank bound is asserted here for the combined weighted Hermitian and bilinear expression. Shared moments can therefore be evaluated without constructing every pair separately. An invertible change of coordinates preserves the negative direction; low rank does not give positivity.

4.3 Two octonionic extensions and their different traces

For a reader unfamiliar with octonions, it suffices here to regard $\mathbb{O}$ as two copies of $\mathbb{H}$, with a specified multiplication. Write $(a,b) = a + \ell b$, where $\ell = (0,1)$, and adopt the Cayley–Dickson convention

$$(a,b)(c,d) = (ac - d\bar{b}, \bar{a}d + cb), \qquad \overline{(a,b)} = (\bar{a}, -b). \tag{4.15}$$

Then $|(a,b)|^2 = |a|^2 + |b|^2$. This is the convention in [6, Section 2.2]. Octonionic multiplication is nonassociative; we keep parentheses whenever three octonionic factors occur. All component products on the right of (4.15) are associative quaternion products.

Let $r = y_0 + \mathbf{i}y_1 + \mathbf{j}y_2 + \mathbf{k}y_3$, so that $p = q + \ell r$ is the eight-dimensional argument. Put $e_0 = 1, e_1 = \mathbf{i}, e_2 = \mathbf{j}, e_3 = \mathbf{k}$, and define the left operator

$$\mathcal{D}_8 = \sum_{\mu=0}^{3} e_\mu \partial_{x_\mu} + \sum_{\mu=0}^{3} (\ell e_\mu)\partial_{y_\mu}. \tag{4.16}$$

We call a field *left monogenic* if $\mathcal{D}_8\mathcal{O} = 0$. The pair convention gives the component formula

$$\mathcal{D}_8(A,B) = \left(D_q A - \sum_{\mu=0}^{3} B_{y_\mu}\bar{e}_\mu, \overline{D}_q B + \sum_{\mu=0}^{3} A_{y_\mu} e_\mu \right). \tag{4.17}$$

This follows by applying (4.15) to each basis term of (4.16). General octonionic Cauchy–Riemann systems are discussed in [7]; their regularity classes must be distinguished from the distinct Clifford and slice formulations [15, Proposition 3.5 and Section 4].

Proposition 4.2 (Reflected extension of the physical trace)**.** *For every $m \geq 0$, the field*

$$\widetilde{\mathcal{O}}_m(t,q,r) = (G_m(t,q), G_{m+2}(t,\bar{q})) \tag{4.18}$$

is globally real analytic and left monogenic. It satisfies

$$\partial_{x_0}\widetilde{\mathcal{O}}_m = \widetilde{\mathcal{O}}_{m+1}, \qquad \partial_t\widetilde{\mathcal{O}}_m = -\widetilde{\mathcal{O}}_{m+2}, \qquad \widetilde{\mathcal{O}}_m(t,x,0) = (G_m(t,x), G_{m+2}(t,x)). \tag{4.19}$$

Proof. Reflection reverses all three imaginary quaternion coordinates, hence

$$\bar{D}_q[B(\bar{q})] = (D_q B)(\bar{q}).$$

Both r-derivatives in (4.17) vanish, and both remaining components vanish by $D_q G_k = 0$. Reflection commutes with ∂_{x_0} and ∂_t, proving the derivative statements. Real analyticity follows from that of the G_k, and $\bar{x} = x$ gives the physical trace. □

This extension is constant in the four r coordinates. Its second component uses $z_0 = x_0 - \mathbf{i}x_1$ and $\tau = t - x_2 - \mathbf{i}x_3$, whereas the first uses the original arguments. A nonzero such field is not in whole-space $L^2(\mathbb{R}^8)$; its local square integrability on bounded domains is unaffected. For $m = 0$, its values use derivatives of H only through order three, evaluated at the original and reflected arguments. Its derivatives still generate higher jets. If the full quaternionic trace $(G_m(t,q), G_{m+2}(t,q))$ at $r = 0$ is required, the following extension retains that stronger prescription.

Proposition 4.3 (Extension with the original quaternionic trace)**.** *For each $m \geq 0$, the field*

$$\mathcal{O}_m(t,q,r) = \left(G_m(t,q) - \tfrac{1}{2} G_{m+3}(t,q)\overline{r}, G_{m+2}(t,q) \right) \tag{4.20}$$

is globally real analytic and satisfies

$$\mathcal{D}_8\mathcal{O}_m = 0, \qquad \partial_{x_0}\mathcal{O}_m = \mathcal{O}_{m+1}, \qquad \partial_t\mathcal{O}_m = -\mathcal{O}_{m+2}, \qquad \mathcal{O}_m(t,q,0) = (G_m, G_{m+2}). \tag{4.21}$$

Proof. For (4.20), B is independent of r. Moreover $D_q(G_{m+3}\overline{r}) = (D_qG_{m+3})\overline{r} = 0$, since the component algebra is associative and r is held fixed. The first component of (4.17) is therefore zero. The second is

$$2G_{m+3} - \tfrac{1}{2}\sum_{\mu=0}^{3} G_{m+3}\overline{e}_\mu e_\mu = 2G_{m+3} - 2G_{m+3} = 0.$$

The derivative identities follow directly from (4.3), and the explicit formula proves global real analyticity. □

The uncorrected pair with the same q argument, held constant in r, has defect $(0, 2G_{m+3})$. This does not apply to the reflected pair. More generally, two left Fueter-regular fields $A(q), B(q)$ admit the extension $(A - \frac{1}{2}B_{x_0}\overline{r}, B)$, by the same calculation. Thus the construction imposes no new condition on the original pair. For $m = 0$, its value away from $r = 0$ involves G_3, hence h_4; it is not a closed evolution of only four jets.

Both extensions have the same physical trace $(q, r) = (x, 0)$:

$$\mathcal{O}_0 = (h_0 + \mathbf{j}h_1) + \ell(h_2 + \mathbf{j}h_3) \in \operatorname{span}_{\mathbb{R}}\{1, \mathbf{j}, \ell, \ell\mathbf{j}\}.$$

This span is an associative quaternion subalgebra, as can be checked from (4.15). Its total norm is

$$|\mathcal{O}_0|^2 = E + h_2^2 + h_3^2. \tag{4.22}$$

At a nondegenerate double zero, $E = 0$ but $h_2 \neq 0$; the total octonionic norm is then positive. A collision criterion must retain the distinguished first pair.

4.4 Oriented associators preserve the signed observations

The *associator* measures the failure of associativity:

$$[A, B, C] = (AB)C - A(BC), \qquad \mathbf{u} = \ell\mathbf{j}.$$

For embedded quaternions $a, b \in \mathbb{H}$, the pair multiplication gives

$$[a, b, \ell] = \ell(\overline{ab} - \overline{ba}) = -\ell(ab - ba). \tag{4.23}$$

The last equality holds because a quaternion commutator has zero real part. The untwisted physical jets lie in the commutative plane $\mathbb{R} + \mathbf{j}\mathbb{R}$, so their associator with ℓ vanishes. Fixed right multiplication by $\mathbf{i}$ and $\mathbf{k}$ produces the needed signed quantities while preserving quaternion left regularity.

Lemma 4.4 (Dot products and determinants as associators)**.** *For $a = A + \mathbf{j}B$, $b = C + \mathbf{j}D$ with real A, B, C, D,*

$$[a\mathbf{i}, b\mathbf{i}, \ell] = -2\mathbf{u}(AD - BC), \qquad [a\mathbf{i}, b\mathbf{k}, \ell] = 2\mathbf{u}(AC + BD). \tag{4.24}$$

Proof. One has $a\mathbf{i} = A\mathbf{i} - B\mathbf{k}$, $b\mathbf{i} = C\mathbf{i} - D\mathbf{k}$, and $b\mathbf{k} = D\mathbf{i} + C\mathbf{k}$. Their quaternion commutators are, respectively, $2\mathbf{j}(AD - BC)$ and $-2\mathbf{j}(AC + BD)$. Substitute these into (4.23). □

In particular define, on the physical slice,

$$\mathcal{A}_E = [F\mathbf{i}, F\mathbf{k}, \ell] = 2\mathbf{u}E, \tag{4.25}$$

$$\mathcal{A}_j = [G_j\mathbf{i}, G_{j+1}\mathbf{i}, \ell] = 2\mathbf{u}\mathcal{L}_j \quad (j = 0, 1), \tag{4.26}$$

$$\mathcal{A}_t = [F\mathbf{i}, F_t\mathbf{k}, \ell] = \mathbf{u}E_t. \tag{4.27}$$

These are immediate from Lemma 4.4 and (4.8). In contrast to (4.22), $\mathcal{A}_E = 0$ is equivalent to $F = 0$. Equation (4.9) becomes

$$h_1 = 0 \implies \mathcal{A}_t = \mathcal{A}_0, \qquad h_2 = 0 \implies \mathcal{A}_t = \mathcal{A}_1.$$

There is no order on $\mathbb{O}$ being invoked. Specify the oriented ray $\mathcal{C}_{\mathbf{u}} = \{s\mathbf{u} : s \geq 0\}$. The two stationary sign conditions are exactly

$$\mathcal{A}_t \in \mathcal{C}_{\mathbf{u}} \quad \text{on} \quad \{h_1 h_2 = 0\}. \tag{4.28}$$

Likewise the locally bounded loss condition is exactly

$$\mathcal{A}_t + K\mathcal{A}_E \in \mathcal{C}_{\mathbf{u}} \quad \text{on} \quad \{h_1 h_2 = 0\}, \qquad K \geq 0, \tag{4.29}$$

with K locally bounded through every candidate collision. Indeed the coefficient on this ray is $E_t + 2KE$. Thus the associators preserve both the collision detector and the required orientation. Their norms alone would replace the signed quantities by absolute values.

4.5 What calibration, symmetry, and logarithmic norms retain

For imaginary octonions, the Harvey–Lawson identity is

$$\varphi(a,b,c)^2 + \tfrac{1}{4}|[a,b,c]|^2 = |a \wedge b \wedge c|^2, \qquad \varphi(a,b,c) = \langle a, bc\rangle. \tag{4.30}$$

Here the wedge denotes exterior volume, and $\langle a, b\rangle = \mathrm{Re}(\bar{a}b)$. See [12, Chapter IV, Theorem 1.6] and [13, Corollary 3.52]. For the twisted triples in Lemma 4.4, $\varphi = 0$, since the first two arguments lie in $\mathbb{H}$ and the third is ℓ. Thus (4.30) reduces to the planar identity (4.10). In particular $F\mathbf{i}$ and $F\mathbf{k}$ are orthogonal with norm $|F|$, and

$$|[F\mathbf{i}, F\mathbf{k}, \ell]| = 2E.$$

Whenever $F \neq 0$, this triple spans the fixed three-plane $\mathrm{span}_{\mathbb{R}}\{\mathbf{i}, \mathbf{k}, \ell\}$. Its volume is a useful consistency check, but its norm does not retain a signed coefficient. No calibrated-submanifold condition on a varying tangent plane has been established by this construction.

An octonion automorphism $g \in G_2$ satisfies $[gA, gB, gC] = g[A, B, C]$. Transforming both the observation and the oriented ray therefore leaves (4.29) unchanged. The larger triality symmetry acts through three related representations, rather than a single algebra automorphism [6, Section 2.4][14]. Any use of it here must also transport the distinguished first-pair projection and the stationary constraints. Mixing that pair with the second pair without transporting the projection changes the collision detector, as (4.22) already shows.

The special holomorphic structure does give an exact logarithmic-norm identity. Put

$$\widehat{E}(t,q) = |H(\tau, z_0)|^2 + |H_z(\tau, z_0)|^2, \qquad \Delta_q = \sum_{\mu=0}^{3} \partial_{x_\mu}^2.$$

At a physical point with $E > 0$,

$$\Delta_q \log \widehat{E}\Big|_{q=x} = \frac{4\{\mathcal{L}_0^2 + (h_0 h_3 - h_1 h_2)^2\}}{E^2}. \tag{4.31}$$

To verify it, write $v = (H, H_z)$. In either complex coordinate $\zeta = z_0, \tau$, holomorphy gives

$$\partial_\zeta \partial_{\bar\zeta} \log |v|^2 = \frac{|v|^2 |v_\zeta|^2 - |\langle v, v_\zeta \rangle_{\mathbb{C}}|^2}{|v|^4} = \frac{|\det(v, v_\zeta)|^2}{|v|^4}.$$

The complex inner product is Hermitian. The same formula in any complex direction proves that $\log \widehat{E}$ is plurisubharmonic away from the common zero set. On the physical slice the two determinants are $-\mathcal{L}_0$ and $-(h_0 h_3 - h_1 h_2)$, while $\Delta_q = 4(\partial_{z_0}\partial_{\bar z_0} + \partial_\tau \partial_{\bar\tau})$. This proves (4.31). The nonnegative right side has squared away the required orientation and is defined only where $E > 0$; it supplies no sign for the external derivative E_t.

This logarithmic positivity comes from the holomorphic pair, not generic octonionic monogenicity. The monogenic Cauchy kernel $\bar p/|p|^8$, away from the origin in eight dimensions, has norm $|p|^{-7}$, and

$$\Delta_8 \log |p|^{-7} = -42/|p|^2 < 0.$$

Likewise a frame chosen using $F/|F|$ is singular at a potential collision. A source-prescribed frame can be used, with its derivative terms retained, but cannot assume the desired first-pair nonvanishing.

4.6 Why a source-specific estimate remains necessary

The constructions above apply to every jointly entire backward-heat family real on the physical slice. The following control shows that even strict positivity and analytic decay of its cosine-transform source do not supply the missing sign.

Proposition 4.5 (A positive entire source with a positive-time collision)**.** *There exist strictly positive, even, entire functions Φ_*, decaying superexponentially on the real axis, whose heat cosine transforms are jointly entire and have a nondegenerate real double zero at a positive time. Their quaternionic and octonionic lifts satisfy all the preceding identities.*

Proof. Set $\rho(v) = 2e^{-v^4}/\Gamma(1/4)$, so that $\int_{\mathbb{R}} \rho(v)\, dv = 1$, and $\rho_\epsilon(v) = \epsilon^{-1}\rho(v/\epsilon)$ for real $\epsilon > 0$. Use

$$\Phi_\epsilon(u) = 2\{\rho_\epsilon(u-1) + \rho_\epsilon(u+1)\} + \rho_\epsilon(u-2) + \rho_\epsilon(u+2).$$

This source is strictly positive, even, and entire, with quartic decay on the real axis. Its half-line heat cosine transform is

$$\begin{aligned}\widetilde{H}_\epsilon(t,z) = 2\int_{\mathbb{R}} \rho(v) e^{t(1+\epsilon v)^2} \cos\big(z(1+\epsilon v)\big)\, dv \\ + \int_{\mathbb{R}} \rho(v) e^{t(2+\epsilon v)^2} \cos\big(z(2+\epsilon v)\big)\, dv.\end{aligned}$$

Quartic decay supplies an integrable majorant on every compact set of complex (t, z, ϵ). Hence this formula extends jointly entirely through $\epsilon = 0$, obeys backward heat, and there becomes

$$\widetilde{H}_0(t,z) = 2e^t \cos z + e^{4t} \cos(2z).$$

At $T_0 = (\log 2)/3$, $x_0 = \pi$, it satisfies $\widetilde{H}_0 = (\widetilde{H}_0)_x = 0$ and $(\widetilde{H}_0)_{xx} = -6e^{T_0} \neq 0$. By the heat equation the Jacobian of $(\widetilde{H}_\epsilon, (\widetilde{H}_\epsilon)_x)$ with respect to (t, x), at this point and $\epsilon = 0$, has determinant $-((\widetilde{H}_0)_{xx})^2 \neq 0$. The real analytic implicit-function theorem therefore gives a nearby nondegenerate real double zero at a positive time for every sufficiently small $\epsilon > 0$. □

The failure is visible in the signed observation. At such a collision (T, a), put $c = H_{xx}(T, a) \neq 0$. The implicit-function theorem gives a real stationary branch $H_x(t, \chi(t)) = 0$ through it. For $\delta = T - t \downarrow 0$, backward heat gives

$$H(t, \chi(t)) = c\delta + O(\delta^2), \qquad E(t, \chi(t)) = c^2\delta^2 + O(\delta^3), \qquad \mathcal{L}_0(t, \chi(t)) = -c^2\delta + O(\delta^2).$$

Thus (4.29) would require $K \geq \delta^{-1} + O(1)$, contradicting local boundedness at the collision. The lift and associator identities alone cannot provide that bound. For the theta family, its particular arithmetic structure must enter a further signed estimate.

For the genuine theta family, Theorem 1.1 supplies the additional sign information in its declared outer region: $\mathcal{A}_0$ and $\mathcal{A}_1$ lie strictly on $\mathcal{C}_{\mathbf{u}}$ there. At genuine stationary points the same is true of $\mathcal{A}_t$. This is the geometric consequence of the common arithmetic amplitude, not a consequence of monogenicity alone.

5 Centered residuals and the common theta carrier

The outer theorem uses oscillatory cancellation to reach a nonzero zeta comparison amplitude. To study the remaining region, we return to the exact theta source. The aim is to retain the same four derivatives and their signed correlations before estimating any error.

5.1 A contour identity for the physical derivatives

Proposition 5.1 (Complete-source contour shift)**.** *For real $t \geq 0$, $x \geq 0$, $0 < a < \pi/8$, and every integer $k \geq 0$,*

$$h_k(t, x) = e^{-ax}\Re \int_0^\infty (i(u + ia))^k e^{t(u+ia)^2} \Phi(u + ia) e^{ixu}\, du. \tag{5.1}$$

Proof. The theta series is normally convergent in the strip $|\Im z| < \pi/8$ because $\Re(e^{4z}) > 0$ there. Analytic continuation of the real reflection identity makes Φ even throughout the strip; Schwarz symmetry then makes $\Phi(iy)$ real. On the left edge of the contour shift, every factor of $(iz)^k e^{tz^2}\Phi(z)e^{ixz}$ is real at $z = iy$, while $dz = i\,dy$. Its contribution has zero real part. On the right edge, $\cos(4y) \geq \cos(4a) > 0$ gives uniform double-exponential decay. Cauchy's theorem proves the formula. The evenness argument applies to the complete source; it must precede any separation into individual theta summands. □

The all-index and integration bounds in Appendix A make truncation of (5.1) rigorous. They control absolute errors at the gamma exponential scale. Obtaining a sign still requires information about the retained, correlated jets.

5.2 A residual adapted to genuine stationarity

Freeze real a, b with $0 < a < \pi/8$ and $b \geq 0$, and put $\mu = a^2 + b^2$. For every integer $j \geq 0$, define

$$C_j = e^{-ax}\Re \int_0^\infty (u^2 - b^2)(i(u + ia))^j e^{t(u+ia)^2} \Phi(u + ia) e^{ixu}\, du. \tag{5.2}$$

The required stationary forms use $j = 0, 1$; higher residuals will enter exact heat-transport identities. The polynomial identity $u^2 - b^2 = -(i(u + ia))^2 - 2a\, i(u + ia) - \mu$ gives

$$C_j = -h_{j+2} - 2ah_{j+1} - \mu h_j, \qquad h_{j+1} = 0 \implies \mathcal{L}_j = \mu h_j^2 + h_j C_j. \tag{5.3}$$

Thus the contour drift disappears at exactly the original stationary constraint. No derivative of an exponentially normalized observation has replaced that constraint.

Proposition 5.2 (A sufficient residual cone)**.** *If $\eta > 0$ and*

$$\Delta := \eta^2 E - C_0^2 - C_1^2 \geq 0, \tag{5.4}$$

then the respective stationary sets satisfy

$$\mathcal{L}_0 \geq (\mu - \eta)E \quad (h_1 = 0), \qquad \mathcal{L}_1 \geq -\frac{\eta^2}{4\mu}E \quad (h_2 = 0). \tag{5.5}$$

A sufficient common negative-loss coefficient is $K = \eta^2/(4\mu)$. It is locally bounded wherever the prescribed parameters are locally bounded and μ stays positive.

Proof. At $h_1 = 0$, $E = h_0^2$ and $|C_0| \leq \eta\sqrt{E}$, so the first inequality follows from (5.3). At $h_2 = 0$, complete the square in $\mu h_1^2 - \eta|h_1|\sqrt{E}$ to obtain the second. Finally $\eta - \mu \leq \eta^2/(4\mu)$ because $(\eta - 2\mu)^2 \geq 0$. No division by E occurs. □

For use with exact algebra or interval certificates, the same argument has the following identities before stationarity is imposed:

$$\begin{aligned} 2\eta[\mathcal{L}_0 + (\eta - \mu)E] &= (C_0 + \eta h_0)^2 + C_1^2 + \Delta \\ &\quad + 2\eta h_1[(1 - \mu + \eta/2)h_1 + 2ah_0], \end{aligned} \tag{5.6}$$

$$4\mu\mathcal{L}_1 + \eta^2 E = (2\mu h_1 + C_1)^2 + C_0^2 + \Delta + 4\mu h_2(h_2 + 2ah_1). \tag{5.7}$$

Each final term is a multiple of the genuine stationary derivative. The cone is sufficient, but a one-sided bound for $h_j C_j$ can be substantially weaker and therefore more useful.

For example, suppose paid normalized approximations give $Z_j = e^{\pi x/8}h_j = p + \delta$ and $e^{\pi x/8}C_j = q + \epsilon'$, where $|\delta| \leq \varepsilon_0$, $|\epsilon'| \leq \zeta_0$. At $h_{j+1} = 0$, expansion of (5.3), retaining $\mu\delta^2 \geq 0$, gives

$$e^{\pi x/4}\mathcal{L}_j \geq \mu p^2 + pq - |2\mu p + q|\varepsilon_0 - |p|\zeta_0 - \varepsilon_0\zeta_0. \tag{5.8}$$

Every truncation, quadrature, and parameter error must be included in the two bounds. The retained product pq preserves its sign.

5.3 One carrier for all four derivatives

There is an exact arithmetic reduction behind the centered residual. Put $d_n = \frac{1}{2}\log n$. The source summands obey

$$\phi_n(z) = n^{-1/2}\phi_1(z + d_n). \tag{5.9}$$

For $v \geq d_n$, define

$$A_n(t, x; v) = n^{-1/2}\exp\{td_n^2 - 2td_n(v + ia) - ixd_n\}. \tag{5.10}$$

Changing variables $v = u + d_n$ in the complete shifted source yields

$$h_j = e^{-ax}\Re\int_0^\infty e^{t(v+ia)^2}\phi_1(v + ia)e^{ixv}\sum_{n \leq e^{2v}} A_n(t, x; v)[i(v + ia - d_n)]^j\, dv. \tag{5.11}$$

In particular $j = 0, 1, 2, 3$ uses only the first four logarithmic moments of the same arithmetic weights. The staircase cutoff $n \leq e^{2v}$ is part of the identity. It is not an independently chosen truncation.

To justify the rearrangement, let $0 \le t \le \tau$. Since $0 \le d_n \le v$, we have $|A_n| \le n^{-1/2}$ and $\sum_{n \le e^{2v}} n^{-1/2} \le 2e^v$. Therefore the absolute integrand in (5.11), before e^{-ax}, is bounded by

$$2(2\pi^2 + 3\pi)(v+a)^j e^{\tau v^2 + 10v - \pi \cos(4a) e^{4v}}. \tag{5.12}$$

This is integrable. It also gives a full integration tail, stated in Appendix A; no unretained index is omitted from that bound.

The common oscillatory exponential in the original n-th channel has phase $(x+2ta)u - \pi n^2 \sin(4a) e^{4u}$. Its critical location is

$$b_n = \frac{1}{4} \log \frac{x+2ta}{4\pi n^2 \sin(4a)} = b_1 - d_n. \tag{5.13}$$

Only nonnegative locations belong to the integration half-line. This observation supplies a centering choice; it does not assert first-channel dominance or a full stationary-phase approximation. Taking $b = b_1$, the multiplier defining C_j becomes

$$(v-d_n)^2 - b^2 = \underbrace{-b \log n + \tfrac{1}{4} \log^2 n}_{\delta_n} + 2(b-d_n)(v-b) + (v-b)^2. \tag{5.14}$$

Thus the remaining signed residual involves logarithmic weights of the same arithmetic sum, rather than unrelated errors for each derivative. At $v = b$ those weights have the transparent form

$$A_n = n^{-1/2 - tb - i(x/2+ta)} e^{t \log^2 n/4}. \tag{5.15}$$

The factors A_n are complex and their products retain arithmetic phases. A positive-weight covariance inequality does not apply to them.

For perspective, if $b > 0$ the frozen-parameter balanced quaternion $W = h_0 + \mathbf{j}(h_1 + ah_0)/b$ obeys

$$W_t = (b^2 - a^2 - 2ab\mathbf{j})W + R, \qquad R = C_0 + \mathbf{j}(C_1 - aC_0)/b. \tag{5.16}$$

The skew term rotates W without changing its norm. Equation (5.16) separates the expected oscillatory motion from the residual, whereas (5.3) keeps the precise scalar conditions needed for a sign. If a, b vary with time, the additional term is $\mathbf{j}(a_t \Re W/b - b_t \Im_{\mathbf{j}} W/b)$; all cell computations below freeze them before differentiating.

5.4 Staircase derivatives and exact entry terms

Put $T = x/2 + ta$ and, for integers $r \ge 0$,

$$B_r(v) = \sum_{n \le e^{2v}} (\log n)^r A_n(t,x;v), \qquad D_j(v) = \sum_{n \le e^{2v}} A_n(t,x;v)[i(v+ia-d_n)]^j.$$

For $r = 0$ the logarithmic weight is one, including at $n = 1$. On $v > 0$, distributional differentiation gives

$$B_r' = -tB_{r+1} + \sum_{n \ge 2} (\log n)^r A_n(t,x;d_n)\delta(v-d_n). \tag{5.17}$$

Between entry points,

$$D_j' = -2t(v+ia)D_j - 2itD_{j+1} + ijD_{j-1}, \tag{5.18}$$

where the last term is omitted for $j = 0$. The jump at d_n is $A_n(t,x;d_n)(-a)^j$. If $K(v) = e^{t(v+ia)^2}\phi_1(v+ia)e^{ixv}$, its full contribution is

$$K(d_n)A_n(t,x;d_n)(-a)^j = e^{-ta^2}\phi_n(ia)(-a)^j. \tag{5.19}$$

Indeed $\partial_v A_n = -2td_nA_n$ and $d_n = v + ia + i[i(v + ia - d_n)]$ give the smooth recurrence; direct substitution and (5.9) give the jump formula. At $v = 0$ the index $n = 1$ is already present and belongs to the lower endpoint of any integration by parts. Every entering-index phase in (5.19) cancels in x. These terms cannot be assigned an artificial oscillatory gain.

The oscillatory block estimate in Section 2 can also be applied to this exact carrier. On a block $M < n \le R(v)$, where $R(v) = \min(2M, e^{2v})$, the logarithmic-moment weight is

$$w_r(y) = (\log y)^r e^{t\log^2 y/4} y^{-1/2-tv}.$$

For $M \ge e^{2r}$ it is decreasing, since $y(\log w_r)' \le r/\log y - 1/2 \le 0$; for $r = 0$ this holds for all $M \ge 1$. Abel summation therefore gives, when $|T| \ge 1$,

$$\left| \sum_{M<n\le R(v)} w_r(n) n^{-iT} \right| \ll w_r(M)(|T|^{1/6} M^{1/2} + |T|^{-1/6} M).$$

For other blocks use the full total variation of w_r, and for $|T| < 1$ use the trivial bound. The actual shortened endpoint remains $R(v)$; these complex block estimates do not imply a signed product bound.

5.5 The complete paired observations

Define the staircase measure space and its coordinates by

$$\begin{aligned} &\Omega = \{(v,n) : n \ge 1,\ v \ge d_n\}, \qquad d\nu = dv \times \text{counting measure}, \\ &y_\alpha = v - d_n, \qquad q_\alpha = -a + iy_\alpha, \\ &W_\alpha = e^{t(v+ia)^2} \phi_1(v + ia) e^{ixv} A_n(t, x; v), \qquad \alpha = (v, n). \end{aligned} \tag{5.20}$$

Then $h_k = e^{-ax}\Re \int_\Omega W_\alpha q_\alpha^k \, d\nu(\alpha)$. The two factors in the following integrals have independent coordinates and independent indices.

Proposition 5.3 (Paired Laguerre and residual forms)**.** *For $j = 0, 1$,*

$$\begin{aligned} \mathcal{L}_j = \frac{e^{-2ax}}{4} \Re \iint_{\Omega^2} [&(y_\alpha + y_\beta)^2 W_\alpha \overline{W_\beta} (q_\alpha \overline{q_\beta})^j \\ &+ (y_\alpha - y_\beta)^2 W_\alpha W_\beta (q_\alpha q_\beta)^j] \, d\nu(\alpha) d\nu(\beta). \end{aligned} \tag{5.21}$$

Writing

$$J_j(\alpha, \beta) = W_\alpha \overline{W_\beta} (q_\alpha \overline{q_\beta})^j + W_\alpha W_\beta (q_\alpha q_\beta)^j, \tag{5.22}$$

the centered product is

$$h_j C_j = \frac{e^{-2ax}}{4} \Re \iint_{\Omega^2} (y_\alpha^2 + y_\beta^2 - 2b^2) J_j(\alpha, \beta) \, d\nu(\alpha) d\nu(\beta). \tag{5.23}$$

Before imposing stationarity, the exact difference is

$$\mathcal{L}_j - \mu h_j^2 - h_j C_j = h_{j+1}(h_{j+1} + 2a h_j). \tag{5.24}$$

Proof. Use $\Re A \Re B = \frac{1}{2}\Re(A\overline{B} + AB)$ and symmetrize in α, β. The Hermitian difference in the Laguerre form is

$$q_\alpha \overline{q_\beta} - \tfrac{1}{2}(q_\alpha^2 + \overline{q_\beta}^2) = -\tfrac{1}{2}(q_\alpha - \overline{q_\beta})^2 = \tfrac{1}{2}(y_\alpha + y_\beta)^2;$$

the bilinear difference is $-\frac{1}{2}(q_\alpha - q_\beta)^2 = \frac{1}{2}(y_\alpha - y_\beta)^2$. Conjugate-swapping the Hermitian term preserves its real integral. The same polarization applied to the real multiplier $y^2 - b^2$ gives (5.23), and (5.3) gives (5.24). The majorant (5.12) has every polynomial moment integrable. Together with $(y \pm z)^2 \le 2(y^2 + z^2)$, this proves absolute convergence of each displayed integral and justifies Fubini. □

The squared frequencies in (5.21) do not give the phase of either amplitude product a sign. As a finite algebra control, two real signed atoms at frequencies $1, 2$, with weights $1, -1/2$ and $a = 0$, give $(h_0, h_1, h_2) = (1/2, 0, 1)$ and $\mathcal{L}_0 = -1/2$. The diagonal contributions sum to 2 and the ordered cross terms to $-5/2$. This control lies outside the theta class and positive-contour hypotheses; it explains why retaining only diagonal pairs loses the observation.

5.6 A complete omitted-pair payment

For a measurable retained set $A \subset \Omega$, denote its jets by h_k^A and set $\mathcal{L}_j^A = (h_{j+1}^A)^2 - h_j^A h_{j+2}^A$. Define absolute masses before the exterior exponential normalization:

$$M_k^A = \int_A |W_\alpha||q_\alpha|^k \, d\nu, \qquad T_k^A = \int_{\Omega\setminus A} |W_\alpha||q_\alpha|^k \, d\nu.$$

Then

$$|\mathcal{L}_j - \mathcal{L}_j^A| \le e^{-2ax}(M_j^A T_{j+2}^A + M_{j+2}^A T_j^A + T_j^A T_{j+2}^A). \tag{5.25}$$

Indeed, including the factor $1/4$ in (5.21), the absolute sum of its two kernels is at most

$$\tfrac{1}{2}(|W_\alpha||q_\alpha|^{j+2}|W_\beta||q_\beta|^j + |W_\alpha||q_\alpha|^j|W_\beta||q_\beta|^{j+2}),$$

using $(y+z)^2 + (y-z)^2 = 2(y^2+z^2)$ and $y^2 \le |q|^2$. Integrating over $A \times A^c$, $A^c \times A$, and $A^c \times A^c$ gives the claim, including all omitted mixed and omitted–omitted pairs. No bound for T_{j+1}^A is required.

For $A = \{(v, n) \in \Omega : v \le V,\ n \le N\}$, use the complete index payment and the shared-coordinate integration payment in Appendix A, removing their e^{-ax} normalization before using them for T_k^A. Overlap may be overpaid. Retained quadrature and parameter variation require their own errors. The absolute payment (5.25) is not a relative bound through an unknown zero of E.

5.7 Signed staircase cutoffs on the closed outer domain

Fix $a = \pi/16$ and retain exactly

$$\Omega_N = \{(v, n) \in \Omega : 0 \le v \le V_N\}, \qquad V_N = \tfrac{1}{2}\log N, \tag{5.26}$$

where the intrinsic restriction $n \le e^{2v}$ remains part of Ω. There is no additional rectangular approximation to that staircase. Write $h_k^{(N)}$ for the restricted common-source moments and $\mathcal{L}_j^{(N)} = (h_{j+1}^{(N)})^2 - h_j^{(N)} h_{j+2}^{(N)}$. All cross terms in these products are retained.

To avoid confusing source constants with the centered residual C_0, use the notation

$$\begin{gathered}\beta_0=\pi\cos(4a)=\frac{\pi}{\sqrt{2}},\qquad \mathsf{C}_0=2(2\pi^2+3\pi),\qquad c=\beta_0/2,\qquad \mathsf{D}_0=\frac{\mathsf{C}_0}{2\beta_0},\\ \mathsf{M}_k=\mathsf{C}_0\int_0^\infty (v+a)^k\exp\{v^2/2+10v-\beta_0e^{4v}\}\,dv,\qquad 0\le k\le 3,\\ B_j=(\mathsf{M}_j+\mathsf{M}_{j+2})\mathsf{D}_0+\mathsf{D}_0^2,\qquad j=0,1.\end{gathered}\tag{5.27}$$

These finite constants depend only on the fixed complete source and a, not on t,x,N.

Corollary 5.4 (Signed staircase cutoff estimates)**.** *For all integers $N\ge 8$, $j=0,1$, and (t,x) in (1.8),*

$$\mathcal{L}_j^{(N)}\ge -A_j^*(t,x)E,\qquad \mathcal{L}_j^{(N)}\ge -e^{-2ax}B_je^{-cN^2},\tag{5.28}$$

where

$$A_j^*(t,x)=\frac{e^{-2ax}\mathsf{M}_j\mathsf{M}_{j+2}\zeta(9/8)^2}{\Gamma(t,x)^2},\qquad \Gamma(t,x)=|M_t((1+ix)/2)|.\tag{5.29}$$

The coefficient is prescribed independently of N and of the measured physical energy; it is locally bounded but is not asserted uniformly bounded as $x\to\infty$. The second estimate has zero energy loss and height-uniform Gaussian tail constants. The observations and stationary sets remain those of the full function.

Proof. For $0\le t\le 1/2$, (5.12) bounds every retained absolute k-th moment by M_k. Summing the absolute values of the Hermitian and bilinear kernels in (5.21), using $(y+z)^2+(y-z)^2=2(y^2+z^2)$ and its factor $1/4$, gives

$$|\mathcal{L}_j^{(N)}|\le e^{-2ax}\mathsf{M}_j\mathsf{M}_{j+2}.\tag{5.30}$$

The energy envelope (2.33) and (5.29) therefore prove the first inequality in (5.28), with a spare factor two. Its denominator is the explicitly nonzero analytic carrier, not E; this proves the asserted independence and local boundedness.

For the second inequality, the entire omitted source region is $v>V_N$. For $v\ge 1$,

$$\log(v+a)\le v,\qquad v^2/2+13v\le(\beta_0/2)e^{4v}.$$

The latter follows because $e^{-4v}(v^2/2+13v)$ is decreasing there and its value at one is less than $\beta_0/2$. Since $V_N\ge 1$ for $N\ge 8$, every omitted moment of order $k\le 3$ satisfies

$$T_k^{(N)}\le\frac{\mathsf{C}_0}{2\beta_0N^2}e^{-cN^2}\le\mathsf{D}_0e^{-cN^2}.\tag{5.31}$$

Indeed substitute $q=e^{4v}$ in the integral of $e^{-ce^{4v}}$ and use $q^{-1}\le N^{-2}$. Apply the complete payment (5.25), including both mixed sectors and the omitted–omitted sector. With the constants in (5.27) it gives

$$|\mathcal{L}_j-\mathcal{L}_j^{(N)}|\le e^{-2ax}B_je^{-cN^2}.\tag{5.32}$$

The independently proved full sign $\mathcal{L}_j>0$ establishes the second inequality in (5.28). □

The absolute and tail estimates in the proof hold for every $x\ge 0$ and $0\le t\le 1/2$; the signed conclusions use the independent outer positivity theorem and are asserted only in its domain. They do not assert positivity of every truncated form. Also $h_{j+1}=0$ does not imply $h_{j+1}^{(N)}=0$; no such substitution is made. At $E=0$ on the corresponding full stationary set, an unpaid inequality

$\mathcal{L}_j^{(N)} \geq -AE$ would require $\mathcal{L}_j^{(N)} \geq 0$ for every cutoff, which does not follow merely from $\mathcal{L}_j = 0$. The Gaussian additive allowance avoids imposing that extra cutoff condition.

One exact alternative retaining the complete amplitude is

$$\mathcal{J}_j^{(N)} = -h_j h_{j+2}^{(N)} \quad \text{on } h_{j+1} = 0, \qquad j = 0, 1.$$

It obeys

$$\mathcal{J}_j^{(N)} - \mathcal{L}_j = h_j(h_{j+2} - h_{j+2}^{(N)}), \qquad |\mathcal{J}_j^{(N)} - \mathcal{L}_j| \leq \sqrt{E}\, e^{-ax} T_{j+2}^{(N)}. \tag{5.33}$$

Thus $\mathcal{J}_j^{(N)}$ vanishes at $E = 0$ without imposing positivity of a raw truncated quadratic form. This is an exact error identity, not a new signed estimate in the remaining cusp.

6 Complete-source correlation kernels

The paired observations can also be organized on the complete real source. This gives a scalar correlation family containing both Laguerre forms and the physical energy. The construction is related to the associated kernels of [26, Theorem 3.7] and the Wronskian identities of [27, Theorem 1.3]. We derive the formulas with the present normalization rather than importing additional log-concavity hypotheses for the deformed source. They preserve the signed problem and do not establish a new sign region.

6.1 The correlation identities and their heat hierarchy

Use the even extension of the complete source and put

$$\begin{aligned} w_t(u) &= e^{tu^2}\Phi(u), \qquad H(t,x) = \frac{1}{2}\int_{\mathbb{R}} w_t(u) e^{ixu}\, du, \\ \mathscr{K}_m(t,s) &= \int_{\mathbb{R}} r^{2m} w_t(s+r) w_t(s-r)\, dr, \qquad m \geq 0. \end{aligned} \tag{6.1}$$

Here t, x, s are real and m is an integer. The following complete-source identities hold at every real time; their domination is uniform when time is restricted to a bounded interval.

Proposition 6.1 (Complete scalar correlations)**.** *Each $\mathscr{K}_m(t,\cdot)$ is real, even, positive, and integrable with every polynomial weight. The two Laguerre observations are*

$$\mathcal{L}_0(t,x) = \int_{\mathbb{R}} \mathscr{K}_1(t,s)\cos(2xs)\, ds, \tag{6.2}$$

$$\mathcal{L}_1(t,x) = \int_{\mathbb{R}} \{\mathscr{K}_2(t,s) - s^2\mathscr{K}_1(t,s)\}\cos(2xs)\, ds. \tag{6.3}$$

The exact time identities are

$$\mathscr{K}_m(t,s) = e^{2ts^2}\int_{\mathbb{R}} r^{2m} e^{2tr^2}\Phi(s+r)\Phi(s-r)\, dr, \tag{6.4}$$

$$\partial_t^\ell \mathscr{K}_m(t,s) = 2^\ell \sum_{k=0}^{\ell} \binom{\ell}{k} s^{2(\ell-k)} \mathscr{K}_{m+k}(t,s) \qquad (\ell \geq 0). \tag{6.5}$$

In particular,

$$(\partial_t + \partial_x^2)\mathcal{L}_0 = 2\mathcal{L}_1. \tag{6.6}$$

Proof. By evenness, (1.7) gives a double-exponential majorant in $|u|$. On $|t| \le T$, multiplying that majorant by e^{Tu^2} still gives double-exponential decay. Thus the source's absolute moments of every order are finite. Under

$$u = s + r, \qquad v = s - r, \qquad du\,dv = 2\,ds\,dr,$$

every polynomial in s, r is a polynomial in u, v. The double integrals below therefore converge absolutely, and the same argument pays every finite time or Fourier derivative. Evenness and positivity of $\mathscr{K}_m$ follow directly from evenness and strict positivity of w_t; for $m > 0$ the integrand vanishes only at $r = 0$.

The full-line formula for h_k has multiplier $(iu)^k/2$. Symmetrizing the product defining $\mathcal{L}_j$, for $j = 0, 1$, gives

$$\mathcal{L}_j(t,x) = \frac{1}{8}\iint_{\mathbb{R}^2} (-uv)^j (u-v)^2 w_t(u) w_t(v) e^{ix(u+v)}\,du\,dv.$$

The change of variables has $(u-v)^2 = 4r^2$, $-uv = r^2 - s^2$, and phase e^{2ixs}. Its Jacobian cancels the remaining factor $1/2$, proving (6.2)–(6.3) after taking real parts. The imaginary parts vanish by evenness in s.

Since $(s+r)^2 + (s-r)^2 = 2(s^2 + r^2)$, differentiation of (6.4) and the binomial theorem give (6.5). At $\ell = 1, m = 1$,

$$\partial_t \mathscr{K}_1 = 2s^2 \mathscr{K}_1 + 2\mathscr{K}_2.$$

Inserting this into (6.2), and using $\partial_x^2 \cos(2xs) = -4s^2 \cos(2xs)$, proves (6.6). □

The last identity has the independent jet verification

$$\partial_t \mathcal{L}_0 = h_2^2 - 2h_1 h_3 + h_0 h_4, \qquad \partial_x^2 \mathcal{L}_0 = h_2^2 - h_0 h_4.$$

It does not make the second stationary observation redundant: a nonnegative function need not have a nonnegative value of $\partial_t + \partial_x^2$.

Proposition 6.2 (The energy in the same correlation family)**.** *Define*

$$\mathscr{E}(t,s) = \frac{1}{2}\{(1-s^2)\mathscr{K}_0(t,s) + \mathscr{K}_1(t,s)\}. \tag{6.7}$$

Then

$$E(t,x) = \int_{\mathbb{R}} \mathscr{E}(t,s)\cos(2xs)\,ds. \tag{6.8}$$

Writing $\mathscr{A}_0 = \mathscr{K}_1$ and $\mathscr{A}_1 = \mathscr{K}_2 - s^2\mathscr{K}_1$, a prescribed coefficient $\kappa_j(t,x) \ge 0$ satisfies the stationary loss $\mathcal{L}_j \ge -\kappa_j E$ if and only if

$$\int_{\mathbb{R}} \{\mathscr{A}_j(t,s) + \kappa_j(t,x)\mathscr{E}(t,s)\}\cos(2xs)\,ds \ge 0 \quad \text{on } S_j = \{h_{j+1} = 0\}. \tag{6.9}$$

Proof. The product expression for $h_0^2 + h_1^2$ has multiplier $(1-uv)/4$ in the u, v coordinates. The same substitution gives

$$E = \frac{1}{2}\iint_{\mathbb{R}^2} (1 - s^2 + r^2) w_t(s+r) w_t(s-r) e^{2ixs}\,dr\,ds,$$

which proves (6.8). Subtracting the target inequalities and using (6.2)–(6.3) proves the equivalence. □

In (6.9), stationarity and energy are still those of the full physical function. The coefficient must be obtained independently of an unknown energy floor and have the local boundedness required by (3.3). Because κ_j may depend on x, this is generally not the Fourier transform of one fixed kernel as x varies.

The positivity $\mathscr{K}_1 > 0$ is not positive definiteness. For a continuous integrable kernel, positive definiteness imposes nonnegativity of its Fourier transform; pointwise positivity alone does not. In particular the associated-kernel criteria in [26] retain a substantive hypothesis. Requiring a positive-definite decomposition everywhere would be stronger than the stationary bounded-loss target (6.9). All complete source correlations have been retained, but their oscillatory sign is still to be proved.

Remark 6.3 (Gaussian normalization and a signed kernel). For the control $w(u) = e^{-\alpha u^2}$, $\alpha > 0$,

$$H(x) = \frac{\sqrt{\pi}}{2\sqrt{\alpha}} e^{-x^2/(4\alpha)}, \qquad \mathscr{K}_0(s) = \sqrt{\frac{\pi}{2\alpha}} e^{-2\alpha s^2},$$
$$\mathscr{K}_1 = \frac{\mathscr{K}_0}{4\alpha}, \qquad \mathscr{K}_2 = \frac{3\mathscr{K}_0}{16\alpha^2}.$$

The formulas above give

$$\mathcal{L}_0 = \frac{\pi}{8\alpha^2} e^{-x^2/(2\alpha)} = \frac{H^2}{2\alpha}, \qquad \mathcal{L}_1 = \mathcal{L}_0 \left(\frac{1}{2\alpha} + \frac{x^2}{4\alpha^2} \right).$$

Both agree with direct differentiation. Nevertheless $\mathscr{K}_2 - s^2 \mathscr{K}_1 = \mathscr{K}_1(3/(4\alpha) - s^2)$ changes sign. Thus a demand that this correlation integrand be pointwise positive would already reject this favorable control.

6.2 The existing staircase cutoff as a cross-correlation

We now retain exactly the cutoff (5.26), with $a = \pi/16$, integer $N \geq 8$, $V_N = \frac{1}{2} \log N$, $0 \leq t \leq 1/2$, and $x \geq 0$. No new source truncation is introduced. In the original, rather than shared, integration coordinate $y = v - d_n$, define

$$\psi_{a,N}(t,y) = \begin{cases} e^{t(y+ia)^2} \displaystyle\sum_{\substack{n \geq 1 \\ y + d_n \leq V_N}} \phi_n(y+ia), & 0 \leq y \leq V_N, \\ 0, & \text{otherwise.} \end{cases} \tag{6.10}$$

The staircase is precisely $n \leq N e^{-2y}$. Endpoint values at its finitely many jumps do not affect any integral. The source translation (5.9) gives $W_\alpha = e^{t(y+ia)^2} \phi_n(y+ia) e^{ixy}$ when $y = v - d_n$. Reversing that change of variables therefore gives

$$h_k^{(N)}(t,x) = e^{-ax} \Re \int_0^{V_N} (-a+iy)^k \psi_{a,N}(t,y) e^{ixy}\, dy. \tag{6.11}$$

This reversal takes place in the retained, already shifted integral; it does not shift individual theta summands across an unpaid contour.

Proposition 6.4 (Full-amplitude hybrid cross-correlation). *For $j = 0, 1$, put*

$$\mathscr{C}_{j,N}^{a}(t,s) = -\int_{s-V_N}^{s} [i(s+r)]^j [-a + i(s-r)]^{j+2} \cdot w_t(s+r) \psi_{a,N}(t, s-r)\, dr. \tag{6.12}$$

The full-amplitude observation, defined at every point by $\mathcal{J}_j^{(N)} = -h_j h_{j+2}^{(N)}$, *has the exact representation*

$$\mathcal{J}_j^{(N)}(t,x) = e^{-ax}\Re\int_{\mathbb{R}} \mathscr{C}_{j,N}^{a}(t,s)e^{2ixs}\,ds. \tag{6.13}$$

On the full stationary set S_j *it obeys*

$$\left|\mathcal{J}_j^{(N)} - \mathcal{L}_j\right| \le \sqrt{E}\,e^{-ax}T_{j+2}^{(N)} \le \mathsf{D}_0 e^{-ax}\sqrt{E}\,e^{-cN^2}, \tag{6.14}$$

with c, D_0 *from* (5.27) *and the complete omitted moments* (5.31).

Proof. Multiply the full-line formula for h_j by (6.11). Since h_j is real, its full-line integral can be placed inside the real part:

$$-h_j h_{j+2}^{(N)} = -\frac{e^{-ax}}{2}\Re\int_{\mathbb{R}}\int_0^{V_N} (iu)^j(-a+iy)^{j+2}w_t(u)\psi_{a,N}(t,y)e^{ix(u+y)}\,dy\,du.$$

The finite retained source is bounded on its compact integration interval, and w_t has every absolute moment. Thus Fubini and the change of variables $u = s + r,\ y = s - r$ apply. The Jacobian is two and the interval $0 \le y \le V_N$ becomes $s - V_N \le r \le s$, proving (6.13).

Before stationarity,

$$\mathcal{J}_j^{(N)} - \mathcal{L}_j = h_j(h_{j+2} - h_{j+2}^{(N)}) - h_{j+1}^2.$$

On S_j the square vanishes and $|h_j| \le \sqrt{E}$. The complete omitted-moment bound proves (6.14). □

The discontinuities and entering-index boundaries remain in $\psi_{a,N}$; the cross-correlation formula has not removed them. Any integration by parts must include their boundary or distributional terms. There is also no replacement of the full stationary derivative by its retained approximation. At $E = 0$ both $\mathcal{J}_j^{(N)}$ vanish by their full-amplitude factorization, independently of the sign of the retained source. An all-N signed estimate for (6.13) is therefore another way to address the existing cutoff transfer, with exactly the same tail obligation.

6.3 An optional common observation for the two stationary sets

The two hybrid products can be combined if a common scalar observation is useful. Put

$$\mathcal{B}_N = -h_0 h_2^{(N)} - h_1 h_3^{(N)}, \qquad \varepsilon_k^{(N)} = e^{-ax}T_k^{(N)}. \tag{6.15}$$

This is an optional bookkeeping device; the separate products can give sharper estimates.

Corollary 6.5 (Common cutoff and its full energy derivative)**.** *At every point in the preceding cutoff domain,*

$$\left|\mathcal{B}_N - \frac{1}{2}E_t\right| \le \sqrt{E}\,\sqrt{(\varepsilon_2^{(N)})^2 + (\varepsilon_3^{(N)})^2}. \tag{6.16}$$

Moreover,

$$\mathcal{B}_N = \mathcal{J}_0^{(N)} \qquad \text{on } S_0, \tag{6.17}$$

$$\mathcal{B}_N - \mathcal{J}_1^{(N)} = h_0(h_2 - h_2^{(N)}) \qquad \text{on } S_1. \tag{6.18}$$

Consequently, for $j = 0, 1$,

$$|\mathcal{B}_N - \mathcal{L}_j| \le \sqrt{2}\,\mathsf{D}_0 e^{-ax}\sqrt{E}\,e^{-cN^2} \quad \text{on } S_j. \tag{6.19}$$

Proof. Subtract (1.3) before imposing stationarity. The difference is

$$h_0(h_2 - h_2^{(N)}) + h_1(h_3 - h_3^{(N)}).$$

Cauchy–Schwarz gives (6.16). Setting $h_1 = 0$ or $h_2 = 0$ proves (6.17)–(6.18). The complete Gaussian moment bounds and (1.3) then give (6.19). □

In the associator notation of Lemma 4.4, set $G_2^{(N)} = h_2^{(N)} + \mathbf{j}h_3^{(N)}$ and $\mathbf{u} = \ell\mathbf{j}$. Then

$$-\frac{1}{2}[F\mathbf{i}, G_2^{(N)}\mathbf{k}, \ell] = \mathbf{u}\,\mathcal{B}_N.$$

The coefficient, rather than its absolute value, is still the observation. Both the scalar and octonionic forms vanish when $E = 0$. On S_1, the combined cutoff pays an additional second-derivative tail which the separate $\mathcal{J}_1^{(N)}$ avoids.

For example, if a new source estimate were to prove $\mathcal{B}_N \geq -A_j N^p E$ on S_j for every $N \geq N_0 \geq 8$, with the prescribed coefficients and local boundedness required by Lemma 9.7, Young's inequality would give

$$\mathcal{L}_j \geq -(A_j N^p + 1/2)E - \mathsf{D}_0^2 e^{-2ax} e^{-2cN^2}.$$

This is within the existing conditional logarithmic-loss transfer. It is not a proof of the proposed lower bound for $\mathcal{B}_N$. Neither the combined notation nor the correlation transform strengthens the arithmetic input already available.

6.4 What modular reflection and exceptional arithmetic would add

For $\Theta(y) = \sum_{n\in\mathbb{Z}} e^{-\pi n^2 y}$, $y > 0$, Poisson summation gives

$$\Theta(y) = y^{-1/2}\Theta(1/y), \qquad P(u) = e^u\Theta(e^{4u}) = P(-u), \qquad \Phi = (P'' - P)/16.$$

The factor e^{tu^2} preserves this reflection. In the complete pair $w_t(s+r)w_t(s-r)$, inversion of either theta variable amounts to exchanging s, r, up to signs, after the prefactors and differentiations are included. That exchange is already a consequence of evenness for any source. It therefore cannot by itself distinguish the theta family from the collision controls of Proposition 4.5.

The two theta arguments $e^{4(s+r)}$ and $e^{4(s-r)}$ have product e^{8s} and are reciprocal only when $s = 0$. There is no automatic diagonalization of the full correlation. Nor can the heat deformation be suppressed in a modular identity: under $y = e^{4u}$ it is $\exp\{t(\log y)^2/16\}$. It is invariant under $y \mapsto 1/y$, but is not a fixed modular weight. With $\tau = ie^{4u}$, even the modular translation $\tau \mapsto \tau + 1$ changes the logarithmic variable to $\frac{1}{4}\log(e^{4u} - i)$, on a consistent branch, and changes both the heat multiplier and the Fourier phase. A useful additional theta identity must retain these factors and, for a filtered source, every transformed boundary contribution.

Integral octonions offer genuine additional arithmetic, but through a different source. The Coxeter order has the E_8 quadratic lattice [28, Section 1]; in the standard lattice normalization its theta series is

$$\Theta_{E_8}(\tau) = E_4(\tau) = 1 + 240\sum_{n\geq 1}\sigma_3(n)e^{2\pi i n\tau}.$$

Termwise Mellin integration and $n = dm$ in the divisor sum give

$$\int_0^\infty (\Theta_{E_8}(iy) - 1)y^{s-1}\,dy = 240(2\pi)^{-s}\Gamma(s)\zeta(s)\zeta(s-3), \qquad \Re s > 4. \tag{6.20}$$

Thus replacing the scalar theta source by the octonionic lattice theta series changes both its multiplicities and its Mellin transform. Positive representation counts do not by themselves give the signed products in (6.9).

Exceptional theta forms and their classical restrictions are developed in [29, 30, 31, 32]. They justify investigating an arithmetic bridge, rather than identifying the two sources merely because both involve theta functions. For example, [32, Theorem B] proves nonvanishing of the stated global lift for a cuspidal Hecke eigenform in $S_{2n+12}(\Gamma_0(N))$, $n \geq 0$, in its PGL_2 setting. Its Theorem E spans $S_{2n+12}(\mathrm{SL}_2(\mathbb{Z}))$ by the specified weighted exceptional theta series for $n > 0$. These are genuine classical modular bridges, with their stated weight and level restrictions.

For the present problem such a bridge would have to reproduce H, its four physical derivatives, or the complete correlations above; preserve the heat deformation and genuine stationary constraints; and yield a signed estimate with all errors paid at the physical energy scale. The modular Fourier-sign constructions of [33] and the modular inequalities of [34, 35] give a useful model for the last step: retain a specifically identified transformed expression and prove its sign. The polynomial/quasimodular inequalities of [36] and Fourier-eigenfunction estimates of [37] offer further tools when their hypotheses can be verified. None applies solely from the presence of the theta source or the number eight. No required identification or new stationary sign estimate is established by this section.

6.5 Analytic tools and the signed physical observation

The available octonionic function theories provide integral representations, boundary projections, and regularity estimates for the fields above. Their application to collision exclusion must preserve the distinguished first pair and estimate the signed radial observation at its energy scale. Proposition 4.5 supplies a control family against which any claimed consequence of generic monogenicity can be tested.

For differential calculations there is an associative real-matrix realization. Let $L_a v = av$ on $\mathbb{O} \cong \mathbb{R}^8$, and let $\epsilon_1, \ldots, \epsilon_7$ be an orthonormal basis of imaginary octonions. Alternativity gives

$$L_{\epsilon_i} L_{\epsilon_j} + L_{\epsilon_j} L_{\epsilon_i} = -2\delta_{ij} I \qquad (1 \leq i, j \leq 7),$$

so the imaginary left-multiplication operators form a Clifford representation; see [6, Section 2.3]. With the declared basis, $\mathcal{D}_8$ is therefore a real matrix Dirac operator and its conjugate composition is $\Delta_8 I$. This is useful for elliptic estimates and symbolic checks. It is not a multiplicative embedding of the octonion algebra: generally $L_{ab} \neq L_a L_b$. The actual associators and their orientation must still be retained in the physical observations. The component conventions are treated in [7].

The Cauchy formulas of Li–Peng [16] depend on parenthesization. Colombo–Kraußhar–Sabadini [17, Proposition 2.2 and Sections 3–4] distinguish monogenic and slice-monogenic reproducing spaces and construct the relevant Hardy/Bergman framework using real parts of appropriately weighted para-linear inner products. Constales–Kraußhar [18] develop boundary projection methods in the same nonassociative setting. The Teodorescu and projection calculus of [19] includes the necessary associator compensation and can be useful when an approximate field has a quantified monogenic defect. These tools provide norm and evaluation estimates. To yield the present criterion, a boundary or defect estimate must additionally control the adverse radial component relative to E; an absolute error does not give such control near a candidate zero. The present lifts are already exactly monogenic, so an integral reconstruction needs a specific further estimate to justify its use.

The regularity class is another substantive hypothesis. Octonionic slice regularity [20], quaternionic-slice Fueter regularity [21], and the partial-slice theory of [22] have distinct defining equations. The latter's Cauchy–Pompeiu, maximum-modulus and extension results organize

these classes but do not imply that a distinguished two-component trace of a nonzero field is nonvanishing. The integral Fueter mapping in [23] transfers slice data to monogenic functions in the Clifford/quaternionic setting; such a transfer changes the field and requires preservation or reconstruction of the collision observation. Differential transforms can annihilate low-degree collision controls, so the transformed norm cannot be substituted for E without that step.

Refined Kato inequalities [24] concern norm gradients in the ambient variables. Frequency and nodal-set results, including [25], describe the structure and dimension of possible Dirac zeros; they do not exclude an intersection with the physical slice. Moreover, an ordinary double collision has $E = 0$ and $h_2 \neq 0$, so the doubled four-jet field need not vanish there. The explicit ambient log-energy calculation above identifies the stronger property of this particular holomorphic lift, while also showing the loss of orientation in the resulting squared determinants.

The phase/Wronskian description similarly connects the physical signs with familiar complex-analytic positivity questions. A Hermite–Biehler or reproducing-kernel hypothesis must nevertheless be proved for the specific source. Conrey–Li [38] give counterexamples to particular proposed de Branges shift-positivity conditions associated with zeta and Dirichlet L-functions; their result concerns those conditions, not every de Branges approach. Here the operative next step is a complete theta identity or factorization yielding the declared signed energy-relative estimate, with both stationary conditions and all cutoff payments preserved.

7 The bounded region: complete certificates through height 112

The bounded calculation supplies a genuine norm gap, so it excludes collisions directly. Its role is to join the central analytic estimates to the large-height theorem, while leaving the intervening and small-$t \log|x|$ regions explicitly visible.

Theorem 7.1 (Certified closed bounded region)**.** *For the complete theta source,*

$$|F(t,x)| = \sqrt{h_0(t,x)^2 + h_1(t,x)^2} > 10^{-16} \quad (0 \le t \le 1/2,\ |x| \le 104). \tag{7.1}$$

On this rectangle, $h_1 = 0$ implies $\mathcal{L}_0 > 0$, and $h_2 = 0$ implies $\mathcal{L}_1 > 0$. The norm bound combines an analytic central estimate with the validated interval records described below.

7.1 A central estimate with no theta truncation

Proposition 7.2 (Central norm gap)**.** *For $0 \le t \le 1$ and $|x| \le 8$, $|F(t,x)| > 1/120000$. Both stationary signs in Theorem 7.1 hold on this larger time interval.*

Proof. Put $w_t(u) = e^{tu^2}\Phi(u)$ and $M_k(t) = \int_0^\infty u^k w_t(u)\,du$. The complete source obeys $0 < \Phi(u) < 4\pi^2 e^{9u-\pi e^{4u}}$ for $u \ge 0$: after dropping each summand's negative term, the ratio of successive $n^4 e^{-\pi e^{4u} n^2}$ terms is at most $16e^{-3\pi} < 1/2$. The following elementary lower cores and upper tails hold uniformly for $0 \le t \le 1$:

core $[0,d]$	$\Phi(u) > c_d$	tail $u \ge b$	$w_t(u) < A_b e^{-q_b(u-b)}$
$d = 1/10$	$6/25$	$b = 7/25$	$(1/20)e^{-28(u-b)}$
$d = 3/20$	$3/25$	$b = 3/10$	$(1/49)e^{-32(u-b)}$
$d = 1/5$	$1/20$		

(7.2)

For completeness, the lower bounds follow from $\phi_1(u) = e^u g(\pi e^{4u})$ with $g(q) = q(2q-3)e^{-q}$ and $g'(q) = -(2q-1)(q-3)e^{-q}$. On the three cores the respective arguments are less than $33/7, 6, 36/5$.

The bounds $e < 11/4$, $e^{2/7} > 9/7$, and $e^{1/5} < 5/4$ give the three stated rational lower bounds for g. For the tails, $\psi_t(u) = 9u + tu^2 - \pi e^{4u}$ is strictly concave. At $b = 7/25$ one has $\psi_t(b) < -69/10$, $\psi_t'(b) < -28$; at $b = 3/10$ one has $\psi_t(b) < -38/5$, $\psi_t'(b) < -32$. Together with $4\pi^2 < 40$, $e^{69/10} > 800$, and $e^{38/5} > 1960$, these prove the full upper tails in (7.2). All comparisons use rational Taylor bounds for the indicated exponentials; in particular they do not truncate the theta source.

Integrating $u^2(1 - 25u^2/2)$ over the first positive core and paying its entire negative tail with $b = 7/25$ gives

$$M_2 - \frac{25}{2}M_4 > \frac{37}{500000} - \frac{5}{8}\frac{3497259}{33614000000} = \frac{2413193}{268912000000} > \frac{1}{120000}. \tag{7.3}$$

Indeed the polynomial is nonnegative up to b and is bounded below on the tail by $-(25/2)u^2(u^2-b^2)$. Likewise, with $f(u) = u^4(1-32u^2/3)$ and $b = 3/10$, the three successive cores and the entire negative tail give

$$M_4 - \frac{32}{3}M_6 > \frac{111799}{33600000000} - \frac{32}{147}\frac{85837761}{6710886400000} = \frac{2092283587}{3853516800000000} > \frac{1}{2000000}. \tag{7.4}$$

The positive term is $(6/25)\int_0^{1/10} f + (3/25)\int_{1/10}^{3/20} f + (1/20)\int_{3/20}^{1/5} f$; the tail integral is $\int_b^\infty u^4(u^2 - b^2)e^{-32(u-b)}\,du$. Thus both displayed differences are exact rational integrations.

The global inequalities $\sin y \geq y - y^3/6$ for $y \geq 0$ and $\cos y \geq 1 - y^2/2$ now imply, for $0 < x \leq 8$,

$$-h_1 \geq x(M_2 - x^2M_4/6) > x/120000, \qquad h_3 \geq x(M_4 - x^2M_6/6) > x/2000000.$$

For $|x| \leq 1$, integrate the first core using $\cos(xu) \geq 199/200$, discard the nonnegative middle interval $[1/10, 7/25]$, and pay the entire remaining tail. This gives

$$H(t,x) > \frac{597}{25000} - \frac{1}{560} = \frac{7733}{350000} > \frac{1}{50}.$$

The value bound covers $|x| \leq 1$, and the slope bound covers $1 \leq |x| \leq 8$, proving the norm gap by evenness. The first stationary set is exactly $x = 0$, where $\mathcal{L}_0 = H(t,0)M_2 > 0$. For $0 < |x| \leq 8$, $\mathcal{L}_1 = h_2^2 - h_1h_3 > x^2/240000000000$; at $x = 0$ one has $h_2 = -M_2 < 0$, so the second stationary set is absent. □

7.2 What the interval certificates enclose

The records use Arb/Acb interval arithmetic to enclose the complete physical jets on closed space–time cells. Each enclosure includes the returned finite-quadrature radius, every omitted theta index, the complete integration tail, rounding, and spatial and temporal Taylor remainders. The older compensated calculation also pays the complete Gaussian transforms and their truncation remainders. Finite compensated channels are transported using the raw heat equation followed by their explicit time-dependent compensation; they are not assigned a homogeneous heat equation.

For a heat-transported sequence Y_r, a cell with half-widths d_t, d_x uses

$$Y_r(t_c + \tau, x_c + \eta) = \sum_{\ell=0}^{q-1}\sum_{m=0}^{p-1}\frac{(-\tau)^\ell\eta^m}{\ell!m!}Y_{r+2\ell+m}(t_c, x_c) + \mathcal{R}_r.$$

If M_k bounds the complete kth derivative on the cell, Taylor's theorem first in time and then in space gives

$$|\mathcal{R}_r| \leq \sum_{\ell=0}^{q-1}\frac{d_t^\ell}{\ell!}\frac{d_x^p}{p!}M_{r+2\ell+p} + \frac{d_t^q}{q!}M_{r+2q}. \tag{7.5}$$

The higher bands sum the retained theta coefficients before applying the common Taylor monomials. This preserves cancellation between different indices while adding every positive remainder payment.

Stationary decisions concern genuine derivatives. If $h_k = p_k + r_k$ with $|r_k| \le \varepsilon_k$, then at $h_{j+1} = 0$ one has

$$\mathcal{L}_j \ge p_{j+1}^2 - p_j p_{j+2} - (\varepsilon_{j+1}^2 + |p_j|\varepsilon_{j+2} + |p_{j+2}|\varepsilon_j + \varepsilon_j\varepsilon_{j+2}). \tag{7.6}$$

Here $p_{j+1}^2 \le \varepsilon_{j+1}^2$ follows from genuine stationarity; the square payment has a negative sign. The products retain all cross terms between retained channels. A cell is discharged either by exclusion of the stationary derivative or by a strictly positive paid signed lower bound. Monotone endpoint exclusions, where used, employ the next physical derivative and uniform endpoint signs. No zero of an ordinary derivative of an exponentially rescaled function is substituted.

Independently, if I_0, I_1 enclose the genuine h_0, h_1 throughout a cell, then

$$E \ge \operatorname{dist}(0, I_0)^2 + \operatorname{dist}(0, I_1)^2. \tag{7.7}$$

Exact rational comparisons and downward square-root rounding turn these energy floors into the reported norm floors. No division by an unproved norm is used.

7.3 Coverage, the centered-residual pilot, and reproducibility

Every numerical band covers the full closed time interval $[0, 1/2]$ using eight slabs $[r/16, (r+1)/16]$. The records give the following conservative norm bounds; the central row is analytic.

Positive-height band	Cells	Precision	Proved lower bound for $\lvert F\rvert$
$[0, 8]$	—	exact estimates	$1/120000$
$[7, 41]$	272	256 bits	$9/5000000$
$[40, 100]$	510	256 bits	10^{-16}
$[100, 104]$	32	192 bits	$4 \cdot 10^{-16}$

The 510 cells comprise 160, 170, and 180 cells in $[40, 60]$, $[60, 80]$, and $[80, 100]$. Their exact norm floor is

$$\frac{22271745300542300149807}{85070591730234615865843651857942052864} > 10^{-16}.$$

Exact rational cover checks include the spatial subdivisions, all shared boundaries, and both time endpoints. The spatial overlaps $[7, 8]$, $[40, 41]$, and the shared boundary $x = 100$, together with evenness, prove (7.1).

The last band also certifies the centered-residual mechanism. On each of its 32 cells, freeze

$$a = \frac{\pi}{16}, \qquad b = \frac{1}{4}\log\frac{x_c + 2t_c a}{4\pi\sin(4a)}, \qquad \mu = a^2 + b^2, \qquad C_j = -h_{j+2} - 2a h_{j+1} - \mu h_j.$$

The complete even source is shifted before theta indices are separated. With eight retained indices, integration cutoff $3/2$, spatial order 12 and time order 8, the computation encloses all center jets through order 28 and complete derivative majorants through order 29. Centered coefficients are combined before Taylor transport, with the parameters frozen on the cell. The cell-constant normalization $e^{\pi x_c/8}$ rescales each physical jet equally. For normalized enclosures P_0, P_1, Q_0, Q_1 of h_0, h_1, C_0, C_1, the actual decisions are

$$E_{\mathrm{low}} = \operatorname{dist}(0, P_0)^2 + \operatorname{dist}(0, P_1)^2 > 0, \qquad E_{\mathrm{low}}/9 - \sup|Q_0|^2 - \sup|Q_1|^2 > 0.$$

Thus $C_0^2 + C_1^2 \le E/9$ and $\mu \ge 2/5$ throughout every cell. The exact stationary identity $\mathcal{L}_j = \mu h_j^2 + h_j C_j$ at $h_{j+1} = 0$ yields

$$\mathcal{L}_0 \ge E/15 \quad (h_1 = 0), \qquad \mathcal{L}_1 \ge -5E/72 \quad (h_2 = 0).$$

The second strict stationary sign in Theorem 7.1 is a separate validated signed decision; it is stronger than the bounded-loss conclusion supplied by this cone. At each level the band has 24 derivative-exclusion cells and eight strict signed cells, with no claim that a signed cell alone proves existence or uniqueness of a stationary branch.

Appendix C identifies the retained drivers, records, exact auditors, and earlier independent internal replays. In particular, the higher-height auditor checks 6682 assertions and the centered-band auditor checks 547. Those checks concern arithmetic, decisions, complete coverage, and source bindings. The complete-source contour bounds used for the latest band are proved in Appendix A. The computations establish the stated closed bounded region; they do not extend it to arbitrary heights.

7.4 Extension through height 112 and all earlier-time comparisons

Theorem 7.3 (Extended closed bounded certificate)**.** *For the complete theta source,*

$$\sqrt{h_0(t,x)^2 + h_1(t,x)^2} > 10^{-17} \qquad (0 \le t \le 1/2,\ |x| \le 112).$$

Both stationary implications $h_{j+1} = 0 \Rightarrow \mathcal{L}_j > 0$, $j = 0, 1$, hold on this rectangle. With the fixed centering $a = \pi/16$, $b = 1/4$, $\mu = a^2 + b^2$, the following stronger delayed statements hold on the positive-height extension $104 \le x \le 112$: at every current stationary point and every $0 \le s \le t$,

$$h_j(t,x)C_j(s,x) > 0, \qquad h_j(t,x)\{C_j(t,x) - C_j(s,x)/2\} > 0. \tag{7.8}$$

There is exactly one $h_1 = 0$ point in $(107, 108)$ and exactly one $h_2 = 0$ point in $(109, 110)$ at each time in $[0, 1/2]$; there are no other stationary points of these respective observations in $[104, 112]$.

Proof from the complete-source interval certificates. The two band records in the numerical archive use the complete source and error bounds already specified above:

`numerics/FIXED_CUSP_BAND_104_108_20261014.json`
`numerics/FIXED_CUSP_BAND_108_112_20261014.json`.

Each has four unit spatial cells and eight closed time slabs $[k/16, (k+1)/16]$, $0 \le k \le 7$. Their fixed numerical parameters are 192-bit arithmetic in python-flint 0.9.0, $a = \pi/16$, retained index $N = 8$, integration endpoint $U = 3/2$, spatial order 12, and time order 8. Center physical jets through order 28 and complete moment payments through order 29 include quadrature, full-index, integration, rounding, spatial, and temporal errors.

The exact audit proves the whole-cell physical norm bound and the normalized lower bound $e^{\pi x/4}E > 3000$ on both new bands. The stated physical floor follows from the stronger cell-specific norm enclosures and a rational exponential estimate using $\pi < 22/7$; the displayed coarse normalized bound alone is not used to obtain that floor. The 128 stationary decisions are

Band	Level	Complete closed-cell decisions
$[104, 108]$	$j = 0$	24 exclusions; 8 strict signs in $(107, 108)$
$[104, 108]$	$j = 1$	32 exclusions
$[108, 112]$	$j = 0$	32 exclusions
$[108, 112]$	$j = 1$	24 exclusions; 8 strict signs in $(109, 110)$.

In each candidate spatial cell, paid endpoint signs of h_{j+1} are opposite and the full-cell h_{j+2} enclosure has one strict sign. This proves existence and uniqueness throughout the closed time slab. The other cells exclude stationarity, so the implications cover branch creation as well as regular branches.

The shared boundaries at 104 and 108 and the preceding certificate leave no spatial gap. Physical parity reflects the norm and stationary signs to negative heights.

For the delayed products the auditor reconstructs the residual from the physical jets with $b = 1/4$, rather than the saddle centering used for the initial stationary test. For current slab k, the earlier slabs $p = 0, \dots, k$ cover all $0 \le s \le t$, conservatively including some $s > t$ in slab k. The current and earlier time displacements $d, e \in [-1/32, 1/32]$ are independent; the spatial displacement $\eta \in [-1/2, 1/2]$ is the same in both observations. If their paid Taylor polynomials are $T_k(\eta, d)$ and $T_p(\eta, e)$, form the coefficients of $T_k(\eta, d) - T_p(\eta, e)/2$ before ranging the common η. Explicitly, for coefficients $c_{k,r,m}$, the polynomial is

$$\sum_{m=0}^{11} \left[c_{k,0,m} - \frac{1}{2} c_{p,0,m} + \sum_{r=1}^{7} \left(c_{k,r,m} d^r - \frac{1}{2} c_{p,r,m} e^r \right) \right] \eta^m.$$

Add the full current residual payment plus half the earlier payment; only at the current time intersect with $C_j = -h_{j+2} - \mu h_j$. No earlier stationarity is imposed. Taking the union over all earlier slabs and multiplying the complete integrated observations preserves all source cross terms.

In the common constant normalization $Z_j = e^{\pi x_c/8} h_j$, with the same factor for each residual, the final lower bounds for all earlier times are

Current branch	$Z_j \widehat{C}_j(s, x)$	$Z_j(\widehat{C}_j(t, x) - \widehat{C}_j(s, x)/2)$
$h_1 = 0$, $x_c = 107.5$	> 15511	> 7559
$h_2 = 0$, $x_c = 109.5$	> 6910	> 3371.

These enclosures prove (7.8). They use complete earlier fields, so they do not apply a multiplier bound restricted to $\delta \le 1/4$ when $t - s$ is larger. □

The earlier uniform residual-cone enclosure remains undecided on 24 cells in the first band and 16 in the second, all outside the candidate stationary cells. Those original verdicts are retained. The successful norm and stationary decisions do not turn the failed cone enclosures into cone proofs or negative physical signs. Likewise the delayed products in (7.8) are asserted for the positive-height protocol actually computed; fixed drift in C_j prevents an automatic unchanged-drift reflection of that numerical claim.

The generating driver is `numerics/certify_fixed_cusp_band_extension.py`, run with `-x-left 104` and `-x-left 108`, each with an `-output` path. The driver `numerics/audit_fixed_cusp_band_extension.py` takes both records through repeated `-record` arguments and an `-output` path. Its retained `numerics/FIXED_CUSP_BAND_EXTENSION_AUDIT_20261014.json` contains 1321 passing checks, all 128 stationary observations, and all 16 genuine-branch comparisons. Fresh 192-bit replays reproduced both generator records after omitting only `elapsed_seconds`; the exact audit was reproduced in full. Source bindings and these historical replay claims are documented in `numerics/PHASE_CONTROLLED_CUSP_CONTINUATION_CHECKS_20261014.json` and Review 19. They are internal validations, not formal verification of the analytic source lemmas or the interval library.

8 One-sided stationary estimates and their obstructions

The outer theorem and the finite cover give complementary information. The unresolved positive-time domain is contained in the transition strip $112 < |x| < X_*$ and the high-height inner region

$$0 < t \le \tfrac{1}{2}, \qquad |x| \ge X_*, \qquad t \log(|x|/(4\pi)) < \frac{61}{24}. \tag{8.1}$$

The closed-boundary assertion is asymptotic: it does not supply a usable numerical value of X_*. The separate explicit threshold-five bound is not a numerical implementation of this sharper theorem. Even the conditional completion down to $5/2$ leaves a smaller unbounded cusp. The remaining analytic problem is a signed estimate in this domain, with coefficients locally bounded through a candidate positive-time collision. The following obstructions delimit that problem.

8.1 Absolute dominance and pointwise positivity are too strong

At the exact common-carrier coordinate v, put $M = \lfloor e^{2v} \rfloor$. For $\lceil M/2 \rceil \le n \le M$,

$$td_n^2 - 2td_n v = t[(d_n - v)^2 - v^2] \ge -tv^2.$$

Consequently

$$\sum_{n \le e^{2v}} |A_n| \ge (M/2 - 1) M^{-1/2} e^{-tv^2}. \tag{8.2}$$

With fixed a, $v = b_1$, and $t = \vartheta / \log x$ for fixed $0 \le \vartheta < 4$, this grows as a positive constant times $x^{1/4 - \vartheta/16}$, whereas $|A_1| = 1$. Absolute first-channel dominance therefore fails in this scaling. Cancellation within the sum and after integration remains available; indeed it is essential to the improved outer theorem.

Pointwise positivity is also unavailable. For fixed $v = b$, write

$$D_j = \sum_{n \le e^{2b}} A_n(t, x; b)[-a + i(b - d_n)]^j, \qquad p_j = \Re(U D_j),$$

where U is the unit phase of $e^{t(b+ia)^2} \phi_1(b + ia) e^{ixb}$. Put

$$\mathfrak{H}_j = |D_{j+1}|^2 - \Re(D_j \overline{D_{j+2}}), \qquad \mathfrak{W}_j = D_{j+1}^2 - D_j D_{j+2}.$$

Then

$$2(p_{j+1}^2 - p_j p_{j+2}) = \mathfrak{H}_j + \Re(U^2 \mathfrak{W}_j). \tag{8.3}$$

The stronger phase-independent condition $\mathfrak{H}_j \ge |\mathfrak{W}_j|$ fails for the actual common-carrier sums. With $a = \pi/16$ and $b = b_1$ from (5.13), retained Arb/Acb enclosures give

x	t	$\mathfrak{H}_0 - \lvert\mathfrak{W}_0\rvert$ (approximate value)
211	$1/\log 211$	-0.0277198
4898.7	$2/\log 4898.7$	-1.26105

At the second point the actual phase also gives $p_1^2 - p_0 p_2 \approx -0.01099197 < 0$. The stored enclosures are strictly negative; Appendix C identifies their replay. These are pointwise integrand forms, not the integrated Laguerre forms, and no physical stationarity is asserted at these points. The paired identities in Section 5 retain correlations between distinct coordinates as well as distinct indices. A proof requiring every integrand to be positive would discard that possibility. The staircase boundary identity (5.19) must likewise be retained: entering terms have an exact cancellation of their arithmetic x-phase, not a freely discardable endpoint error.

8.2 A sufficient split that preserves favorable curvature

Throughout this section let $0 < a < \pi/8$, prescribe a real b, and put $\mu = a^2 + b^2$. At each point the centered observations are

$$C_k = -h_{k+2} - 2ah_{k+1} - \mu h_k, \qquad k \geq 0. \tag{8.4}$$

Whenever these observations are differentiated or compared at different times below, a, b are held fixed. The exact identity is

$$\mathcal{L}_j = \mu h_j^2 + h_j C_j + h_{j+1}(h_{j+1} + 2ah_j).$$

Thus the remaining target can be stated as

$$\boxed{h_j C_j \geq -\mu h_j^2 - K_j E \quad \text{on } h_{j+1} = 0, \qquad j = 0, 1,} \tag{8.5}$$

where $K_j \geq 0$ is prescribed without dividing by an unknown energy and is locally bounded through every candidate positive-time collision. This is precisely the stationary loss condition in Section 3.

Lemma 8.1 (Favorable curvature and a residual square)**.** *Prescribe an exact split $C_j = G_j + R_j$ and parameters $0 < \kappa \leq \mu$, $\rho \geq 0$. Define*

$$\Delta_j = 4\kappa h_j G_j + \rho^2 E - R_j^2. \tag{8.6}$$

Without imposing stationarity,

$$\begin{aligned} 4\kappa\mathcal{L}_j + \rho^2 E = (R_j + 2\kappa h_j)^2 + \Delta_j + 4\kappa(\mu - \kappa)h_j^2 \\ + 4\kappa h_{j+1}(h_{j+1} + 2ah_j). \end{aligned} \tag{8.7}$$

In particular, $\Delta_j \geq 0$ on $h_{j+1} = 0$ implies $\mathcal{L}_j \geq -\rho^2 E/(4\kappa)$. This loss is locally bounded if ρ is locally bounded and κ is locally bounded below by a positive constant.

Proof. Expand the square and use $G_j + R_j = C_j$. All terms are polynomial in the physical jets, so neither $E > 0$ nor division by a jet is required. On stationarity the last term vanishes and the other displayed squares and $4\kappa(\mu - \kappa)h_j^2$ are nonnegative. □

The split must expose an independently estimable source contribution. For the exact domain and fields of (5.20), choose a prescribed real multiplier $m_j(\alpha)$ with at most polynomial growth in y_α, and set

$$\begin{aligned} G_j &= e^{-ax}\Re \int_\Omega m_j(\alpha) W_\alpha q_\alpha^j \, d\nu(\alpha), \\ R_j &= e^{-ax}\Re \int_\Omega r_j(\alpha) W_\alpha q_\alpha^j \, d\nu(\alpha), \qquad r_j(\alpha) = y_\alpha^2 - b^2 - m_j(\alpha). \end{aligned} \tag{8.8}$$

A proper source filter takes $m_j = (y_\alpha^2 - b^2)\mathbf{1}_A$, with the retained set A intersected with the original staircase. The filter and its parameters are fixed before evaluating the physical observations. For real m_j, r_j, the polarization (5.22) gives the exact scalar obligation

$$\begin{aligned} \Delta_j = \frac{e^{-2ax}}{2}\Re \iint_{\Omega^2} \big[\{2\kappa(m_j(\alpha) + m_j(\beta)) - r_j(\alpha) r_j(\beta)\} J_j(\alpha, \beta) \\ + \rho^2 (J_0(\alpha, \beta) + J_1(\alpha, \beta))\big] \, d\nu(\alpha) \, d\nu(\beta). \end{aligned} \tag{8.9}$$

Indeed h_jG_j is $e^{-2ax}/4$ times the real part of the integral of $(m_j(\alpha)+m_j(\beta))J_j$, whereas R_j^2 has coefficient $e^{-2ax}/2$ and integrand $r_j(\alpha)r_j(\beta)J_j$. The complete source majorants justify these identities and their tail payments; they do not prove the sign in (8.9). The formula requires real multipliers. A complex filter requires its own conjugated polarization or an independent physical-field identity.

Even a quotient-free split can be vacuous: choosing $m_j = y_\alpha^2 - b^2 + 2\kappa$ makes $R_j = -2\kappa h_j$ and cancels the first square in (8.7). With $\kappa = \mu$, the resulting slack condition only restates the target and supplies no theta estimate.

8.3 Collision controls and the limitation of the square criterion

For the backward-heat polynomial $P(t,y) = y^2 - d^2 - 2t$, put $D = d^2 + 2t > 0$. Its Laguerre forms are $2y^2 + 2D > 0$ and 4. At $y = 0$, $h_0 = -D$, $h_1 = 0$, $E = D^2$, $C_0 = -2 + \mu D$, and $C_1 = -4a$. The whole-residual cone would require

$$\eta \geq \frac{\sqrt{(2-\mu D)^2 + 16a^2}}{D}. \tag{8.10}$$

For bounded a, μ, this grows without bound as the pair closes, although the required negative loss is zero. The prescribed split $G_0 = -2$, $R_0 = \mu D$ instead gives

$$\Delta_0 = 8\kappa D + (\rho^2 - \mu^2)D^2.$$

It accepts $\rho = 0$ whenever $0 < D \leq 8\kappa/\mu^2$, preserving the large favorable curvature. This control is not a theta counterexample; growth as $t \downarrow 0$ is also compatible with local boundedness at each positive time.

The same algebra rejects adverse collision predecessors. For $P = y^2 - 2(t-T)$ at $y = 0$, $t = T - s < T$, the split $G_0 = -2$, $R_0 = -2\mu s$ yields

$$\Delta_0 = -16\kappa s + 4(\rho^2 - \mu^2)s^2, \qquad \mathcal{L}_0 = -4s, \qquad E = 4s^2.$$

For $P = y^3 - 6(t-T)y$ at $y = 0$, $t = T - s < T$, the second stationary set $h_2 = 0$ has $h_1 = 6s$, $h_3 = 6$. Taking $G_1 = -6$, $R_1 = -6\mu s$ gives

$$\Delta_1 = -144\kappa s + 36(\rho^2 - \mu^2)s^2, \qquad \mathcal{L}_1 = -36s, \qquad E = 36s^2.$$

Both slacks are negative for sufficiently small $s > 0$ when ρ is bounded and κ stays positively bounded below. The second stationary set cannot be omitted.

There is also a precise restriction at the candidate collision itself: if $h_0 = h_1 = 0$, then $\Delta_0 = -R_0^2$. The square criterion therefore forces $R_0 = 0$ exactly; if also $h_2 = 0$, it forces $R_1 = 0$. A small absolute bound on an omitted source does not imply this vanishing. Thus Lemma 8.1 is a sufficient criterion with an extra cancellation obligation, not a solution of the remaining theta problem.

Nor can its sign be justified by arbitrary-source positivity. On $h_1 = 0$ the arbitrary-jet form $\mathcal{L}_0 + KE = -h_0h_2 + Kh_0^2$ is indefinite for every finite K; on $h_2 = 0$ the form is $-h_1h_3 + K(h_0^2 + h_1^2)$ and is likewise indefinite. Terms in the stationary ideal vanish there. Any useful positivity argument must therefore use relations of the actual theta source.

8.4 Direct product absorption

The residual-square condition can be replaced by a one-sided product condition which imposes no residual vanishing at $E = 0$.

Lemma 8.2 (Fixed-fraction absorption)**.** *For a prescribed split $C_j = G_j + R_j$, let $0 \le \lambda \le 1$ and $K_j \ge 0$. On the genuine stationary set $h_{j+1} = 0$, suppose*

$$h_j G_j \ge 0, \qquad h_j(R_j + \lambda G_j) \ge -K_j E. \tag{8.11}$$

Then $\mathcal{L}_j \ge -K_j E$.

Proof. On this set,

$$\mathcal{L}_j = \mu h_j^2 + (1-\lambda) h_j G_j + h_j(R_j + \lambda G_j) \ge -K_j E.$$

□

At a first-pair zero the displayed products vanish automatically, but the estimate must still hold with a locally bounded prescribed loss in a neighborhood of that point. A favorable retained product alone does not suffice. The finite results below use $\lambda = 1/2$ and $K_0 = K_1 = 0$, without fitting a loss to an observed energy floor.

9 Proper source filters and complete theta correlations

We first test the preceding mechanisms on populated stationary branches inside the existing height-104 cover. We then prove why fixed sharp filters cannot support uniformly small separate payments at arbitrary heights, and derive a complete-source alternative. The finite tests do not extend the bounded cover or settle (8.1).

9.1 Two certified stationary branches and a signed comparison

Fix

$$a = \pi/16, \qquad b = 1/4, \qquad \mu = a^2 + b^2, \qquad \lambda = 1/2, \qquad K_0 = K_1 = 0. \tag{9.1}$$

Use the constant normalization $Z_k = e^{\pi x_c/8} h_k$, with $x_c = 101.5$ on $[101, 102]$ and $x_c = 103.5$ on $[103, 104]$. Hats denote the same normalization of centered observations. The complete-source certificate and paid endpoint Taylor transport give, for every $t \in [0, 1/2]$,

$$\begin{aligned} &Z_1(t, 101) > 37, \qquad Z_1(t, 102) < -20, \qquad Z_2 < 0 \quad (101 \le x \le 102), \\ &Z_2(t, 103) < -26, \qquad Z_2(t, 104) > 14, \qquad Z_3 > 0 \quad (103 \le x \le 104). \end{aligned} \tag{9.2}$$

The intermediate value theorem and strict monotonicity therefore prove exactly one $h_1 = 0$ point in $(101, 102)$ and exactly one $h_2 = 0$ point in $(103, 104)$ at each time. The implicit function theorem gives smooth branches. These are stationary points of the physical derivatives, not derivatives of a variable normalization.

A first signed comparison uses the full residual on the first branch. Exact rational extraction from the eight stored closed time slabs gives

$$\begin{aligned} &Z_0 > 0, \qquad \widehat{C}_0 > 0, \qquad Z_0 \widehat{C}_0 > 6000, \qquad \widehat{C}_0 - \mu Z_0 > 7/5, \\ &\mu Z_0^2 + Z_0 \widehat{C}_0 > 9400, \qquad \mu^2 Z_0^2 - \widehat{C}_0^2 - \widehat{C}_1^2 < -78. \end{aligned} \tag{9.3}$$

Thus the signed condition holds with $K_0 = 0$, whereas every prescribed whole-residual cone budget $\eta \le \mu$ fails on this branch. The previously certified saddle-centered $\eta = 1/3$ cone still passes; the comparison concerns the different fixed centering (9.1), not a failure of every norm cone. Taking $G = C$, $R = 0$ here is a full-residual comparison, not a proper source split. The next experiment supplies such a split.

9.2 A proper filter absorbs an adverse complement on 64 cells

Before evaluating partial integrals, prescribe the four filters

$$(N,U)\in\{(1,1),(1,3/2),(2,1),(2,3/2)\}.$$

For $z=u+ia$, define

$$g_k(t,x)=e^{-ax}\Re\sum_{n=1}^{N}\int_0^U (iz)^k e^{tz^2+ixu}\phi_n(z)\,du,$$
$$G_j=-g_{j+2}-2ag_{j+1}-\mu g_j,\qquad R_j=C_j-G_j. \tag{9.4}$$

This is precisely the real source multiplier $(u^2-b^2)\mathbf{1}_{n\le N,\,0\le u\le U}$. Its shared-carrier domain is $n\le N$, $d_n\le v\le d_n+U$, so all entry boundaries are retained. Each of the eight time slabs $[r/16,(r+1)/16]$, $0\le r\le 7$, is tested on both branches: four filters, two branches, and eight slabs give 64 closed-cell tests. Even-index slabs were designated development and odd-index slabs holdout before evaluation; no filter, parameter, or loss was changed between them. Their interiors are disjoint and their common boundary times are covered.

Both values of U give the following conservative bounds on every corresponding slab. Products are normalized by $e^{\pi x_c/4}$.

N	j	$Z_j\widehat{G}_j$	$Z_j\widehat{R}_j$	$Z_j(\widehat{R}_j+\widehat{G}_j/2)$
1	0	>0	<0	$<-1{,}200{,}000$
1	1	>0	<0	$<-250{,}000$
2	0	>7800	$(-212,-120)$	>3800
2	1	>3800	$(-46,-25)$	>1900

Here $j=0$ means the $h_1=0$ branch and $j=1$ the $h_2=0$ branch. The two-index filters therefore pay a genuinely adverse complementary product with the prescribed half of a favorable retained product. Lemma 8.2 applies with the fixed zero loss. For one index the same prescribed absorption test actually fails on nonempty stationary sets: the negative upper bounds are not an interval precision failure. The large one-index terms nearly cancel in the complete residual.

For completeness, the certification encloses retained center derivatives through order 28 with 192-bit Arb quadrature, then uses space order 12 and time order 8 for transport. Fixed endpoints give $\partial_x g_k=g_{k+1}$ and $\partial_t g_k=-g_{k+2}$ exactly. The complete-source moment payments remain valid for the three masks $\mathbf{1}_A$, $1-\mathbf{1}_A$, and $1-\mathbf{1}_A/2$, all of which lie in $[0,1]$. Centered payments are bounded by $P_{j+2}+2aP_{j+1}+\mu P_j$ for each spatial or temporal remainder. The full C arrays already include quadrature error, all omitted theta indices, and the infinite integration tail. The complement is the exact difference $C-G$, and $C-G/2$ is formed before transport to preserve cancellation. On current stationarity its enclosure is intersected with $-h_{j+2}-\mu h_j-G_j/2$; the complement is treated similarly. The product always uses the full physical jet and an integrated observation, retaining all cross terms. No individual theta index is assigned the vanishing vertical contour integral of the complete source. The complete 64-case receipt, including failures, is identified in Appendix C.

9.3 The endpoint terms of a fixed sharp filter

Now fix $t\in(0,1/2]$, $0<a<\pi/8$, $b\ge 0$, $N\ge 1$, and $U>0$, independently of x. Write

$$\Phi_N(z)=\sum_{n=1}^{N}\phi_n(z),\qquad q=-a+iu=iz,\qquad f_{N,j}(u)=(u^2-b^2)q^j e^{tz^2}\Phi_N(z).$$

Then the retained residual in (9.4) is

$$G_j(x) = e^{-ax}\Re \int_0^U f_{N,j}(u)e^{ixu}\,du, \qquad R_j = C_j - G_j. \tag{9.5}$$

These fixed filtered fields still solve backward heat. The following effect comes from source integration by parts, not a heat-equation defect.

Lemma 9.1 (Complete endpoint expansion)**.** *For $x > 0$,*

$$\begin{aligned} e^{ax}G_j(x) &= \frac{B_j(x)}{x} + \mathcal{E}_j(x), \qquad B_j(x) = \Im\{f_{N,j}(U)e^{ixU} - f_{N,j}(0)\}, \\ |\mathcal{E}_j(x)| &\le \frac{M_{N,j}}{x^2}, \qquad M_{N,j} = |f'_{N,j}(U)| + |f'_{N,j}(0)| + \int_0^U |f''_{N,j}(u)|\,du. \end{aligned} \tag{9.6}$$

In particular the lower coefficient is

$$-\Im f_{N,j}(0) = b^2(-a)^j e^{-ta^2}\Im\Phi_N(ia). \tag{9.7}$$

Proof. Twice integrating $\int_0^U fe^{ixu}\,du$ by parts gives

$$\frac{[fe^{ixu}]_0^U}{ix} - \frac{[f'e^{ixu}]_0^U}{(ix)^2} + \frac{1}{(ix)^2}\int_0^U f''e^{ixu}\,du.$$

Taking real parts uses $\Re(-iZ) = \Im Z$ and gives the stated bound. Substitution at $u = 0$ proves (9.7). □

For the complete theta source, evenness and Schwarz symmetry make $\Phi(ia)$ real. More generally its centered integrand obeys $f_j(-u) = \overline{f_j(u)}$. Its even derivatives at zero are real and its odd derivatives purely imaginary, so every lower-endpoint coefficient in the real integration-by-parts expansion vanishes. A finite raw-index sum need not have this symmetry. If instead a common cutoff $v \le V$ is used, retain $U_n = V - d_n > 0$ for every included index; the leading term becomes

$$\Im\left\{\sum_{\substack{n\le N\\ d_n<V}} f_{n,j}(U_n)e^{ixU_n} - \sum_{\substack{n\le N\\ d_n<V}} f_{n,j}(0)\right\}.$$

Here $f_{n,j}$ is defined with the individual term ϕ_n in place of Φ_N. The remainder is the sum of the bounds in (9.6). An index with $d_n = V$ has an empty interval and contributes zero, not a single unmatched endpoint. This is the same entry cancellation as (5.19).

The upper endpoint is provably nonzero for the pilot family, without a genericity assumption. More generally suppose $a = \pi/16$, $U \ge 1$, and $U \ne b$. Put $D_4 = e^{4U} > 50$ and $\beta = \pi\cos(4a)D_4 > 100$. For $n \ge 2$,

$$\left|\frac{\phi_n(U+ia)}{\phi_1(U+ia)}\right| \le n^2\frac{2\pi n^2 D_4 + 3}{2\pi D_4 - 3}e^{-\beta(n^2-1)} \le 2n^4 e^{-100(n^2-1)}.$$

Here $\phi_1(U+ia) \ne 0$ since $|2\pi e^{4(U+ia)} - 3| \ge 2\pi D_4 - 3 > 0$. Using $n^4 \le e^{n^2-1}$ and $n^2 - 1 \ge 3 + 5(n-2)$ gives

$$\sum_{n=2}^{\infty} 2n^4 e^{-100(n^2-1)} \le \frac{2e^{-297}}{1 - e^{-495}} < \tfrac{1}{2}. \tag{9.8}$$

Thus $\Phi_N(U + ia) \ne 0$ for every finite N, and $f_{N,0}(U) \ne 0$. This includes $U = 1, 3/2$, $b = 1/4$.

Theorem 9.2 (Adverse sharp-filter products on genuine stationary sets)**.** *Fix the parameters of* (9.5) *and assume* $f_{N,0}(U) \neq 0$*. For each* $j = 0, 1$ *there is a sequence* $x_k \to \infty$ *of genuine stationary points* $h_{j+1}(t, x_k) = 0$ *such that* $h_j G_j > 0$*,* $h_j R_j < 0$*, and*

$$\frac{-h_j R_j}{E}(t, x_k) = \exp\{(\pi/8 - a)x_k + o(x_k)\}. \tag{9.9}$$

For every fixed $0 \leq \lambda < 1$*, along the same sequence eventually*

$$h_j(R_j + \lambda G_j) < 0, \qquad \frac{-h_j(R_j + \lambda G_j)}{E} = \exp\{(\pi/8 - a)x_k + o(x_k)\}. \tag{9.10}$$

There are also stationary sequences with $h_j R_j > 0$ *and the same exponential rate for its absolute ratio. All these points eventually belong to the proved outer region, where* $E > 0$ *and both complete Laguerre forms are positive.*

Proof. At fixed $t > 0$, the outer four-jet estimates apply for all sufficiently large x. With $A(x) = 2|Q_t(x)|$, $\omega = \log(x/(4\pi))/4$, and a continuous phase $\varphi = \arg Q_t$, their fixed-time consequences are

$$\begin{aligned} h_k &= A\omega^k\{\cos(\varphi + k\pi/2) + O(1/\log x)\}, \qquad 0 \leq k \leq 3, \\ \varphi'(x) &= \omega + O(1) > 0, \qquad \log A(x) = -\pi x/8 + O_t((\log x)^2). \end{aligned} \tag{9.11}$$

For $j = 0$, take phase windows centered at $\varphi = k\pi$ with half-width $\pi/6$. For all sufficiently large windows, h_1 has opposite endpoint signs and h_2 has a fixed nonzero sign throughout. There is exactly one $h_1 = 0$ point in each window, and the sign of h_0 alternates. There $|h_0| \asymp A$ and $E = h_0^2 \asymp A^2$.

For $j = 1$, center the windows at $\varphi = \pi/2 + k\pi$. Now h_2 has opposite endpoint signs and h_3 has a fixed nonzero sign. There is exactly one $h_2 = 0$ point in each window and h_1 alternates sign. At these points $|h_1| \asymp A\omega$, while $h_2 = 0$ in (9.11) gives $h_0 = O(A/\log x)$; therefore $E \asymp A^2\omega^2$.

Since $q(U) \neq 0$, the hypothesis makes each B_j a nonconstant sinusoid plus a constant. It has infinitely many fixed-width intervals on which its sign is fixed and $|B_j| \geq c_j > 0$. The spacing of the stationary points just constructed is $O(1/\log x)$. Each such interval at sufficiently large height therefore contains stationary points of both signs of h_j. By (9.6), G_j has the sign of B_j there and $|G_j| \asymp e^{-ax}/x$. In contrast, C_j is a fixed linear combination of complete physical jets, so (9.11) makes it exponentially smaller than G_j. Hence $R_j = -G_j(1 + o(1))$ on these intervals.

Choose the stationary points with $h_j G_j > 0$ or with $h_j G_j < 0$. The energy estimates above give (9.9) and its opposite-sign version: all powers of $x, \log x$ and the $O_t((\log x)^2)$ amplitude correction contribute $o(x)$ to the logarithm. Finally $h_j(R_j + \lambda G_j) = h_j C_j - (1 - \lambda)h_j G_j$, and the same comparison proves (9.10). □

Thus no constant, polynomial, or other budget $K_j(x) = \exp(o(x))$ can make the second inequality in (8.11) hold globally for this fixed sharp filter and a fixed $\lambda < 1$. This conclusion does not exclude losses locally bounded at every finite positive-time point: exponential growth in x is still locally bounded. It is a fixed-positive-time obstruction in an already favorable region, not an obstruction to an adaptive filter in the unresolved moving cusp. Neither $\lambda = 1$ nor a fraction tending exponentially fast to one is covered. In the complete product $h_j C_j$, the artificial endpoint terms cancel exactly.

9.4 A compact filter has no finite rational jet closure

Proposition 9.3. *For any fixed* t, a, b, N, U *as in* (9.5)*, no finite family of real rational functions* $r_0, \ldots, r_m$ *can satisfy*

$$R_j(x) = \sum_{k=0}^{m} r_k(x) h_k(t, x) \tag{9.12}$$

on a nonempty open real interval free of their poles, for $j = 0$ or 1.

Proof. Since C_j is a constant-coefficient combination of physical jets, clearing denominators would give

$$p(x)G_j(x) = \sum_{k=0}^{m} p_k(x)h_k(t,x), \qquad p \neq 0,$$

with polynomials. Both sides extend to entire functions, so this identity holds for all real x. Complete evenness and Schwarz symmetry give

$$e^{ax}h_k(t,x) = \frac{1}{2}\int_{\mathbb{R}} e^{t(u+ia)^2}\Phi(u+ia)[i(u+ia)]^k e^{ixu}\,du. \tag{9.13}$$

The Fourier density is real analytic on all of $\mathbb{R}$, and it and all its derivatives decay rapidly: holomorphy follows from the open theta strip, positive-end decay from the theta series, and negative-end decay from the complete evenness. Polynomial multiplication in x corresponds to finite derivatives in u, preserving real analyticity.

By contrast, $e^{ax}G_j$ is the Fourier transform of the compactly supported L^1 density equal to $f_{N,j}(u)/2$ on $(0,U)$, $\overline{f_{N,j}(-u)}/2$ on $(-U,0)$, and zero elsewhere. Multiplication by $p(x)$ gives a compactly supported distribution. Fourier injectivity for tempered distributions equates this compact distribution to a real analytic function. That function vanishes on (U,∞), hence everywhere. Therefore $pG_j = 0$ and $G_j = 0$.

This is impossible. The finite sum Φ_N is a nonzero holomorphic function, since every ϕ_n is positive on the positive real axis. The other factors in $f_{N,j}$ have only isolated zeros, so its compact Hermitian extension is not zero. Fourier injectivity implies G_j is not identically zero. The injectivity used here follows, for example, by Gaussian regularization and the approximate-identity limit; it requires no information about zeros of H. □

The proposition excludes rational-coefficient finite-jet identities for these fixed filters, including constant and polynomial coefficients. It does not exclude general transcendental coefficients, merely locally bounded coefficients, or moving filters. Nor can complete symmetry be restored by naively adding $\Phi_N(-z)$: as real $u \to \infty$,

$$\Phi_N(-u) = -3\pi\Big(\sum_{n=1}^{N} n^2\Big)e^{-5u} + O_N(e^{-9u}),$$

so multiplication by e^{tu^2} destroys integrability at every positive time. Complete theta evenness is an infinite-index identity. A vertical contour correction alone does not restore this raw-index symmetry. Smooth compact cutoffs still require matched endpoint derivatives and a new relative signed estimate.

9.5 Complete-source Gaussian transport and a 14-cell pilot

There is an exact alternative without sharp source endpoints. For fixed $\delta \geq 0$, multiply the complete source on the shifted contour by $e^{-\delta z^2}$, $z = u + ia$. With the same frozen a, b at both times, its retained observation and complement are

$$G_j(t,x) = C_j(t-\delta,x), \qquad R_j(t,x) = C_j(t,x) - C_j(t-\delta,x). \tag{9.14}$$

This follows directly from $e^{tz^2}e^{-\delta z^2} = e^{(t-\delta)z^2}$ and the complete contour identity. The multiplier preserves the symmetry on the imaginary segment and introduces no raw-index or upper cutoff boundary. It is generally complex on the horizontal contour:

$$e^{-\delta(u+ia)^2} = e^{\delta a^2 - \delta u^2} e^{-2i\delta au}.$$

Accordingly (8.9), which assumes real multipliers, is not applied unchanged. Formula (9.14) supplies the physical products directly. Numerical truncation errors still require full payment.

For the parameters (9.1), fix $\delta = 1/16$ before testing. The two current-time branches in (9.2) and the seven closed time slabs $[r/16, (r+1)/16]$, $1 \le r \le 7$, give 14 predeclared cases. All satisfy $h_j G_j > 0$, $h_j R_j > 0$, and $h_j(R_j + G_j/2) > 0$. Conservative normalized margins are

j	$Z_j\widehat{G}_j$	$Z_j\widehat{R}_j$	$Z_j(\widehat{R}_j + \widehat{G}_j/2)$
0	> 8300	> 54	> 4200
1	> 4200	> 7	> 2100

Thus this second finite pilot has favorable retained and complementary products, with the same prescribed zero loss. Stationarity is imposed only at (t, x), never at $(t - \delta, x)$.

The proof is exact rational extraction from the stored full-source certificate and requires no new quadrature. Previous and current slabs have the same spatial center and time displacements. Center coefficient differences are formed before transport. The complete spatial and temporal payments are P_{prior} for G, $P_{\text{current}} + P_{\text{prior}}$ for R, and $P_{\text{current}} + P_{\text{prior}}/2$ for $R + G/2$. All full center arrays already contain the complete index and integration tails. Intersecting with the current stationary identity $C_j = -h_{j+2} - \mu h_j$ preserves that constraint, and products of full integrated observations retain every cross term. The receipt in Appendix C records all 14 cases. It proves the stated finite comparison on $1/16 \le t \le 1/2$, not persistence of the correlation near an unknown collision or suitability of a fixed delay as $t \downarrow 0$.

The distinct real Gaussian multiplier $(u^2 - b^2)e^{-\delta u^2}$ also retains the complete source. Denote its retained residual by $G^{\text{real}}_{\delta,j}$. It satisfies

$$G^{\text{real}}_{\delta,j}(t, x) = e^{a^2\delta} C_j(t - \delta, x + 2a\delta). \tag{9.15}$$

Indeed $tz^2 - \delta u^2 = (t - \delta)z^2 + 2i\delta au - \delta a^2$. After including the physical prefactor e^{-ax}, the corresponding filtered jet is $e^{a^2\delta} h_k(t - \delta, x + 2a\delta)$. Applying $-\partial_x^2 - 2a\partial_x - \mu$ proves (9.15). Both identities converge for all real times; $0 \le \delta \le t$ is imposed when using a theorem restricted to nonnegative time. The real mask obeys the hypotheses of the paired multiplier formula, but no numerical certificate for this additional mask is asserted. These are standard Fourier/heat identities specialized to the complete theta source, not consequences of an unproved positivity property.

Their exact complementary remainders expose the next analytic obligation. Since the frozen observations satisfy $\partial_t C_j = -C_{j+2}$,

$$R_j(t, x) = -\int_0^\delta C_{j+2}(t - s, x)\, ds, \tag{9.16}$$

$$C_j(t, x) - G^{\text{real}}_{\delta,j}(t, x) = -\int_0^\delta e^{a^2 s}[C_{j+2} + 2aC_{j+1} + a^2 C_j](t - s, x + 2as)\, ds. \tag{9.17}$$

The earlier or displaced point need not be stationary.

9.6 Complex delay polarization and moving-parameter identities

The Gaussian comparison uses a complex multiplier on the shifted contour. Its paired formula must therefore conjugate the multiplier in the second Hermitian factor. To state this without identifying source coordinates, use $\Omega_0 = [0,\infty)\times\mathbb{N}$. For $\alpha = (u,n)$ set

$$z_\alpha = u + ia, \quad q_\alpha = iz_\alpha, \quad D_\alpha = u^2 - b^2, \quad W_\alpha = e^{tz_\alpha^2 + ixu}\phi_n(z_\alpha),$$

and put $\mathcal{F}_j(k) = e^{-ax}\Re\int_{\Omega_0} W_\alpha q_\alpha^j k_\alpha \, d\nu(\alpha)$. The complete-source contour identity gives $h_j = \mathcal{F}_j(1)$, $C_j = \mathcal{F}_j(D)$. For $G_j = C_j(t-\delta,x)$, the multipliers for $P_j = h_jG_j$ and $Q_j = h_j(C_j - G_j/2)$ are respectively

$$k_\alpha^P = D_\alpha e^{-\delta z_\alpha^2}, \qquad k_\alpha^Q = D_\alpha(1 - e^{-\delta z_\alpha^2}/2).$$

The complete source is shifted before its arithmetic terms are separated. In particular no individual raw theta term is assumed even.

Lemma 9.4 (Complex polarization and full pair-tail payment)**.** *For the above multipliers, with independent α,β,*

$$h_j\mathcal{F}_j(k) = \frac{e^{-2ax}}{4}\Re\iint_{\Omega_0^2}\big[(k_\alpha + \overline{k_\beta})W_\alpha\overline{W_\beta}(q_\alpha\overline{q_\beta})^j + (k_\alpha + k_\beta)W_\alpha W_\beta(q_\alpha q_\beta)^j\big]d\nu(\alpha)d\nu(\beta). \tag{9.18}$$

For $a = \pi/16$, $b = 1/4$, $0 \le t \le 1/2$ and $0 \le \delta \le \min(t, 1/4)$, put $M = e^{a^2/4}$. If A is a retained set, let $M_{A,k} = \int_A |W||q|^k$ and $T_k = \int_{A^c}|W||q|^k$. The absolute multiplier masses satisfy

$$B_A^P \le M(M_{A,j+2} + b^2M_{A,j}), \qquad B_A^Q \le (1 + M/2)(M_{A,j+2} + b^2M_{A,j}),$$

and the same bounds hold on A^c with $M_{A,k}$ replaced by T_k. Truncating both factors of either product has error at most

$$e^{-2ax}\{M_{A,j}B_{A^c} + T_jB_A + T_jB_{A^c}\}. \tag{9.19}$$

Proof. Use $\Re U\Re V = \frac{1}{2}\Re(U\overline{V} + UV)$ and then average under interchange of the two source coordinates. This gives the two sectors and the conjugated multiplier in (9.18). Absolute convergence follows, with $c_a = \cos(4a)$ and $\rho_0 = 16e^{-3\pi c_a} < 1$, from

$$\sum_n |W_{(u,n)}||q_{(u,n)}|^k \le \frac{2\pi^2 + 3\pi}{1 - \rho_0}(u+a)^k e^{u^2/2 + 9u - \pi c_a e^{4u}},$$

which has every polynomial moment. The inequality $|e^{-\delta(u+ia)^2}| \le M$ and $|u^2 - b^2| \le |q|^2 + b^2$ prove the multiplier bounds. Expanding a product into its retained and omitted factors proves (9.19), including both mixed sectors and the omitted–omitted sector. The complete index and integration bounds of Appendix A supply the T_k on their declared rectangles. Quadrature and parameter-transport errors remain additional. □

The real horizontal Gaussian $e^{-\delta u^2}$ is a different mask; its time-and-position transport identity above must not replace the complex mask in this lemma. At a fixed observation point the latter has the exact time identity

$$G_j = C_j(t-\delta,x), \qquad C_j(t,x) - G_j = -\int_0^\delta C_{j+2}(t-s,x)\,ds.$$

Earlier stationarity is never imposed. If δ depends on t, x, its derivatives also enter transport. With superscript $-$ denoting evaluation at $(t-\delta(t,x),x)$ and fixed a, b,

$$\begin{aligned} G_{j,t} &= -(1-\delta_t)C^-_{j+2}, & G_{j,x} &= C^-_{j+1} + \delta_x C^-_{j+2}, \\ (\partial_t + \partial_x^2)G_j &= (\delta_t + \delta_{xx})C^-_{j+2} + 2\delta_x C^-_{j+3} + \delta_x^2 C^-_{j+4}. \end{aligned} \tag{9.20}$$

These identities follow by the chain rule and $\partial_t C_k = -C_{k+2}$. All needed derivatives are justified by the complete-source moment majorant, not by differentiating a four-jet asymptotic. On a regular current branch $x = \xi(t)$,

$$\xi' = h_{j+3}/h_{j+2}, \qquad \dot{h}_j = -h_{j+2}, \qquad \dot{G}_j = \xi' C^-_{j+1} - (1-\delta_t - \xi'\delta_x)C^-_{j+2}.$$

Thus $\dot{P}_j = -h_{j+2}G_j + h_j\dot{G}_j$ and

$$\dot{Q}_j = -h_{j+2}(C_j - G_j/2) + h_j[-C_{j+2} + \xi' C_{j+1} - \dot{G}_j/2].$$

The denominator-free tangent operator $\mathcal{T}_j = h_{j+2}\partial_t + h_{j+3}\partial_x$ satisfies

$$\mathcal{T}_j h_j = -h_{j+2}^2, \quad \mathcal{T}_j G_j = -[h_{j+2}(1-\delta_t) - h_{j+3}\delta_x]C^-_{j+2} + h_{j+3}C^-_{j+1}.$$

It may vanish at a singular point. None of these transport identities alone proves continuation or positivity through branch creation.

9.7 Why every fixed proportional delay fails its exact retained sign

Theorem 9.5 (Proportional-delay phase obstruction)**.** *Fix $t_0 \in (0, 1/2]$, $0 < \rho < 1$, and finite real centering parameters a, b. Put $\delta = \rho t_0$, $\tau = (1-\rho)t_0$. For each $j = 0, 1$, the genuine current set $h_{j+1}(t_0, x) = 0$ contains sequences $x \to +\infty$ on which $h_j(t_0, x)C_j(\tau, x) < 0$, and other sequences with the opposite strict sign. Nevertheless, for every fixed $\lambda \in [0, 1]$,*

$$h_j(t_0, x)\{C_j(t_0, x) - (1-\lambda)C_j(\tau, x)\} > 0$$

at every sufficiently high current stationary point. Both full Laguerre forms are also positive there.

Proof. Use the analytic carrier Q_v in (2.4) at each fixed positive time $v = t_0, \tau$, and put $A_v = 2|Q_v|$, $\phi_v = \arg Q_v$, $L = \log(x/(4\pi))$, $\omega = L/4$. The four-jet theorem applies at both times for all sufficiently large x, giving

$$h_k(v, x) = A_v\omega^k\{\cos(\phi_v + k\pi/2) + O(L^{-1})\}, \quad 0 \le k \le 3, \qquad \phi'_{t_0} = \omega + O(1) > 0.$$

The exact carrier quotient and $\alpha(s) = L/2 + i\pi/4 + O(x^{-1})$, $s = (1+ix)/2$, give

$$D := \phi_{t_0} - \phi_\tau = \frac{\delta\pi L}{16} + o(1), \qquad r_\delta := \frac{A_\tau}{A_{t_0}} = \exp\{-\delta L^2/16 + \delta\pi^2/64 + o(1)\}. \tag{9.21}$$

Indeed the two zeta arguments have real parts tending to infinity, so their Dirichlet series and their analytic logarithms tend to 1 and 0, respectively. Continuous carrier phases may be chosen consistently.

At current stationarity let $\psi = \phi_{t_0} + j\pi/2$; then $\sin\psi = O(L^{-1})$. The lower-order drift and mass terms in the residual contribute only $O(L^{-1})$ after normalization. Hence

$$\frac{h_j(t_0, x)C_j(\tau, x)}{A_{t_0}A_\tau\omega^{2j+2}} = \cos D + O(L^{-1}), \qquad \frac{h_j(t_0, x)C_j(t_0, x)}{A_{t_0}^2\omega^{2j+2}} = 1 + O(L^{-1}).$$

The normalized absorption expression tends to 1 because $r_\delta \to 0$.

To obtain actual stationary points, take each sufficiently high phase window $|\psi - k\pi| \le \pi/6$. The signs of h_{j+1} at its endpoints are opposite, and h_{j+2} has one strict sign throughout, so the window contains exactly one current stationary point. On the logarithmic bands

$$\left| L - \frac{16(2n+1)}{\delta} \right| \le \frac{8}{3\delta},$$

the phase D is within $\pi/6 + o(1)$ of $(2n+1)\pi$ and $\cos D \le -1/2$ eventually. Their lengths in x grow exponentially, whereas the phase windows have width of order $1/L$, so each sufficiently high band contains complete windows of both levels. Bands centered at $L = 32n/\delta$ give the positive sign. All these points lie in the already proved outer-sign region. □

The obstruction also excludes a height-independent rule $\delta(t)$ whenever $0 < \delta(t_0) < t_0$ at one positive time. It concerns the exact retained-sign requirement, not the bounded-loss program: on these proved nonzero fixed-time branches,

$$P_j/E = r_\delta \omega^2 \{\cos D + O(L^{-1})\} \longrightarrow 0.$$

The sign alternates while its energy-relative magnitude vanishes. No first counterexample height is evaluated, and this fixed-time argument does not settle a moving inner cusp.

For the particular half-time rule $\delta = t/2$, a complete-source pilot in $(101, 102)$ on $h_1 = 0$ and $(103, 104)$ on $h_2 = 0$ certifies $P_j, Q_j > 0$ in all 16 branch/slab cases over $0 \le t \le 1/2$. In the constant branch normalizations its lower bounds for (P_j, Q_j) are $(8108, 3999)$ and $(4100, 2025)$. The complement $h_j(C_j - G_j)$ is undecided on the first two time slabs of each branch, and equals zero at $t = 0$; these four undecided intervals are not negative signs. The exact earlier-time map on slab k is $d^- = d/2 + \epsilon_k$, with $\epsilon_k = -1/64$ for even k and $1/64$ for odd k. Substitution in the common Taylor polynomial, before interval ranging and with all original payments retained, proves the pilot. Its record is `numerics/PROPORTIONAL_GAUSSIAN_SOURCE_FILTER_PILOT_20261014.json`. This bounded success is consistent with the asymptotic obstruction.

9.8 A height-dependent delay with a uniform outer comparison

Keep $a = \pi/16$, $b = 1/4$ fixed, and prescribe

$$\ell(x) = \log(1 + x^2), \quad B = 1 + t\ell(x), \quad \delta(t, x) = \frac{t}{2B}, \quad \tau = t - \delta, \quad G_j = C_j(\tau, x). \tag{9.22}$$

For $t \ge 0$, $0 \le \delta \le t/2$ and $t/2 \le \tau \le t$.

Theorem 9.6 (Phase-controlled outer comparison)**.** *There is a finite unevaluated X such that, uniformly for $|x| \ge X$, $0 < t \le 1/2$, and $\theta = t\log(|x|/(4\pi)) \ge 13/4$, the rule (9.22) satisfies at current stationarity*

$$P_j \ge \frac{1}{8} A_t A_\tau \omega^{2j+2} > 0, \qquad Q_j \ge \frac{1}{2} A_t^2 \omega^{2j+2} > 0, \qquad j = 0, 1,$$

where $A_v = 2|Q_v|$ is the analytic carrier amplitude at time v and $\omega = \frac{1}{4}\log(|x|/(4\pi))$.

Proof. Take $x > 4\pi$ and write $L = \log(x/(4\pi))$, $d = \delta L$. Since $\ell(x) > 2L$,

$$0 < d < \frac{\theta}{2 + 4\theta} < \frac{1}{4}, \qquad \tau L > f(\theta) := \theta - \frac{\theta}{2 + 4\theta}.$$

The increasing function f satisfies $f(13/4) = 91/30 = 3+1/30$. Thus the uniform four-jet asymptotic is available at both times with a fixed margin, even when those times decrease with height. Write $s = (1+ix)/2$, $w_v = s + v\alpha(s)/2$. The full relative phase is

$$D := \phi_t - \phi_\tau = \Im\left\{\frac{\delta}{4}\alpha(s)^2 + \int_\tau^t \frac{\alpha(s)}{2}\frac{\zeta'}{\zeta}(w_v)\,dv\right\}. \tag{9.23}$$

The analytic prime-power logarithm is valid throughout this path: $\Re w_v \geq 5/4$ uniformly once X is large. In that half-plane,

$$\left|\frac{\zeta'}{\zeta}(w)\right| \leq \sum_{n\geq 2}(\log n)n^{-5/4} < 18.$$

For the last inequality bound the $n = 2, 3$ terms by 1 each and the decreasing remainder by $\int_3^\infty (\log u)u^{-5/4}du < 16$. Since $\alpha = L/2 + i\pi/4 + O(x^{-1})$ and $d < 1/4$, the gamma and arithmetic parts of (9.23) are bounded by $\pi/64 + o(1)$ and $18/16 + o(1)$, respectively. Increase X until

$$|D| < 5/4, \qquad \cos D \geq 1 - D^2/2 \geq 7/32.$$

At current stationarity, $\psi = \phi_t + j\pi/2$ has $\sin\psi = O(L^{-1})$, so

$$\frac{P_j}{A_t A_\tau \omega^{2j+2}} = \cos\psi\cos(\psi - D) + O(L^{-1}) = \cos D + O(L^{-1}).$$

This proves the bound $1/8$ after a further uniform increase of X.

For absorption, the current jets give $h_j C_j/(A_t^2\omega^{2j+2}) = 1 + O(L^{-1})$. The delayed/current amplitude ratio is exactly

$$r_\delta = \frac{A_\tau}{A_t} = e^{-\delta\Re(\alpha^2)/4}\left|\frac{\zeta(w_\tau)}{\zeta(w_t)}\right|.$$

The second factor is at most $\zeta(5/4)^2 \leq 25$, by the absolutely convergent series for zeta and its reciprocal. Uniformly for $\theta \geq 13/4$, the fact that $\ell/L \to 2$ implies $d \geq 1/5$ eventually (it suffices that $\ell/L \leq 21/10$). Therefore

$$r_\delta \leq 25e^{-L/80+O(L^{-1})}, \qquad \frac{Q_j}{A_t^2\omega^{2j+2}} = 1 + O(L^{-1}) - \frac{r_\delta}{2}\{\cos D + O(L^{-1})\} \geq \frac{1}{2}$$

eventually. For $x = -y < 0$, the delay is unchanged and $h_k(t,-y) = (-1)^k h_k(t,y)$. The drift in the delayed residual changes sign under reflection but remains in the same lower-order $O(L^{-1})$ payment. The leading margins are consequently unchanged. □

This theorem proves a delayed comparison within the known outer-sign region. It does not improve the full-sign threshold. Using only the historical condition $\tau L > 3$ in the same proof gives, for every fixed $\varepsilon > 0$, the comparison on $\theta \geq (11 + \sqrt{217})/8 + \varepsilon$, with an unevaluated ε-dependent height. No claim is made at the limiting value, nor is that historical threshold asserted optimal for the filter.

For completeness, the exact moving derivatives in (9.20) are supplied by

$$\delta_t = \frac{1}{2B^2}, \quad \delta_x = -\frac{t^2\ell'}{2B^2}, \quad \delta_{xx} = -\frac{t^2\ell''}{2B^2} + \frac{t^3(\ell')^2}{B^3},$$

where $\ell' = 2x/(1+x^2)$ and $\ell'' = 2(1-x^2)/(1+x^2)^2$. The pointwise theorem requires no regular stationary branch; its transport formulas do not themselves handle branch creation. The finite

all-earlier-time result (7.8) includes this rule and the half-time rule on the two new positive-height branches.

The exact checkers `numerics/check_proportional_delay_identities.py` and `numerics/check_phase_controlled_delay_identities.py` verify 20 and 21 finite algebraic statements, respectively. The affine transport audit performs 1200 rational point comparisons. Their receipts and Reviews 18–19 distinguish these checks from the analytic phase proof and the complete-source interval calculations. No check evaluates the asymptotic height or proves an unbounded inner-cusp estimate.

9.9 The remaining cross-time estimate

One optional sufficient target is to prescribe a delay rule $0 < \delta(t,x) \le t$, a fraction $0 \le \lambda < 1$, and losses locally bounded through a candidate collision, and prove on the *current* set $h_{j+1}(t,x) = 0$ that

$$\begin{aligned} h_j(t,x)C_j(t-\delta,x) &\ge 0, \\ h_j(t,x)\{C_j(t,x) - (1-\lambda)C_j(t-\delta,x)\} &\ge -K_j(t,x)E(t,x), \qquad j = 0,1. \end{aligned} \tag{9.24}$$

Lemma 8.2 would then supply the required stationary loss. This is a theta-specific correlation statement: neither backward heat, complete-source symmetry, nor the transport identities prove its sign. The finite comparisons establish it only on their declared cells; Theorem 9.6 gives the separate outer comparison. Any adaptive parameters remain prescribed independently of the unknown energy; differentiating an adaptive filter additionally requires all chain-rule and moving-boundary terms. The paired source formulas and the geometric formulation identify what must be controlled, while (9.24) remains an open all-height analytic estimate. The combined signed observation can be bounded directly without requiring both separate products to have favorable signs; Section 10 develops that less restrictive target.

9.10 Polynomial cutoff losses as a quantitative target

The useful role of logarithmic slack is to permit deterioration with source resolution while retaining a collision-excluding estimate. The following transfer is independent of any unproved theta sign.

Lemma 9.7 (Polynomial cutoff transfer)**.** *On a declared stationary domain, suppose independently established $A, B \ge 0$, $c > 0$, a fixed finite $p \ge 0$, and an integer $N_0 \ge 1$ give, for every integer $N \ge N_0$,*

$$\mathcal{L}_j \ge -AN^pE - Be^{-cN^2}. \tag{9.25}$$

Assume A, B, N_0, c^{-1} are locally bounded through candidate events. For every fixed $M > 0$ there is an independently prescribed, locally bounded C such that

$$\mathcal{L}_j \ge -CE[1 + \log_+(M/E)]^{p/2}, \tag{9.26}$$

with value zero on the right at $E = 0$.

Proof. For $E > 0$ set $\ell_E = 1 + \log_+(M/E)$, $d_B = \log_+(B/M)$ if $B > 0$ and $d_B = 0$ otherwise. Take

$$N = \left\lceil \max\left\{ N_0, \sqrt{c^{-1}\log_+(B/E)} \right\} \right\rceil,$$

where the logarithm is defined as zero for $B = 0$. Then $Be^{-cN^2} \le E$ and $\log_+(B/E) \le d_B + \log_+(M/E) \le (1 + d_B)\ell_E$. Consequently $N \le D\sqrt{\ell_E}$ for

$$D = N_0 + 1 + c^{-1/2}\sqrt{1 + d_B}, \qquad C = AD^p + 1.$$

This proves (9.26) and its local uniformity. At $E = 0$, let $N \to \infty$ in the hypothesis. □

The same conclusion follows if a retained observation $\mathcal{L}_j^{(N)}$ obeys the corresponding inequality and its difference from $\mathcal{L}_j$ has a separately paid Gaussian tail. This is an all-N family with fixed coefficient rules: proving one cutoff or choosing coefficients from an unknown energy floor does not suffice. The final optimization may depend on E because all members of the family have already been proved with the same prescribed constants.

For the common-carrier source, use the exact staircase cutoff

$$\Omega_N = \{(v, n) : 0 \le v \le \tfrac{1}{2}\log N, \quad n \le e^{2v}\}.$$

The complete-source majorant in (5.12) implies Gaussian omitted moments $O(e^{-cN^2})$ at bounded time, after reducing $c > 0$ to absorb polynomial and logarithmic factors. The paired error (5.25) pays both mixed sectors and the omitted–omitted sector. Thus a sufficient new arithmetic target on an unresolved domain is

$$\mathcal{L}_j^{(N)} \ge -A_j(t, x)N^pE - B_j(t, x)e^{-cN^2} \quad \text{on the full set } h_{j+1} = 0, \quad N \ge N_0, \quad j = 0, 1. \tag{9.27}$$

All retained jet cross terms must remain, and both E and stationarity refer to the full physical jets. The coefficients must be derived independently of E and locally bounded through candidate events. A formulation with retained energy and a paid stationary defect is also possible: the tail implies $E_N \le 2E + O(e^{-2cN^2})$ and $|h_{j+1}^{(N)}| = O(e^{-cN^2})$ at full stationarity. None of these tail estimates proves the signed target.

9.11 A cumulative signed moment and its stronger sufficient bound

An alternative exposes the signed correlation directly on the original real source. For $j = 0, 1$, put

$$a_j(u) = e^{tu^2}\Phi(u)u^j\cos(xu + j\pi/2), \quad W = h_j = \int_0^\infty a_j(u)\,du, \quad F_j(r) = \int_0^r a_j(u)\,du.$$

At full stationarity, $\mathcal{L}_j = W\int_0^\infty u^2 a_j(u)\,du$. An integration by parts, with the full tail retained, gives

$$\mathcal{L}_j = \int_0^R 2r\{W^2 - WF_j(r)\}\,dr + W\int_R^\infty (u^2 - R^2)a_j(u)\,du. \tag{9.28}$$

Indeed integrating $u^2\,dF_j(u)$ on $[0, R]$ gives $R^2F_j(R) - \int_0^R 2rF_j(r)dr$, and $W - F_j(R) = \int_R^\infty a_j$ supplies the displayed tail.

Suppose, as an additional unproved source estimate, that

$$WF_j(r) \le W^2 + A_jE \qquad (r \ge 0) \tag{9.29}$$

with a prescribed locally bounded $A_j \ge 0$. For $0 \le t \le 1/2$, $0 \le k \le 3$, and $R \ge 1$, the complete real-source envelope yields

$$\int_R^\infty u^k e^{tu^2}\Phi(u)\,du \le 2\pi e^{-4R}e^{-(\pi/2)e^{4R}} \le 2\pi e^{-(\pi/2)e^{4R}}. \tag{9.30}$$

To verify this, use $\Phi(u) < 4\pi^2 e^{9u-\pi e^{4u}}$ and, for $u \ge 1$, $k\log u + tu^2 + 9u \le u^2/2 + 12u \le (\pi/2)e^{4u}$. Integrating the remaining exponential using its derivative gives the first bound. Since $|W| \le \sqrt{E}$ for both levels, (9.28) implies

$$\mathcal{L}_j \ge -A_j R^2 E - 2\pi\sqrt{E}\, e^{-(\pi/2)e^{4R}}.$$

At $E = 0$ the stationary identity gives $\mathcal{L}_j = 0$ directly. For $E > 0$, set $\ell_E = 1 + \log_+(M/E)$, $R = 1 + \frac{1}{4}\log \ell_E$, and $c_0 = (\pi/2)e^4$. Then $R^2 \le 25\ell_E/16$ and $\sqrt{E}e^{-c_0\ell_E} \le e^{-c_0}E/\sqrt{M}$, both for $E \le M$ and for $E \ge M$. The conditional consequence is

$$\mathcal{L}_j \ge -\left(\frac{25}{16}A_j + \frac{2\pi e^{-c_0}}{\sqrt{M}}\right) E[1 + \log_+(M/E)].$$

Keeping $R^2 = (1 + \frac{1}{4}\log \ell_E)^2$ gives the sharper squared iterated-log factor. This is a deduction from (9.29), not a proof of that target. An estimate on the integrated expression in (9.28) would demand less than pointwise control for every r.

Positive source weights alone cannot establish the needed alignment. For the positive two-frequency heat family $H_*(t,x) = e^t \cos x + e^{-3T}e^{4t}\cos(2x)$, there is a double collision at (T,π). At $t = T - s$, $x = \pi$, $W = -3e^T s + O(s^2)$, $E = 9e^{2T}s^2 + O(s^3)$, and $\mathcal{L}_0 = -9e^{2T}s + O(s^2)$. For a source cutoff $1 < r < 2$, $F_0(r) = -e^{T-s}$ and $WF_0(r) = 3e^{2T}s + O(s^2)$, contradicting (9.29) for bounded A_0. This atomic-source control is outside the theta class. It shows why an arithmetic relation among the actual theta contributions is needed; evenness, rapid decay, or an absolute source-tail payment does not supply an energy-relative signed estimate.

10 Conditional transfers and the remaining research program

The preceding outer theorem does not cover the entire positive-time collision domain. This section records three further deductions that clarify possible next steps. Their transfer implications are proved; the arithmetic hypotheses proposed below are not established here. In particular, this section adds no unconditional sign region. Write $S_j = \{h_{j+1} = 0\}$ for $j = 0, 1$, and retain $E = h_0^2 + h_1^2$. All stationary conditions below refer to the full physical jets.

10.1 A weighted remainder sufficient for the closed strip

Retain the branches $s_\pm, w_\pm, Q_\pm$ of (2.2)–(2.4) and the analytic model $\mathcal{P}_t = Q_+ + Q_-$. For a real center x, put

$$L = \log(x/(4\pi)), \qquad D_x = \{z : |z - x| \le L^{-1}\}, \qquad \Gamma(t,x) = |M_t(s_+(x))|.$$

For $z = u + iv \in D_x$, define

$$\ell = \log(u/(4\pi)), \qquad Y = e^{19\ell/100}, \qquad N_{\mathrm{RS}} = \left\lfloor \sqrt{u/(4\pi) + t/16} \right\rfloor.$$

The notation N_{RS} distinguishes this cutoff from the staircase source cutoff used below.

Proposition 10.1 (Conditional weighted-remainder transfer)**.** *Suppose that, as $x \to \infty$, for both signs one has*

$$\sum_{Y < n \le N_{\mathrm{RS}}} e^{t\log^2 n/4} n^{-w_\pm(z)} = o(1) \tag{10.1}$$

uniformly for real centers satisfying

$$0 < t \le \tfrac{1}{2}, \qquad \tfrac{5}{2} \le tL \le \tfrac{8}{3}, \qquad z \in D_x.$$

Then, with $Q = Q_+(x) \neq 0$, $\omega = L/4$, *and* $\phi = \arg Q$,

$$\frac{1}{2|Q|}\left(h_0, \frac{h_1}{\omega}, \frac{h_2}{\omega^2}, \frac{h_3}{\omega^3}\right) = (\cos\phi, -\sin\phi, -\cos\phi, \sin\phi) + o(1) \tag{10.2}$$

uniformly on this closed strip. Consequently $E, \mathcal{L}_0, \mathcal{L}_1$ *are strictly positive there at sufficiently large height. The hypothesis alone does not imply an* $O(L^{-1})$ *error in* (10.2).

Proof. The same disc estimates as in the closed-strip argument give $\Re w_\pm \ge 11/10$ at sufficiently large height. On $n \le Y$, $e^q - 1 \le qe^q$ bounds the discrepancy from the ordinary Dirichlet series by

$$\frac{t}{4}\sum_{n\ge 2}(\log n)^2 n^{-\{1/2+(81/400)t\ell - O(L^{-1})\}} = O(L^{-1}).$$

Indeed, at the real lower boundary the exponent tends to $1/2 + (81/400)(5/2) = 161/160$, and disc perturbations preserve the lower bound $1+1/320$. Also $t = O(L^{-1})$ on this fixed strip. The ordinary Dirichlet tail beyond Y is $O(e^{-19\ell/1000})$. Adding (10.1) therefore gives $S_{t,N_{\mathrm{RS}}}(w_\pm) - \zeta(w_\pm) = o(1)$ on the disc.

The complete imported value errors in (2.5) remain negligible: $e_A + e_B = O(L^2 u^{-3/4})$ and $e_{C,0} = O(u^{-2/5})$ suffice here. Carrier magnitudes on D_x are comparable to Γ. The upper-half-disc approximation and Schwarz reflection thus yield

$$H(t,z) - \mathcal{P}_t(z) = o(\Gamma), \qquad z \in D_x.$$

This difference is analytic even though the intermediate cutoffs can jump. Cauchy's estimates, applied only to this analytic difference, give errors $o(\Gamma L^k)$ for $0 \le k \le 3$ at x.

In $\Re w \ge 11/10$, the zeta factors are bounded above and away from zero, and their fixed-order logarithmic derivatives are bounded. The explicit carrier differentiation consequently gives

$$Q_\pm^{(k)}(x) = (\pm i\omega)^k Q_\pm(x) + O(\Gamma L^{k-1}), \qquad 1 \le k \le 3.$$

Together with $Q_-(x) = \overline{Q_+(x)}$ and $|Q| \asymp \Gamma$, this proves (10.2). Substitution into the quadratic forms gives

$$h_0^2 + (h_1/\omega)^2 = 4|Q|^2(1+o(1)), \qquad \mathcal{L}_j = 4|Q|^2\omega^{2j+2}(1+o(1)),$$

so the strict signs follow uniformly in phase. □

Thus a uniform logarithmic saving for the actual weighted remainder can suffice. The hypothesis retains its full disc uniformity and both endpoints; a pointwise real-axis estimate is not a substitute. An estimate obtained only after combining the two branches may also suffice if it directly proves the displayed analytic comparison $H - \mathcal{P}_t = o(\Gamma)$. Neither alternative is proved here.

10.2 A full-amplitude cutoff and its conditional loss transfer

Recall the exact staircase domain

$$\Omega_N = \{(v,n) : 0 \le v \le \tfrac{1}{2}\log N,\ n \le e^{2v}\}, \qquad N \ge 8,$$

and its retained jets $h_k^{(N)}$. Retain $a = \pi/16$, M_k, D_0, and c from (5.27), and put $c_0 = c$. In particular $|h_k|, |h_k^{(N)}| \le e^{-ax}\mathsf{M}_k$ for $0 \le t \le 1/2$ and $x \ge 0$. The complete staircase tail bound is

$$|h_k - h_k^{(N)}| \le e^{-ax}\mathsf{D}_0 e^{-c_0 N^2}, \qquad \mathsf{D}_0 = \frac{\mathsf{C}_0}{2\beta_0}, \quad c_0 = \frac{\beta_0}{2}. \tag{10.3}$$

It includes all omitted source indices and coordinates.

On S_j define the full-amplitude observation

$$J_{j,N} := \mathcal{J}_j^{(N)} = -h_j h_{j+2}^{(N)}, \qquad j = 0, 1. \tag{10.4}$$

This notation is distinct from the two-source integrand $J_j(\alpha, \beta)$. Full stationarity gives the exact identity and estimate

$$J_{j,N} - \mathcal{L}_j = h_j(h_{j+2} - h_{j+2}^{(N)}), \qquad |J_{j,N} - \mathcal{L}_j| \le D_j\sqrt{E}\, e^{-c_0 N^2}, \quad D_j = e^{-ax}\mathsf{D}_0. \tag{10.5}$$

In particular $J_{j,N} = 0$ at $E = 0$ for both levels. No stationarity condition is imposed on the retained jets.

Proposition 10.2 (Conditional full-amplitude cutoff transfer)**.** *Let $\mathcal{D}$ be a declared domain in $0 < t \le 1/2$, $x > 0$. Suppose, on $S_j \cap \mathcal{D}$, that the source-prescribed inequality*

$$J_{j,N} \ge -A_j N^p E - B_j e^{-cN^2} \quad \text{for every integer } N \ge N_0 \ge 8 \tag{10.6}$$

holds with a fixed finite $p \ge 0$, nonnegative A_j, B_j, and $c > 0$. Assume A_j, B_j, c^{-1}, N_0 are locally bounded through candidate positive-time collisions. Then, for each fixed $M > 0$, a source-prescribed locally bounded K_j satisfies

$$\mathcal{L}_j \ge -K_j E\,[1 + \log_+(M/E)]^{p/2} \quad \text{on } S_j \cap \mathcal{D}, \tag{10.7}$$

where the right side is zero at $E = 0$ and $\log_+ v = \max\{0, \log v\}$. If this is proved for both levels on neighborhoods containing the predecessor branches of every candidate collision in $\mathcal{D}$, it excludes those collisions.

Proof. Young's inequality in (10.5) gives

$$D_j\sqrt{E}\, e^{-c_0 N^2} \le \tfrac{1}{2}E + \tfrac{1}{2}D_j^2 e^{-2c_0 N^2}.$$

Set $\widehat{A}_j = A_j + 1/2$, $\widehat{B}_j = B_j + D_j^2/2$, and $\widehat{c} = \min\{c, 2c_0\}$. Since $N^p \ge 1$,

$$\mathcal{L}_j \ge -\widehat{A}_j N^p E - \widehat{B}_j e^{-\widehat{c}N^2}.$$

For $E > 0$ choose

$$N = \left\lceil \max\left\{ N_0, \sqrt{\widehat{c}^{-1}\log_+(\widehat{B}_j/E)} \right\} \right\rceil.$$

Put $\ell_E = 1 + \log_+(M/E)$ and $d_j = \log_+(\widehat{B}_j/M)$. Then the tail is at most E, and

$$N \le \Delta_j\sqrt{\ell_E}, \qquad \Delta_j = N_0 + 1 + \widehat{c}^{-1/2}\sqrt{1 + d_j}.$$

Thus (10.7) holds with $K_j = \widehat{A}_j\Delta_j^p + 1$, which is locally bounded under the stated hypotheses. At $E = 0$, let $N \to \infty$ in the full-form family. All cutoff inequalities precede this optimization; none of their coefficients is defined using a measured adverse quotient.

For the collision conclusion, the predecessor calculation in Lemma 3.3, applied to H or H_x according to the parity of the multiplicity, gives

$$E \asymp (T-t)^{2r}, \qquad -\mathcal{L}_j/E = \frac{r}{T-t} + O(1), \qquad r \geq 1.$$

The locally bounded right coefficient in (10.7) permits at most a fixed power of $1 + |\log(T-t)|$, contradicting this pole. This proves exclusion only in the covered neighborhoods. □

The proposition proves a transfer, not (10.6) for the theta source in the unbounded inner cusp. A bound only for one chosen cutoff, or with coefficients that depend on a sampled energy floor, does not establish its hypothesis.

10.3 Localizing the missing estimate near small energy

The nonzero scale Γ defined from (2.1) and (2.2) contains no zeta factor or physical jet denominator. It can therefore be used even where a zeta-carrier nonvanishing claim has not been established. The following observation separates a simple region from the difficult cancellation regime.

Lemma 10.3 (Small-energy localization)**.** *Let $\mathcal{D} \subset (0, 1/2] \times (0, \infty)$ be a declared domain on which Γ is continuous and strictly positive, and fix $\delta > 0$. For either observation $\mathcal{O}_{j,N} = J_{j,N}$ or $\mathcal{O}_{j,N} = \mathcal{L}_j^{(N)}$, suppose a polynomial-cutoff lower bound*

$$\mathcal{O}_{j,N} \geq -A_j N^p E - B_j e^{-cN^2}$$

has been proved for all $N \geq N_0 \geq 8$ on

$$S_j \cap \mathcal{D} \cap \{E \leq \delta \Gamma^2\}. \tag{10.8}$$

Here $p \geq 0$ is fixed and finite, and the prescribed nonnegative coefficients are defined on $\mathcal{D}$ with the local boundedness conditions of Proposition 10.2. The same type of bound then holds on all of $S_j \cap \mathcal{D}$ after replacing A_j by

$$\widetilde{A}_j = \max\left\{ A_j, \frac{e^{-2ax}\mathsf{M}_j\mathsf{M}_{j+2}}{\delta\Gamma^2} \right\}. \tag{10.9}$$

This new coefficient is locally bounded and independent of N.

Proof. For $J_{j,N}$ the separate complete and retained jet envelopes give $|J_{j,N}| \leq U_j$, where $U_j = e^{-2ax}\mathsf{M}_j\mathsf{M}_{j+2}$. For the raw cutoff the complete paired identity gives the same envelope $|\mathcal{L}_j^{(N)}| \leq U_j$. On $E > \delta\Gamma^2$,

$$\mathcal{O}_{j,N} \geq -U_j \geq -\frac{U_j}{\delta\Gamma^2} E.$$

Use $N^p \geq 1$ and combine this with the assumed bound on (10.8). Local boundedness follows from continuity and positivity of Γ at every finite point. □

For any hypothetical finite collision with $\Gamma > 0$ whose sufficiently nearby predecessor points stay in $\mathcal{D}$, continuity and $E = 0$ at the collision place those points in (10.8). Thus, wherever the declared domain covers these predecessors, the restricted small-energy region contains the essential collision obstruction. The localization assumes no positive energy floor there and does not prove the missing lower bound. Growth of the coefficient as height tends to infinity or as $t \downarrow 0$ is compatible with the local criterion.

10.4 Arithmetic directions and limits of the present coverage

The principal cusp target remains (10.6), preferably on the small-energy region (10.8). The complete correlations in Section 6 give a focused way to investigate it: derive an exact identity for $\mathscr{A}_j+\kappa_j\mathscr{E}$, or for the hybrid kernel $\mathscr{C}^a_{j,N}$, and estimate its adverse transform with source-prescribed coefficients. A positive-definite component with a paid signed remainder could suffice; global positive definiteness is stronger than the stationary target. The first deliverable is the identity with its complete remainder, followed by an all-N estimate on a declared domain. Pointwise positivity of a correlation density is not that estimate.

Poisson reflection alone exchanges the two correlation coordinates and is already shared by arbitrary even sources. An additional theta relation must improve the signed remainder, retain the logarithmic heat factor, and pay every transformed boundary. The hybrid formula (6.13) keeps the original staircase inside $\psi_{a,N}$, so it cannot justify discarding entering-index terms. The reflected lift and planar dictionary are useful finite algebraic reductions; further triality, integral-operator, or exceptional-theta work should proceed only after specifying a source- and observation-preserving identity that could improve this estimate.

The cumulative identity (9.28) provides a related diagnostic. Grouping consecutive oscillatory lobes before estimating the integrated signed overshoot may demand less than (9.29). The complete tail must still be paid at the physical energy scale; a merely absolute error proportional to $|h_j|$ is insufficient near a collision.

The central arithmetic gap is not removed by the published systematic exponent table [11, Table 1]. At block scale $\beta = 2/5$, its bound is

$$\frac{18}{199}+\frac{521}{796}\frac{2}{5}=\frac{701}{1990},\qquad \frac{701}{1990}-\frac{7}{20}=\frac{9}{3980}>0.$$

Here $7/20$ is the exponent allowed by the $\theta = 5/2$ target before its additional saving. This is a limitation of the quoted bounds, not an obstruction to a logarithmic-phase-specific or signed estimate.

The outer program seeks (10.1) through exact partitions, transformed logarithmic-phase sums, or cancellation across neighboring scales. A small diagnostic computation should specify its observation and coefficient rule before validation, retain complete tails, and increase the cutoff and precision when the physical carrier scale requires it. Along a sampling curve $t = \theta/L(x)$ the derivative used to isolate a root of the physical stationary observation is

$$\frac{d}{dx}h_{j+1}(\theta/L(x),x)=h_{j+2}+\frac{\theta}{xL(x)^2}h_{j+3}.$$

An undecided enclosure is not a sign violation; favorable finite cells do not establish an unbounded-domain estimate.

Finally, the nominal zeta argument has real part $1/2+\theta/4$. The elementary nonvanishing input for the present carrier works with a fixed margin when $\theta > 2$. At and below two that justification no longer applies. A different argument may cross the boundary, but dividing by a zeta factor whose nonvanishing has not been proved would import the missing information. This is a limitation of the present method, not an impossibility theorem for all carrier methods.

Even a sequence of outer thresholds $\theta_n \downarrow 0$ does not settle the global problem without controlling its associated starting heights X_n. At a fixed hypothetical collision (T, x_*), choosing n so that $\theta_n < T\log(x_*/(4\pi))$ may also give $X_n > x_*$. A finite-height or other global argument must close that quantifier gap. Effective bridging should therefore target an evaluated reachable analytic cutoff; repeated bounded extensions alone do not address the unbounded small-time cusp.

A Complete source and integration error bounds

The following bounds make the numerical and shared-source constructions independent of an unpaid finite theta truncation. Set

$$S_k(q,d)=\sum_{r=0}^{k}\binom{k}{r}\frac{d^{k-r}r!}{q^{r+1}}=\int_0^\infty (v+d)^k e^{-qv}\,dv \qquad (q>0).$$

Let $0\le t\le\tau\le 1/2$, $0<a<\pi/8$, and $x\ge 0$. For an index cutoff N, put $m=N+1$ and

$$c_a=\cos 4a,\quad \alpha=\pi m^2,\quad \beta=c_a\alpha,\quad q_N=4\beta-9,\quad \rho_N=((m+1)/m)^4e^{-\pi c_a(2m+1)}.$$

If $\rho_N<1$, $q_N>0$, and $8\beta\ge\tau$, the complete omitted index contribution in (5.1) is at most

$$E_{N,k}=e^{-ax}\frac{(2\alpha^2+3\alpha)e^{-\beta}}{1-\rho_N}S_k(q_N,a). \tag{A.1}$$

For a cutoff $U\ge 0$, put

$$\beta_U=\pi c_a e^{4U},\quad \rho_U=16e^{-3\beta_U},\quad q_U=4\beta_U-9-2\tau U,\quad B_U=\tau U^2+9U-\beta_U.$$

If $\rho_U<1$, $q_U>0$, and $8\beta_U\ge\tau$, the entire source on $u\ge U$ contributes at most

$$E_{U,k}=e^{-ax}\frac{(2\pi^2+3\pi)e^{B_U}}{1-\rho_U}S_k(q_U,U+a). \tag{A.2}$$

Thus $E_{N,k}+E_{U,k}$ pays the omitted domain of an N-term integral on $[0,U]$. Validated retained quadrature and cell variation are additional errors.

To prove the bounds, use

$$|\phi_n(u+ia)|\le(2\alpha_n^2+3\alpha_n)e^{9u-c_a\alpha_n e^{4u}},\quad |e^{t(u+ia)^2}|\le e^{\tau u^2},\quad |u+ia|^k\le(u+a)^k.$$

For $n\ge m$, successive coefficient-majorant ratios are bounded by ρ_N uniformly for $u\ge 0$. The inequality $e^{4u}\ge 1+4u+8u^2$ gives

$$\tau u^2+9u-\beta e^{4u}\le-\beta-q_Nu-(8\beta-\tau)u^2\le-\beta-q_Nu.$$

Summing the full geometric index bound and integrating proves (A.1). For $u\ge U$, the exponent $\tau u^2+9u-\pi c_ae^{4u}$ is concave and has slope $-q_U$ at U. Its tangent bound and the geometric ratio ρ_U give (A.2). Adding the bounds overcounts their intersection, which is harmless.

For the centered residual C_j, replace any factor $S_j(q,d)$ in these payments by $S_{j+2}(q,d)+b^2S_j(q,d)$, since $|u^2-b^2|\le(u+a)^2+b^2$. For the shared carrier (5.11), set

$$\widehat{\beta}_V=\pi\cos(4a)e^{4V},\quad \widehat{q}_V=4\widehat{\beta}_V-10-2\tau V,\quad \widehat{B}_V=\tau V^2+10V-\widehat{\beta}_V.$$

If $\widehat{q}_V>0$ and $8\widehat{\beta}_V\ge\tau$, the majorant (5.12) and the same tangent argument give the full tail payment

$$e^{-ax}2(2\pi^2+3\pi)e^{\widehat{B}_V}S_j(\widehat{q}_V,V+a). \tag{A.3}$$

The same degree-two replacement pays the shared expression for C_j, because $0\le v-d_n\le v$.

For completeness, these absolute errors can be made small at the gamma exponential scale at arbitrary height. For $x\ge 41$, $p>0$, and $\ell_x=\log x$, take

$$a_x=\pi/8-1/x,\quad m=\left\lceil\sqrt{\frac{x}{2\pi}(p\ell_x+32)}\right\rceil,\quad N=m-1,\quad U=\frac{\ell_x+2\log(\ell_x+8)}{4}.$$

All preceding payment hypotheses hold, and for fixed k,

$$e^{\pi x/8}E_{N,k} = O_{p,k}(x^{5/2-p}(\log x)^{3/2}), \qquad e^{\pi x/8}E_{U,k} = O_{k,A}(x^{-A}) \quad (A>0). \tag{A.4}$$

Indeed $c_a = \sin(4/x) \geq 2/x$, $\beta \geq p\ell_x + 32$, and $m^2 \geq 2x/\pi$. Hence $-\log\rho_N \geq 4\pi m/x - 4/m \geq 2\pi m/x$ and $(1-\rho_N)^{-1} \leq 1 + x/(2\pi m)$. Together with $e^{-a_x x} = e\,e^{-\pi x/8}$, substitution gives the first bound. For the second, $\beta_U \geq 2\pi(\ell_x+8)^2$, $U \leq \ell_x/2+2$, and $B_U \leq -47\ell_x^2/8 - 181\ell_x/2 - 364$. These are absolute error estimates. They neither establish a norm gap nor replace a signed estimate of the retained source. On a spatial cell, freeze the contour and cutoffs before differentiating or transporting physical jets.

B An explicit height at the condition $tL \geq 5$

The exponent-pair argument in Section 2 gives the stronger asymptotic condition $tL \geq 61/24$, with an unevaluated height. The following elementary absolute-sum calculation supplies an explicit height for $tL \geq 5$. These are distinct quantitative statements.

Proposition B.1 (Explicit outer height)**.** *Let $L = \log(x/(4\pi))$, $\omega = L/4$, and let $Q = Q_+(x)$ be the amplitude in (2.4). If*

$$x \geq X_{\exp} := 4\pi e^{10^9}, \qquad 0 < t \leq \tfrac{1}{2}, \qquad tL \geq 5,$$

then every coordinate error in (1.9) is at most

$$\frac{108000000}{L} \leq 0.108 < \frac{1}{8}. \tag{B.1}$$

In particular, for $P_k = h_k/(2|Q|\omega^k)$,

$$P_0^2 + P_1^2 \geq \tfrac{1}{2}, \qquad P_1^2 - P_0P_2 \geq \tfrac{31}{64}, \qquad P_2^2 - P_1P_3 \geq \tfrac{31}{64}.$$

The same conclusions hold at negative heights after reflection.

B.1 Explicit weights on the complex disc

All bounds until the last sign step hold for $L \geq 1000$, $0 < t \leq 1/2$, and $tL \geq 5$. On D_x from (2.13), write $z = u + iv$ and $\theta_u = tL_u$. Then

$$u \geq x/2 \geq e^L, \qquad |L_u - L| \leq \frac{2}{xL} < 1, \qquad \theta_u \geq 5 - \frac{1}{xL} \geq \frac{19}{4}. \tag{B.2}$$

For $s = s_\pm(z)$, one has $0 \leq \Re s \leq 1$ and $|s|, |s-1| \geq u/2$. Formula (2.2) therefore yields

$$\Re\alpha(s) \geq L_u/2 - 3/u, \qquad |\alpha(s)| \leq L_u/2 + 1 \leq L, \qquad |\alpha'(s)| \leq 1/u + 6/u^2 \leq 2/u. \tag{B.3}$$

For the magnitude estimate, use $|s| \leq (u/2)\sqrt{1+4/u^2}$, $|\arg s| \leq \pi/2$, and the bound $3/u$ for the rational terms. It follows that

$$\sigma_\pm := \Re w_\pm \geq \frac{1}{2} - \frac{1}{2L} + \frac{\theta_u}{4} - \frac{3t}{2u} \geq \frac{3}{2}. \tag{B.4}$$

The cutoff satisfies

$$\tfrac{1}{2}e^{L_u/2} \leq N, \qquad \log N \leq L_u/2 + 1/u.$$

Indeed the upper correction is at most $\pi t/(8u)$, and flooring reduces a number at least two by at most a factor of two. Consequently, for $n \le N$,

$$e^{t\log^2 n/4} n^{-\sigma_\pm} \le n^{-p}, \qquad p = \frac{1}{2} + \frac{\theta_u}{8} - \delta, \qquad \delta = \frac{1}{2L} + \frac{7t}{4u} \le \frac{1}{32}. \tag{B.5}$$

Writing $s_0 = \theta_u - 19/4 \ge 0$, we obtain $p \ge 17/16 + s_0/8$. On the upper half-disc, the same bound holds for the first branch after multiplication by n^v: its leading real part changes from $(1+v)/2$ to $(1-v)/2$, already included in the estimate.

For $\eta > 0$ and an integer $m \ge 0$, comparison on $[n-1, n]$ gives

$$\sum_{n=2}^{\infty} (\log n)^m n^{-1-\eta} \le \int_1^{\infty} (\log y + 1)^m y^{-1-\eta}\, dy. \tag{B.6}$$

At $m = 2$, $\eta = 1/16$, the right side is $2/\eta^3 + 2/\eta^2 + 1/\eta = 8720 < 9000$. Hence

$$\begin{aligned} |S_{t,N}(w_\pm) - \zeta(w_\pm)| &\le \frac{9000}{4L_u}(19/4 + s_0)2^{-s_0/8} + 2N^{-1/2} \\ &\le \frac{29250}{L_u} + 3e^{-L_u/4} \le \frac{31000}{L}. \end{aligned} \tag{B.7}$$

Here $s_0 e^{-s_0 \log 2/8} \le 8/(e\log 2) < 8$, so the time-dependent numerator is less than 13. Moreover, $29250/(L-1) < 30000/L$ for $L \ge 1000$, while $e^{(L-1)/4} \ge (L-1)^2/32 \ge 3L$. Thus the displayed bound includes both the weighted discrepancy and the complete ordinary zeta tail, uniformly even as tL grows.

B.2 All imported approximation errors

On the upper half-disc, (2.6) gives $|\gamma| \le 2$ and

$$|\kappa| \log N \le \frac{tv}{2(u-6)}(L_u/2 + 1/u) < \tfrac{1}{2}, \qquad N^{|\kappa|} \le 2.$$

Since $\zeta(17/16) \le 17$, the full positive weight sum in (2.7) is at most $5 \cdot 17 = 85$. Denote that equation's exponent by

$$q_n = \frac{(t^2/16)\log^2(u/(4\pi n^2)) + 0.626}{u - 6.66}.$$

The estimates $-2/u \le \log(u/(4\pi n^2)) \le L_u$, $(L+1)^2/64 + 1 \le L^2/32$, and $u - 6.66 \ge u/2$ imply $0 \le q_n \le L^2/(16u) < 1$. Using $e^q - 1 \le 2q$ on $[0,1]$ gives

$$e_A + e_B \le \frac{85L^2}{8u} \le \frac{11L^2}{u} \le \frac{1}{L}. \tag{B.8}$$

For the final inequality, $u \ge e^L \ge L^4/24 \ge 11L^3$ suffices.

The positive correction exponent in (2.8) is at most

$$\frac{1.24(3^v + 3^{-v})}{N - 0.125} + \frac{3|L_u + i\pi/2| + 10.44}{u - 12} \le 20e^{-L_u/2} + 7L/u \le 1.$$

To verify the last inequality, use $20e^{-(L-1)/2} \le 160/(L-1)^2 < 1/2$ and $7Le^{-L} \le 168/L^3 < 1/2$. Its negative main exponent therefore gives

$$e_{C,0} \le e\exp\{-(1/4 + 19/64)L_u\} \le 3e^{-L_u/2} \le \frac{1}{L}, \qquad e_A + e_B + e_{C,0} \le \frac{2}{L}. \tag{B.9}$$

All three errors in the imported theorem are included; the constant 0.626 in q_n is independent of t.

B.3 Amplitude comparison and all four derivatives

Let $\Gamma = |M_t(s_+(x))| = |M_t(s_-(x))|$. On D_x,

$$\left|\frac{d}{dz}\log M_t(s_\pm(z))\right| \le \frac{|\alpha|}{2}(1+t|\alpha'|/2) \le L.$$

Integration along a segment of length at most $1/L$ yields $|M_t(s_\pm(z))| \le e\Gamma < 3\Gamma$. Also (B.4) and the zeta and reciprocal-zeta Dirichlet series give $|\zeta(w)|, |1/\zeta(w)| \le \zeta(3/2) \le 3$, hence $\Gamma \le 3|Q|$. Combining (B.7) and (B.9) with (2.5) gives

$$\begin{aligned}|H(t,z) - \mathcal{P}_t(z)| &\le 2(3\Gamma)\frac{31000}{L} + (3\Gamma)\frac{2}{L} = \frac{186006\Gamma}{L}\\ &\le \frac{558018|Q|}{L} \le \frac{560000|Q|}{L}.\end{aligned} \tag{B.10}$$

Schwarz symmetry extends this upper-half-disc estimate to the full disc. Cauchy's inequality now gives

$$\frac{|\partial_x^k(H(t,\cdot) - \mathcal{P}_t)(x)|}{2|Q|\omega^k} \le \frac{280000\,k!\,4^k}{L} \le \frac{107520000}{L}, \qquad 0 \le k \le 3. \tag{B.11}$$

For the analytic carrier, at $s = (1+ix)/2$ direct differentiation of (2.2) gives

$$|\alpha - (L/2 + i\pi/4)| \le 4/x, \quad |\alpha'| \le 2/x, \quad |\alpha''| \le 3/x^2, \quad |\alpha'''| \le 9/x^3. \tag{B.12}$$

The first estimate follows from $s = (ix/2)(1 - i/x)$ and $|\operatorname{Log}(1 - i/x)| \le 2/x$. For the others one uses

$$\alpha' = -\frac{1}{2s^2} - \frac{1}{(s-1)^2} + \frac{1}{2s}, \qquad \alpha'' = \frac{1}{s^3} + \frac{2}{(s-1)^3} - \frac{1}{2s^2}, \qquad \alpha''' = -\frac{3}{s^4} - \frac{6}{(s-1)^4} + \frac{1}{s^3}.$$

Let $J(x) = \log M_t(s(x))$. Then

$$\begin{aligned}J' &= (i/2)\alpha(1 + t\alpha'/2),\\ J'' &= -\tfrac{1}{4}\{\alpha' + (t/2)[(\alpha')^2 + \alpha\alpha'']\},\\ J''' &= -\tfrac{i}{8}\{\alpha'' + (t/2)[3\alpha'\alpha'' + \alpha\alpha''']\}.\end{aligned}$$

Equations (B.3) and (B.12) imply $|J' - i\omega| \le 1$, $|J''|, |J'''| \le 1$ and $|w'|, |w''|, |w'''| \le 1$. A majorant for the first bound is $\pi/8 + 2/x + L/(4x) < 1$; the remaining bounds follow by inserting the stated powers of x^{-1} and $x \ge e^L$.

The prime-power series and (B.6) at $\eta = 1/2$ give

$$|(\log\zeta)'(w)| \le 6, \qquad |(\log\zeta)''(w)| \le 26, \qquad |(\log\zeta)'''(w)| \le 158.$$

Thus for a local logarithm $g = \log Q_+$, the chain rule yields

$$|g' - i\omega| \le 7, \qquad |g''| \le 1 + 26 + 6 = 33, \qquad |g'''| \le 1 + 158 + 3\cdot 26 + 6 = 243.$$

The resulting carrier errors are

k	$\dfrac{\lvert Q_+^{(k)}/Q - (i\omega)^k\rvert}{\omega^k}$
1	$28/L$
2	$56/L + 1312/L^2$
3	$84/L + 3936/L^2 + 81856/L^3$

(B.13)

Every entry in the right-hand column is at most $100/L$ for $L \geq 1000$. At order three, for instance, expanding $(i\omega + d)^3 + 3(i\omega + d)g'' + g'''$ with $|d| \leq 7$ bounds the unnormalized discrepancy by $21\omega^2 + 246\omega + 1279$. The conjugate branch contributes the same normalized bound; the factor two is already included in $2|Q|\omega^k$. The carrier error at order zero is exactly zero.

Adding (B.11) and (B.13), every coordinate error is at most $107520100/L < 108000000/L$. For $L \geq 10^9$ this is less than $1/8$, and (2.32) proves Proposition B.1.

The value X_{exp} is an explicit consequence of the primary approximation and elementary inequalities. Its exponential need not be evaluated numerically. The accompanying exact checker verifies 52 rational budgets and sufficient polynomial inequalities for every $L \geq 1000$ by shifting $L = 1000 + y$ and checking nonnegative coefficients. It audits the arithmetic in this proof, rather than replacing its analytic arguments. The bound supplies no practical finite-strip coverage between 112 and X_{exp}, and it does not evaluate the height in the stronger closed-threshold theorem.

C Reproducibility and scope of the checks

All paths in this appendix are relative to `papers/investigations-2/active/21_quaternion_theta_geometry/` in the accompanying repository. The analytic proofs above are distinct from the finite interval decisions. The latter use complete error bounds, outward rounding, and closed-cell coverage; hashes identify the evaluated sources and records, but do not substitute for their mathematical content.

The bounded results can be regenerated using the drivers `numerics/certify_compensated_stationary_time_extension.py`, `numerics/certify_correlated_source_height_extension.py`, and `numerics/certify_phase_centered_residual.py`. The correlated-height guide `numerics/CORRELATED_SOURCE_HEIGHT_README_20261014.md` gives the parameters for its three bands. The full-time certificate is `numerics/COMPENSATED_STATIONARY_TIME_EXTENSION_20261014.json`; the final centered-band record is `numerics/THETA_CENTERED_RESIDUAL_100_104_20261014.json`. The independent internal replays are documented in Reviews 6, 9, and 11 in `reviews/`. They reproduced the numerical records after removal of runtime metadata.

For compact exact audits, run the following from `numerics/`:

```
python3 audit_correlated_source_height_records.py
python3 audit_theta_centered_residual_record.py
python3 check_octonion_jet_identities.py
python3 check_theta_centered_residual_identities.py
```

The centered-band auditor additionally requires `-record THETA_CENTERED_RESIDUAL_100_104_20261014.json`. Each requires the option `-output` followed by a desired JSON path. The first two inspect the retained records and bound source versions; the latter two check 248 and nine exact algebraic identities, respectively. The octonionic identities, centered residual, and outer analytic argument have separate internal assessments in Reviews 10–12. The diagnostic `numerics/check_outer_cusp_and_phase_obstructions.py`, also run with `-output`, requires python-flint and imports `numerics/check_shifted_contour_payments.py`. Its retained record `numerics/OUTER_CUSP_AND_PHASE_OBSTRUCTION_CHECKS_20261014.json` contains the pointwise counterexamples in Section 8, twelve staircase boundary checks, exact four-jet perturbation algebra, and selected complete-source point enclosures. It does not certify a new rectangle or prove the uniform asymptotic theorem.

The subsequent paired algebra, explicit outer constants, and signed-split controls are audited by `numerics/check_signed_carrier_pair_identities.py`, `numerics/check_explicit_outer_constants.py`, and `numerics/check_signed_curvature_split.py`. They require only the Python standard library and an `-output` path, and check 34, 52, and 14 exact statements respectively. Their records are `numerics/SIGNED_CARRIER_PAIR_IDENTITY_CHECKS_20261014.json`, `numerics/EXPLICIT_OUTER_CONSTANT_CHECKS_20261014.json`, and `numerics/SIGNED_CURVATURE_SPLIT_IDENTITY_CHECKS_20261014.json`. These checks support algebra and explicit inequalities; the infinite source and asymptotic arguments require the proofs in the text.

The full-residual branch comparison is recorded in `numerics/STATIONARY_FAVORABLE_CURVATURE_PILOT_20261014.json`. The proper source-filter and Gaussian pilots are produced by `numerics/certify_predetermined_source_filter_pilot.py` and `numerics/check_gaussian_source_filter_pilot.py`. Each accepts `-record`, `-original-driver`, `-contour-source`, and `-output`; the inputs are the centered-band record, its driver, and `numerics/check_shifted_contour_payments.py`. The proper-filter run uses python-flint 0.9.0 at 192 bits and retains all 64 classifications, including 32 failed absorption conditions. The Gaussian extraction uses exact rational arithmetic and retains all 14 cases. Their records are `numerics/PREDETERMINED_SOURCE_FILTER_PILOT_20261014.json` and `numerics/GAUSSIAN_SOURCE_FILTER_PILOT_20261014.json`. Reviews 14–16 document the analytic audits and independent replays, including the eight full-residual branch slabs. Review 17 audits the earlier manuscript integration; its fresh exact checks are distinct from the preceding full interval replays.

The height-112 extension and its all-earlier-time comparisons use `numerics/certify_fixed_cusp_band_extension.py` and the two records `numerics/FIXED_CUSP_BAND_104_108_20261014.json` and `numerics/FIXED_CUSP_BAND_108_112_20261014.json`. The exact auditor `numerics/audit_fixed_cusp_band_extension.py` accepts both paths through repeated `-record` options and an `-output` path. Its 1321 checks retain all 64 closed cells, 128 stationary observations, and 16 genuine branch/time cases. The source quadratures were independently replayed in Review 19; the preceding manuscript integration reran the exact auditor, not those quadratures.

The newly integrated arithmetic and collision material is audited by `numerics/check_proportional_delay_identities.py` (20 checks), `numerics/check_phase_controlled_delay_identities.py` (21), `numerics/check_collision_behavior.py` (1072), `numerics/check_hybrid_cutoff_lemma.py` (84), and `numerics/check_remaining_strip_lemma.py` (97). The proportional checker takes a positional output path; the phase and strip checkers accept `-output`. The collision driver writes its named receipt beside its source; the integration replay calls its pure check routines without replacing that historical receipt. Together with the finite-band audit, the preceding integration executed 2615 exact checks. Its receipt is `numerics/MANUSCRIPT_PROGRAM_INTEGRATION_CHECKS_20261015.json`. These checks verify finite algebra, rational margins, and recorded interval decisions. They do not prove the imported arithmetic theorems, the unproved middle-block or weighted-remainder hypotheses, or an all-height theta sign estimate. Review 24 records the integrated proof and layout audit, including the conditional deductions from Note 33.

The octonionic simplification and literature integration is documented in Reviews 25–26. The standard-library checker `numerics/check_octonion_reflected_lift.py`, run with `-output`,

verifies 325 exact polynomial assertions: the full eight-dimensional reflected operator, physical trace, heat and jet identities, and the planar bilinear reductions. The original 248-check octonion driver was also replayed. The independent moment and multiplier checks for the correlation formulas are supplied by `numerics/check_theta_correlation_identities.py`; its scope and the consolidated receipt are recorded in Review 26. These finite controls support the general proofs above; they add no theta sign certificate or interval-quadrature coverage.

The main remaining obligation is the one-sided inner-region bound (8.5), with losses locally bounded through candidate positive-time collisions. Equations (6.9) and (6.13) organize that same obligation using complete source correlations. Improving the explicit height and connecting the finite cover remain separate quantitative work. Neither a fixed-time asymptotic nor a ratio divided by unknown physical energy supplies the missing bound.

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